EEBus Initiative e.V.
Deutz-Mülheimer-Straße 183
51063 Cologne
GERMANY

Rue d'Arlon 25
1050 Brussels
BELGIUM

Phone: +49 221 / 715 938 - 0
Fax: +49 221 / 715 938 - 99

info@eebus.org
www.eebus.org

District court: Cologne
VR 17275

EEBus UC Technical Specification

Incentive-Table based Power Consumption Management

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1 Scope of the document

This document describes the Use Case "Incentive-Table based Power Consumption Management" (short-name: ITPCM). Chapter 2 specifies the High-Level Use Case. Chapter 3 details the technical solution for SPINE for this Use Case. Within this document, a top-down approach is used to derive the requirements for the technical solution from the High-Level description. In Annex A, the best practice message flow as well as example XMLs are provided for the technical SPINE solution.

1.1 References

1.1.1 EEBUS documents

[ProtocolSpecification]	*Minimum*: EEBus_SPINE_V1.3.0_Final.zip -> EEBus_SPINE_TS_ProtocolSpecification.pdf	
[ResourceSpecification]	*Minimum*: EEBus_SPINE_V1.3.0_Final.zip -> EEBus_SPINE_TS_ResourceSpecification.pdf	
[SHIP]	*Minimum*: *Recommended*:	EEBus_SHIP_TS_Specification_v1.0.1.pdf EEBus_SHIP_TS_Specification_v1.1.0.pdf

1.1.2 Normative references

[RFC2119]	IETF RFC 2119: 1997, Key words for use in RFCs to indicate requirement levels. Please see section 1.3.1 for details.

1.2 Terms and definitions

Actor

An Actor models a role within a Use Case definition (e.g. an energy manager or an electric vehicle).

CEM

Abbreviation for Customer Energy Manager. The CEM is an energy manager located at the user's home or premises or in a cloud application. The energy manager enables energy-optimized operation of the connected devices by harmonising energy demand and availability.

CHP

Abbreviation for Combined Heat Power.

IT

Abbreviation for Incentive Table.

ITPCM

Incentive-Table based Power Consumption Management (short-name of this Use Case)

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Passive sign convention
An electrical current is positive if the current is flowing into a device or component. In this case, the device or component consumes electrical power and the active power is greater than zero. An electrical current is negative if the current is flowing out of a device or component. In this case, the device or component produces electrical power and the active power is smaller than zero.

PIT
Abbreviation for Preliminary Incentive Table.

PV
Abbreviation for Photovoltaic (system)

Scenario
Part of a Use Case. Splitting a Use Case into Scenarios helps readers understand the Use Case more quickly. Some Scenarios are mandatory for a Use Case, whereas others may be recommended or optional.

Specialization
Reusable data collection for a specific functionality.

SPINE
Smart Premises Interoperable Neutral-message Exchange: Technical Specification of EEBus Initiative e.V.

UTC
Abbreviation for Coordinated Universal Time.

1.3 Requirements

1.3.1 Requirements wording

The following keywords are used:

- SHALL
- SHALL NOT
- SHOULD
- SHOULD NOT
- MAY

Note: They apply only if written in capital letters.

For the meaning of the keywords, please refer to [RFC2119].

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1.3.2 Mapping of High-Level requirements

Within the High-Level Use Case description, the following abbreviation is used:

[ITPCM-xyz]

e.g.: [ITPCM-007]

The abbreviation is used to mark High-Level requirements or rules of this Use Case with a unique number xyz. These requirements are referenced throughout the technical solution to show how each High-Level requirement is realized in the technical part.

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2 High-Level description

2.1 Introduction

Generally, this Use Case aims at adjusting the operation process of an Energy Consumer such that higher-level constraints (also constraints and demands of other devices at the customer's site) or optimization goals are met.

Examples for optimization goals are:

- the reduction of the electricity costs associated with an operation process,
- the reduction of the carbon footprint (CO_2 emission, e.g.) associated with an operation process,
- compliance with constraints of higher grid levels,
- the coordinated realization of demand response setpoints from higher-level aggregators.

To achieve the goals an Actor CEM can influence the consumption of an Actor Energy Consumer with incentives. For this the CEM sends an Incentive Table that defines different tiers with power levels and incentives. Different kinds of incentives are possible, like price, CO_2 emission or renewable energy percentage.

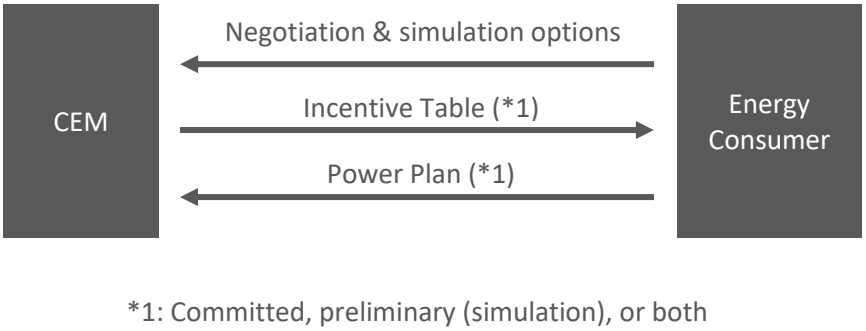


Figure 1: High-Level Use Case functionality overview

Based on the Incentive Table, the Energy Consumer creates a plan of its power consumption (called "Power Plan"), while considering its own constraints resulting from the user-configuration and other technical constraints.

The Use Case is accompanied with an important feature called "simulation phase":

The above-mentioned Incentive Table and Power Plan normally come into effect once they have been issued/sent. They are therefore called "committed data". In order to refine Power Plans iteratively, an additional simulation phase is allowed. Data that is exchanged in the context of this simulation phase is called "preliminary data".

Without simulation phase, the CEM can just submit a Committed Incentive Table - upon request by the Energy Consumer - without much room for renegotiation. Though renegotiations of committed data may be feasible, it can be expected that many types of Energy Consumers need to stick to their Committed Power Plan for a while (esp. due to mechanical or thermal constraints) before they are able to alter it upon a renegotiation.

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With simulation phase, on the other hand, (re)negotiations can occur at higher frequency as the simulation has no immediate physical impact. This gives the CEM the opportunity to increase the overall energy demand optimization through cyclical variations of the Preliminary Incentive Table and observation of the resulting Preliminary Power Plans. I.e. the CEM can use a simulation phase to create a Committed Incentive Table in such way that it can expect the resulting Committed Power Plan to support the CEM's overall energy optimization. The number of simulation cycles to achieve an optimization depends on the capability of the Energy Consumer and the constraints of the CEM.

Note on incentives and Power Plans:
The incentives exchanged in this Use Case are meant as "realistic" values, though they cannot be used for any kind of billing or similar purposes. This means an incentive reflects a sensible estimation of an Energy Consumer's portion of the finally resulting "price" (e.g.) of a building. This holds for the simulation phase as well: Intended variations of incentives need to be considered as "realistic" values under the assumption of a hypothetical (but realistic) set of power consumptions and generations of all devices of the building. Section 2.2.4.1 gives some further background information on "realistic" Incentive Tables. Likewise, Committed Power Plans as well as Preliminary Power Plans shall contain realistic values.

Note on the relation between committed and preliminary data:
It cannot be expected that identical Committed Incentive Tables and Preliminary Incentive Tables lead to identical Committed Power Plans and Preliminary Power Plans. Differences may occur for several reasons (change of environmental conditions, non-deterministic algorithms, etc.).

2.2 Detailed background information

2.2.1 Incentive Tables

2.2.1.1 Structure

An Incentive Table describes incentives over time ([ITPCM-010^{IT_GEN}]). Each incentive ([ITPCM-022^{INCENTIVE}]) can depend on the consumed power ([ITPCM-012^{IT_POWER}]). This dependency is expressed with so-called tiers ([ITPCM-013^{IT_TIER}]). The time is split into slots ([ITPCM-011^{IT_SLOT}]). Figure 2 shows an example.

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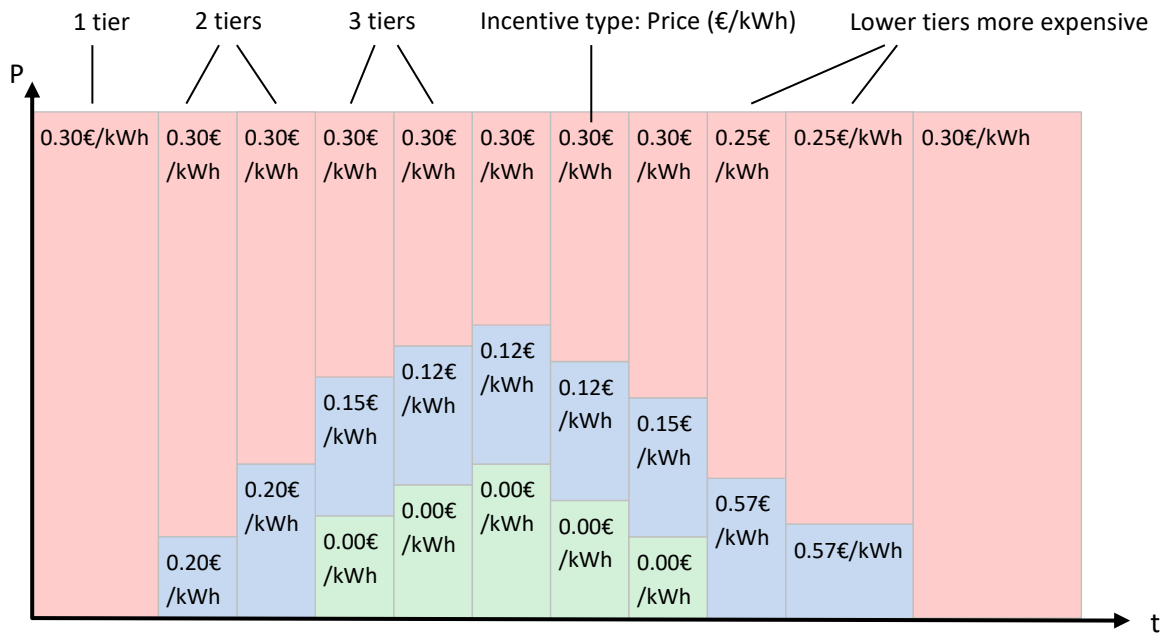


Figure 2: Incentive Table example: Power (P) dependent incentives over time (t)

Each incentive only applies within its power boundary ([ITPCM-012^{IT_POWER}]) and only for the share of the power within these boundaries. This means for the calculation of the total incentive of a given power P, all tiers need to be considered that intersect the range from 0W to the consumed power P. Then, for each relevant tier the respective share of the power P is applied. This will be explained in more detail with some examples in Figure 3.

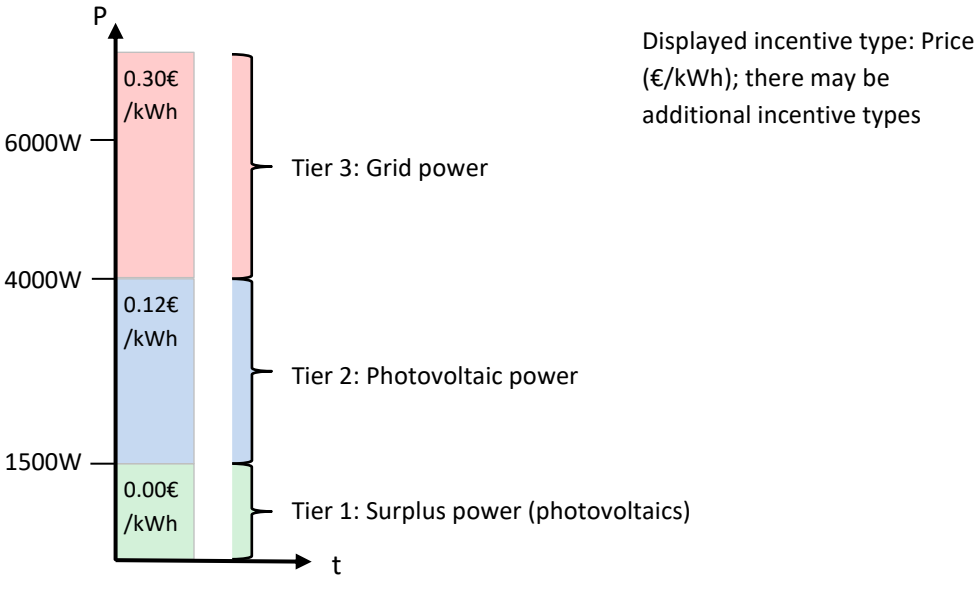


Figure 3: Single slot of an Incentive Table with three tiers

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Example 1: If a power of 5000W is consumed:

- 1500W are for free (tier 1)
- 2500W cost 0.12€/kWh (tier 2)
- 1000W cost 0.30€/kWh (tier 3)

Example 2: If a power of 1000W is consumed:

- 1000W are for free (tier 1)

Example 3: If a power of 2000W is consumed:

- 1500W are for free (tier 1)
- 500W cost 0.12€/kWh (tier 2)

The power boundaries of the tiers SHALL be greater or equal to zero (W) ([ITPCM-035^{IT_CONSUMPTION}]). There SHALL NOT be gaps between power boundaries of adjacent tiers ([ITPCM-040^{IT_P_GAPLESS}]).

The Energy Consumer can limit the number of boundaries per tier ([ITPCM-043^{MAX_BOUNDARIES}]).

Note: This Use Case does not prescribe a particular order of the incentive magnitudes within a slot. With regards to price-based incentives a "lower" tier may be more expensive than an "upper" tier.

The Actor Energy Consumer stores Incentive Tables received by the Actor CEM. The Actor Energy Consumer SHALL support storing Incentive Tables with up to 3 tiers and it SHOULD support storing Incentive Tables with up to 5 tiers. The Actor Energy Consumer SHALL express the number of supported tiers ([ITPCM-009^{MAX_TIERS}]). Section 2.2.4.2 shows an example how a CEM may "compress" an Incentive Table with 5 tiers into an Incentive Table with 3 tiers.

Each tier MAY include a human readable description ([ITPCM-029^{TIER_DESCR}]). The human readable description can be freely chosen by the Actor CEM or its user. The main goal is to enable the Actor Energy Consumer to show the energy mix of a Power Plan to a user. The example in Figure 3 shows some possible descriptions for the three tiers ("Grid power", "Photovoltaic power", "Surplus power (photovoltaics)"). As a tier may also be based on multiple energy sources, a proper description could show this as well ("Photovoltaics & battery", e.g.). Note: The content of the human readable description is not standardized in any way, thus the meaning of the human readable description should only be interpreted by human interaction and not by a device's algorithm.

The examples above only show price-based incentives. In general, each tier can comprise of up to three distinct incentive types. The Energy Consumer can limit the number of incentives per tier ([ITPCM-014^{MAX_INCENTIVES}]). The following incentive types are permitted:

- Absolute price per energy (unit currency/kWh) ([ITPCM-016^{ABSOLUTE_PRICE}])
- Renewable energy percentage ([ITPCM-017^{RENEWABLE_PCT}])
- CO_2 emission per energy (unit kg/kWh) ([ITPCM-018^{CO2}])

Note: Negative values (i.e., profits) for absolute prices are permitted ([ITPCM-020^{SIGNED_PRICES}]).

Each tier SHALL have at least an absolute price ([ITPCM-016^{ABSOLUTE_PRICE}]).

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The set of incentive types per tier SHALL be the same throughout the whole Incentive Table*[1], i.e. across all tiers and all slots ([ITPCM-019^{UNIQUE_I_TYPES}]). This means that if one tier of one slot contains an incentive for an absolute price in €/kWh, and also an incentive for renewable energy percentage, then all other tiers in all other slots of the Incentive Table must have these two incentive types as well. Only the values of the incentives may differ ([ITPCM-023^{INCENTIVE_VALUE}]) (and the power boundaries of the tiers as well).

*[1]: Different Incentive Tables may have different sets of incentive types.

2.2.1.2 Time information

An Incentive Table SHALL NOT begin later than "now". This means that an Incentive Table that is submitted at time "x" is only valid if the table states an incentive for a slot that covers time "x" (i.e., the start time is either in the past or right now and the end time is in the future).

The start time and end time of each slot SHALL be expressed as UTC values ([ITPCM-021^{SLOT_PERIOD}]). An Incentive Table SHALL NOT have time gaps between two slots ([ITPCM-033^{IT_GAPLESS}]). This means the end time of a slot SHALL be determined by the start time of its subsequent slot. The last slot of an Incentive Table SHALL state its end time explicitly as UTC value.

An Incentive Table SHALL NOT be longer than 7 days (168 hours).

If an Incentive Table is submitted by Actor CEM to Actor Energy Consumer at time "x", the following additional rules apply:

1. The Incentive Table SHALL have slots until time "x" + 24 hours at least.
2. The Incentive Table SHOULD have slots until time "x" + 48 hours at least.

Note (informative): An Incentive Table that covers a long time period may have a higher time resolution (i.e. more slots) for the near future, while slots of the distant future have a coarser time resolution. In general, the time resolution should be chosen to express significant changes in the tier (e.g. the power level or an incentive of a tier changes). E.g., if neither the incentive values nor the tier's power boundaries change over the time span of the Incentive Table, only a single time slot is needed.

The Energy Consumer can limit the number of slots per Incentive Table ([ITPCM-044^{MAX_SLOTS}]).

2.2.1.3 Committed vs. Preliminary Incentive Table

Actor Energy Consumer SHALL provide a resource for a Committed Incentive Table.

If Actor Energy Consumer supports a simulation phase, the Actor SHALL provide a resource for a Preliminary Incentive Table.

It SHALL be identifiable whether an Incentive Table is a Committed Incentive Table ([ITPCM-045^{IT_COMMITTED}]) or a Preliminary Incentive Table ([ITPCM-046^{IT_PRELIMINARY}]).

Note: Actor Energy Consumer is expected to calculate a Preliminary Power Plan the same way as it calculates a Committed Power Plan.

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Note: The time information rules in section 2.2.1.2 apply for each type of Incentive Table.

2.2.2 Power Plans

2.2.2.1 Structure

The Actor Energy Consumer can forecast its anticipated power consumption with a Power Plan ([ITPCM-024^{PLAN}]). The Power Plan describes power values over time. The time is split into slots ([ITPCM-025^{PLAN_SLOT}]). For each slot at least the expected (average) power consumption P_{exp} ([ITPCM-027^{PLAN_PEXP}]) SHALL be given. Optionally, the minimum power consumption P_{min} ([ITPCM-026^{PLAN_PMIN}]) or maximum power consumption P_{max} ([ITPCM-028^{PLAN_PMAX}]) MAY be given (or both). Figure 4 shows an example.

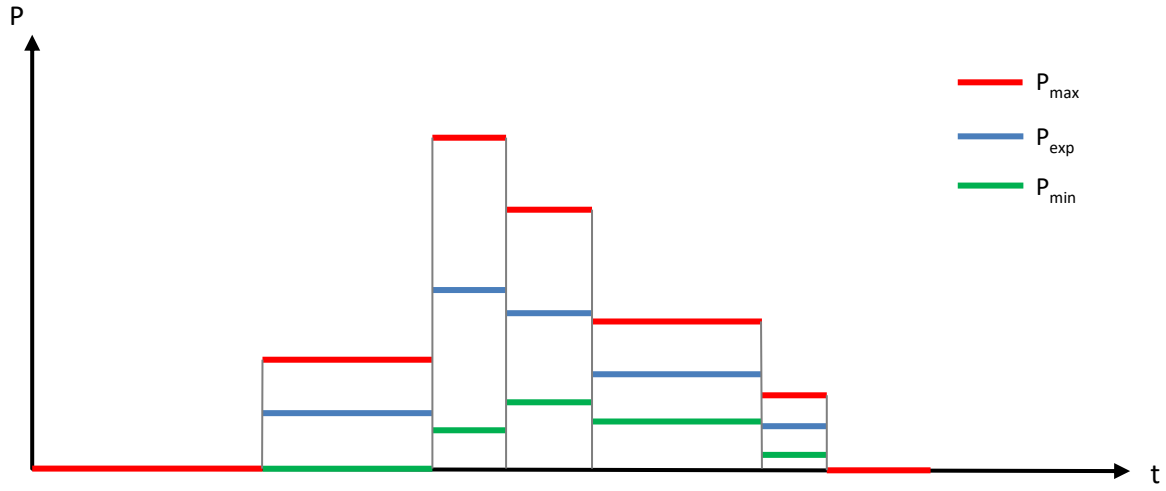


Figure 4: Power Plan example: Power (P) curves over time (t)

2.2.2.2 Time information

A Power Plan SHALL NOT begin later than "now". This means that a Power Plan that is submitted at time "x" is only valid if the plan states a power value for a slot that covers time "x" (i.e., the start time is either in the past or right now and the end time is in the future).

The start time and end time of each slot SHALL be expressed as UTC values ([ITPCM-041^{UTC}]). A Power Plan SHALL NOT have time gaps between two slots ([ITPCM-034^{PLAN_GAPLESS}]).

A Power Plan SHALL NOT be longer than 7 days (168 hours).

Notes (informative):

1. A Power Plan that covers a long time period may have a higher time resolution (i.e. more slots) for the near future, while slots of the distant future have a coarser time resolution. In general, the time resolution should be chosen to express significant changes of the power.
2. It can be expected that announced power values of the distant future are less reliable in comparison to those of the near future in many cases. As a Power Plan always needs to contain realistic values, updates of at least the Committed Power Plan are required if deviations to the

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announced values get too large. However, with regards to the intended workflow of this Use Case (negotiation based upon an Incentive Table) this is usually only feasible within Scenario 2 ([ITPCM-042^{NEGOTIATED_COM_PLAN_UPDATE}]).

2.2.2.3 Committed vs. Preliminary Power Plan

In general, a Committed Power Plan is a result of an accepted negotiation with a Committed Incentive Table. Likewise, a Preliminary Power Plan is a result of an accepted negotiation with a Preliminary Incentive Table. Detailed rules on the relation can be found in the respective sections on committed and preliminary data.

It SHALL be identifiable whether a Power Plan is a Committed Power Plan ([ITPCM-047^{PLAN_COMMITTED}]) or a Preliminary Power Plan ([ITPCM-048^{PLAN_PRELIMINARY}]).

Note: The time information rules in section 2.2.2.2 apply for each type of Power Plan.

2.2.3 Applied sign convention

This Use Case applies the "passive sign convention" (see section 1.2) by default, unless stated otherwise.

2.2.4 Implementation aspects (informative)

The following sub-sections provide supplementary details and examples on the scope and exchanged data of this Use Case. These sub-sections are not normative.

2.2.4.1 Realistic Incentive Tables

As described in section 2.1, an Energy Consumer expects to get "realistic" incentives in order to schedule its operation in a cost-effective way. As a building's electricity consumption is typically billed according to its total energy consumption there is a need to define how "realistic" incentives may be derived per Energy Consumer of a building. The following sub-sections give some explanations on "realistic" values and examples to derive such values. The descriptions hold for Committed Incentive Tables as well as for Preliminary Incentive Tables.

2.2.4.1.1 Realistic energy prices

The energy prices modelled in an Incentive Table should be as realistic as possible, i.e. they should represent the energy prices that are relevant for billing of the building as realistic as possible. The CEM therefore should derive the energy prices from reliable information sources. These could for example be:

- Price information that the customer takes from his electricity contract with his electricity provider and enters into the CEM's user interface.
- Price information that the CEM directly receives from the electricity provider (for example flexible prices that are provided by the electricity provider by means of some protocol).

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- If an energy source is available at the building that can feed its power into the electric system at the building, the feed-in remuneration/generation costs and possible feed-in limitations need to be taken into account.

The CEM should not manipulate those energy prices.

Note: If an additional energy source (battery, e.g.) is available, it is not uniquely possible to assign a fixed energy price. The CEM is responsible for the assignment of a realistic energy price. An example could be a battery storage that has been or will be charged by different energy sources with different energy prices.

2.2.4.1.2 Realistic power boundaries between tiers

The price per kWh (or other incentive types) may depend on the consumed power. This is modeled with so-called tiers: Each tier expresses a price per kWh (or other incentive types) for a given power range. This means an Incentive Table can contain several tiers which are separated by respective power boundaries (see section 2.2.1.1 for details).

Note: The unit used in the technical solution (e.g., SPINE) may differ from "kWh" (e.g., "Wh").

A CEM needs to assign appropriate power boundaries to the "realistic" incentives. This should be based on at least these factors:

1) If an electricity provider communicates grid electricity prices which contain power boundaries, then the CEM should include them into its derivation of power boundary values.
2) If energy sources are available at the building, then forecasted power values of the production should be taken into account when calculating the power boundary values.
3) If energy sources with feed-in limitations are available at the building, these limitations should be taken into account when calculating the power boundary values.

Note: Especially for buildings with photovoltaic systems or batteries or other additional energy sources it is quite likely that the CEM needs to assign power boundaries in its Incentive Table that differ from the power boundaries of the Incentive Table received from the electricity provider.

The order of both, the incentives and the power boundaries, shall comply to the rules stated in section 2.2.1.1 (see esp. the example of Figure 3).

2.2.4.1.3 Base and derived Incentive Tables

This section gives examples to derive Incentive Tables per Energy Consumer.

Base Incentive Table:

The "Base Incentive Table" represents the Incentive Table that is applicable within this Use Case if all consumers that participate in this Use Case would be represented by a single consumer.

The "Base Incentive Table" is the CEM's best estimation of available power and prices that is available within this Use Case. This means that uncontrolled consumers or consumers that are controlled by other Use Cases should already have been excluded from the available power. It also

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means that uncontrolled producers or producers that are controlled by other Use Cases should already have been included to the available power.

Derived Incentive Table:
The CEM may generate Derived Incentive Tables from the Base Incentive Table and provide them (timely in parallel or sequentially) to the different consumers. Two basic approaches are given as examples in the following sub-sections.

2.2.4.1.3.1 Providing Incentive Tables to consumers sequentially

The CEM may provide Incentive Tables to consumers sequentially, i.e. send an Incentive Table "T(1)" to the first consumer, wait for a corresponding Power Plan "P(1)", subtract the Power Plan from the Incentive Table "T(1)" power boundaries, send the adjusted Incentive Table "T(2)" to the next consumer, wait for a corresponding Power Plan "P(2)" and so on. This means the CEM ensures that the available energy of each tier is split uniquely among the energy consumers.

This means the first consumer gets the Base Incentive Table. Energy consumer no. "n+1" gets Incentive Table "T(n+1)" which is Incentive Table "T(n)", but with reduced power boundaries (or even removed tiers that are not available anymore) according to Power Plan "P(n)".

However, this is just a simplified description for ideal conditions. Under real conditions there is a need to consider updates of Power Plans (i.e. unexpected but necessary deviations from the formerly announced Power Plan) and also re-negotiations. Esp. re-negotiations need to be handled with care as they cover a time period that at least partially overlaps with the time period of the previous negotiation. This means that a CEM would need to consider the formerly negotiated Power Plan of the consumer for the derivation of new power boundaries in a different way: The CEM would need to assume that the consumer is willing to discard its formerly announced Power Plan (for present and future times) which means that new power boundaries must not contain the reduction from the former Power Plan.

This leads to the general description that a Derived Incentive Table for consumer "i" is created from the Base Incentive Table, with reduced power boundaries according to the current Power Plans of all other consumers, but not by the Power Plan of consumer "i" itself. More formally, this can be written as follows (please note that notations like T(1) above do not apply here):

$$T_{i,next} = T_{base} \ominus \sum_{k,k\neq i} P_{k,current}$$

with

- T_{base} : Base Incentive Table
- C_i : Consumer no. i
- $T_{i,current}$: Current Incentive Table for C_i
- $P_{i,current}$: Current Power Plan of C_i (the Power Plan according to $T_{i,current}$ or an updated Power Plan of C_i)
- $T_{i,next}$: Next (potential) Incentive Table for C_i (potential successor of $T_{i,current}$)
- $P_{i,next}$: Next (potential) Power Plan of C_i (the Power Plan according to $T_{i,next}$)

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- ⊖ : The expression "T ⊖ P" yields Incentive Table T, but with reduced power boundaries according to P

2.2.4.1.3.2 Providing Incentive Tables to consumers in parallel

The CEM may provide Incentive Tables to consumers in parallel, i.e. send different Incentive Tables to consumers at the same time and receive Power Plans of these consumers at the same time. In this approach the CEM has as well the challenge to avoid that several consumers schedule for the same lower tiers, but finally lead to a power consumption that exceeds the lower tiers' power boundaries. This kind of multiple occupancy of tiers means the consumers would mistakenly assume to operate for lower costs, but in fact the building would be charged at higher costs.

A very primitive approach to avoid multiple occupancy of the (base) lower tiers is to split the lower tiers into unique portions for each energy consumer:

1. Starting point to derive consumer specific Incentive Tables is the Base Incentive Table.
2. All Incentive Tables that are sent out in parallel sum up to the Base Incentive Table, i.e.:
   a. In each Derived Incentive Table, the price values of the different tiers are the same as in the respective tiers of the Base Incentive Table.
   b. For each tier, the power boundary values of all Derived Incentive Tables sum up to the respective power boundary values of the Base Incentive Table.

Similar to section 2.2.4.1.3.1, re-negotiations as well as negotiations with new consumers need to be considered with care as well. If just a single consumer is then involved in a (re-)negotiation, mechanisms as in section 2.2.4.1.3.1 can be applied. If parallel (re-)negotiations with multiple consumers need to be handled, a similar principle as in section 2.2.4.1.3.1 can be applied: From the Base Incentive Table the power boundaries are reduced according to the current Power Plans of all consumers except those that belong to the upcoming negotiation. The result is a "temporary Base Incentive Table" whose lower tiers can be split into unique portions as described above.

2.2.4.1.3.3 Notes on limitations of given example approaches

It should be noted that the basic approaches explained in the previous sub-sections have some restrictions. Depending on the business case, a CEM manufacturer may need to implement more sophisticated algorithms to overcome the restrictions. Some restrictions will be explained briefly.

The sequential approach in section 2.2.4.1.3.1 may not work well in case of "many" energy consumers. This is due to the fact that each sequential negotiation requires some time and blocks a negotiation with other energy consumers. This may then even lead to some timeout results where an energy consumer at least for its current task withdraws its participation in the Use Case and just schedules its consumption on its own.

The parallel approach with the primitive dissection of tiers in section 2.2.4.1.3.2 may lead to "weakly used lower tiers" and it may suggest larger costs than assumed for a consumer: If two consumers "A" and "B" are provided in parallel with Incentive Tables "T_A" and "T_B" (based upon the primitive dissection of tiers), the result might be that the Power Plans "P_A" and "P_B" acquire power consumption at completely different times. In this case, the total cost of "P_A" as well as the total cost of "P_B" (each cost calculated by consumer "A" or "B", resp.) would probably be wrong if the

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respective time frame has more than one tier (with different prices) in a time slot. This is due to the fact that in this case wrong power boundaries are used for the calculation of the total cost. As a result, the sum of both costs would also be wrong. The real cost would instead require the application of the Base Incentive Table for the Power Plans "P_A" and "P_B".

2.2.4.2 Example for reduction of tiers

Figure 5 shows an example with five tiers on the left side. The related example installation, current conditions of battery, PV (photovoltaic), and CHP (Combined Heat Power) systems, as well as contract terms are summarized as follows:

- 2kW CHP production (uncontrolled) with feed-in remuneration of 0.07€/kWh
- PV production of 4kW with feed-in remuneration of 0.09€/kWh
- Battery output of 1kW with assumed price of 0.10€/kWh
- Feed-in limit at 5kW
- Available grid electricity at 0.30€/kWh

The feed-in limit of 5kW yields a surplus power of 1kW from the CHP and PV output.

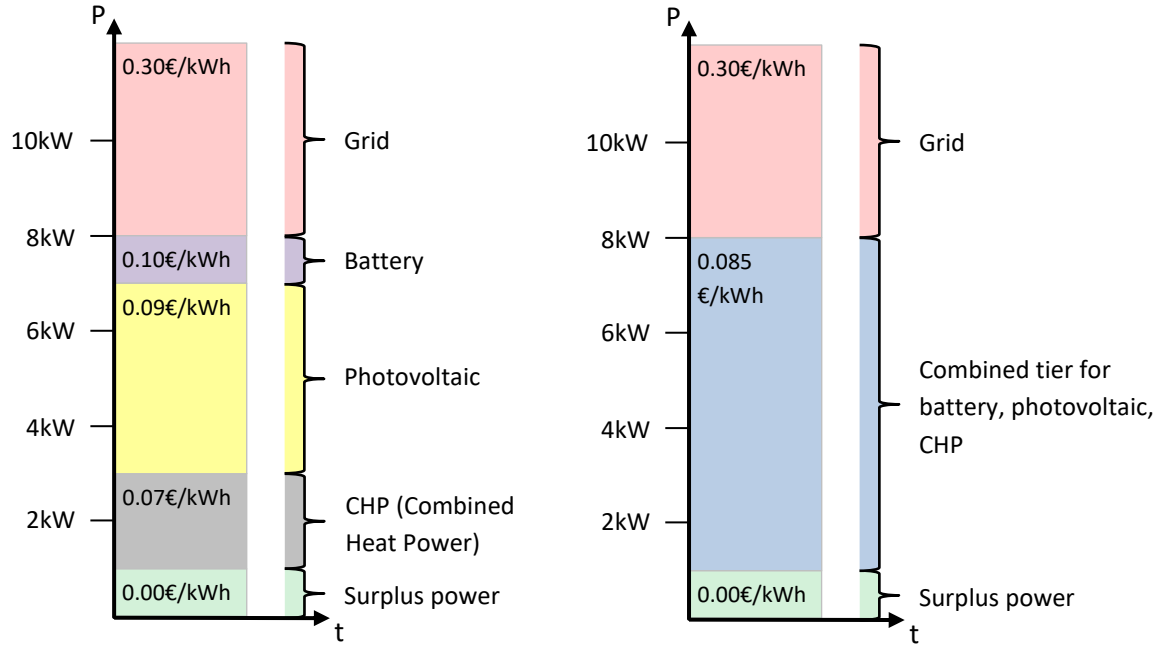


Figure 5: Example for five tiers and "compression" to three tiers

As explained previously, the Actor Energy Consumer can limit the number of supported tiers ([ITPCM-009^{MAX_TIERS}]) to three (even though a support of at least five tiers is recommended). In such cases, a CEM may need to "compress" the Incentive Table, i.e. it may need to create an Incentive Table with just three tiers. The CEM has then to assign "realistic" incentive values and power boundaries to the respective tiers. The right side of Figure 5 shows an example for a "compression" of the left side.

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2.3 Actors

2.3.1 Energy Consumer

The Actor Energy Consumer is the Energy Consumer that wants to cover its energy needs with low environmental or monetary costs. Within this Use Case, Actor Energy Consumer SHALL execute this Use Case with at most one Actor CEM.

2.3.2 CEM

The Actor CEM delivers Incentive Tables. Multiple Energy Consumers MAY be connected to a CEM. However, only one CEM can execute this Use Case with a given Energy Consumer.

2.4 Scenarios

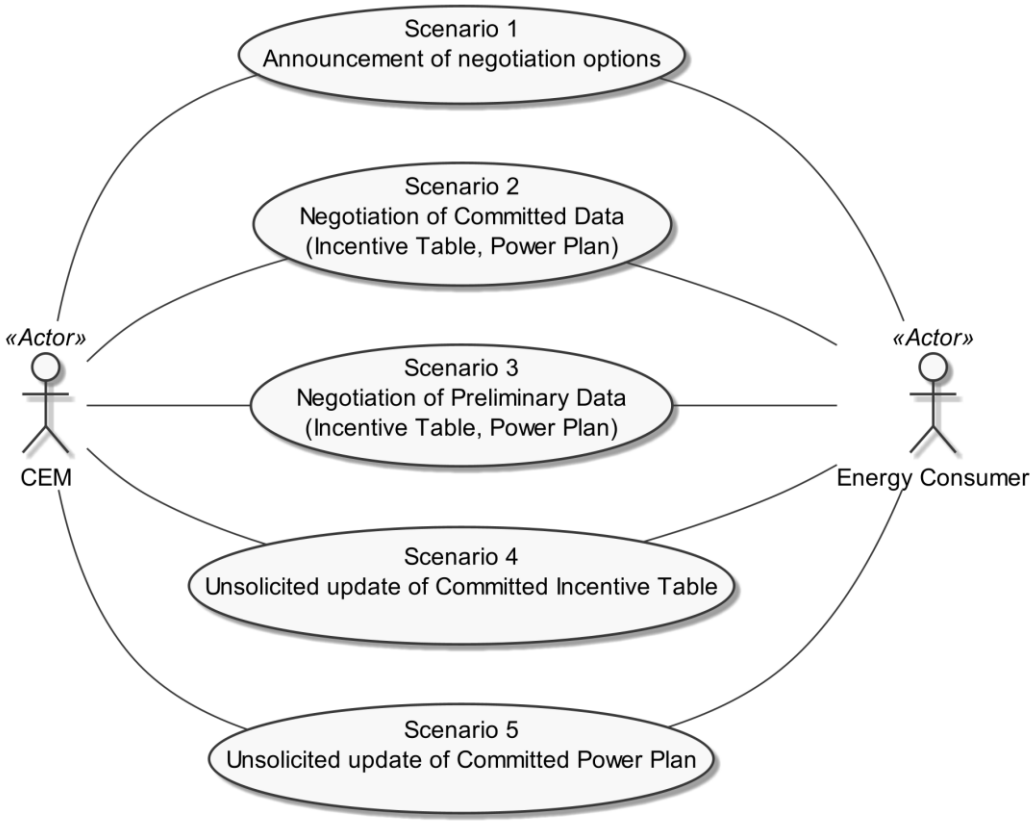


Figure 6: Scenario Overview

Scenario number	Scenario name	Energy Consumer	CEM
1	Announcement of negotiation options	M	M

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2	Negotiation of Committed Data	M	M
3	Negotiation of Preliminary Data	O	O
4	Unsolicited update of Committed Incentive Table	M	O
5	Unsolicited update of Committed Power Plan	O	M

Table 1: Scenario implementation requirements for Actors

Note: The order in which the Scenarios occur is defined by their triggers. For some Scenarios, data of other Scenarios is relevant, while other Scenarios can run in parallel.

2.4.1 Scenario 1 - Announcement of negotiation options

2.4.1.1 Description

The Use Case comprises several processes (Scenarios) with negotiation options of Actor Energy Consumer. These options must be fetched and considered properly by Actor CEM. This Scenario 1 primarily summarizes the negotiation options in Table 2. Rules and further details on the negotiation options can be found in the respective Scenarios or below Table 2.

Data point name	**Data point description [High-Level requirement]**	**Support indication**	**High-Level requirement**
Maximum Simulation Cycles	For Scenario 3: The maximum number of successive simulation cycles (Scenario 3) permitted by Actor Energy Consumer. Once the number of completed successive simulation cycles reaches this maximum, Actor Energy Consumer does not support further simulation cycles unless a negotiation of committed data (Scenario 2) completed successfully. A maximum number of zero denotes that Actor Energy Consumer does not support Scenario 3 at all. The value SHALL be a non-negative integer.	M	[ITPCM-031^{MAX_SIMUL_CYCLES}]
Concurrent Negotiation Capability	For Scenario 3 (and 2): Value "true" = Actor Energy Consumer is capable of executing calculations of Scenario 3 in parallel to Scenario 2; otherwise, the value is "false". See details below table.	O/M*[1]	[ITPCM-030^{CONCURRENT_SIMUL}]
Timeout Incentive Request	For Scenario 2: Maximum waiting period of Actor Energy Consumer to receive a valid update of the Committed Incentive Table. See details in section 2.4.2.	M	[ITPCM-005^{TIMEOUT_IT_REQUEST}]
Wait Incentive Writeable	For Scenario 2: Waiting period of Actor Energy Consumer before it permits execution of Scenario 4. See details in section 2.4.2.	M	[ITPCM-008^{WAIT_IT_MODIFIABLE}]

Table 2: Scenario 1 - Negotiation options

*[1]: Mandatory if Scenario 3 is supported.

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On data point "Concurrent Negotiation Capability" ([ITPCM-030^{CONCURRENT_SIMUL}]**):**
In Scenarios 2 and 3, Actor Energy Consumer computes a Committed Power Plan or Preliminary Power Plan based upon a respective Incentive Table. The Actor Energy Consumer may not have sufficient computing resources to calculate a Preliminary Power Plan and a Committed Power Plan at the same time and deliver results within the required time constraints. There might be further reasons that prevent Actor Energy Consumer from parallel computations of these plans. In these cases, Actor Energy Consumer SHALL set the data point "Concurrent Negotiation Capability" to "false". Otherwise, Actor Energy Consumer SHOULD set the data point "Concurrent Negotiation Capability" to "true".

Note: Scenario 3 also describes how a computation of a Preliminary Power Plan may be aborted by Actor Energy Consumer (see section 2.4.3.1.5).

2.4.1.2 Conditions

Triggering Event:
This Scenario is triggered by Actor CEM to get negotiation options of Actor Energy Consumer to execute further Scenarios of this Use Case properly.

Pre-condition:
The Actor CEM has no or outdated negotiation options of Actor Energy Consumer.

Post-condition:
In case of success, the Actor CEM received updated negotiation options of Actor Energy Consumer.

2.4.2 Scenario 2 - Negotiation of Committed Data

2.4.2.1 Description

2.4.2.1.1 Overview

In this Scenario, the Actor Energy Consumer explicitly requests an update of its Committed Incentive Table. If the Actor Energy Consumer receives a valid Committed Incentive Table in time, it updates its Committed Power Plan. Otherwise, the Actor Energy Consumer operates autonomously (with regards to this Use Case). An overview is shown in Figure 7.

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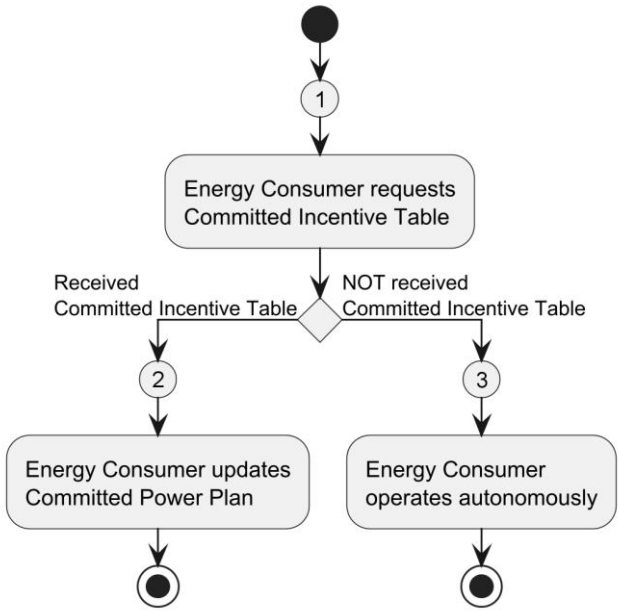


Figure 7: Overview of Scenario 2 activity

The activity is split into several segments which begin at the circled numbers shown in Figure 7:

1. Activity begin ①
2. Activity branch "Update of Committed Power Plan" ②
3. Activity branch "Autonomous operation" ③

Details are described in the subsequent sections.

Note: If Actor Energy Consumer supports Scenario 3, it should trigger Scenario 3 before Scenario 2. This allows the Actor CEM to perform the simulation phase before the committed phase in order to optimize the Committed Incentive Table.

2.4.2.1.2 Activity begin ①

The Actor Energy Consumer is ready to initiate a negotiation of committed data. It requests for a (new or updated) Committed Incentive Table and waits for the receipt. This is shown in Figure 8.

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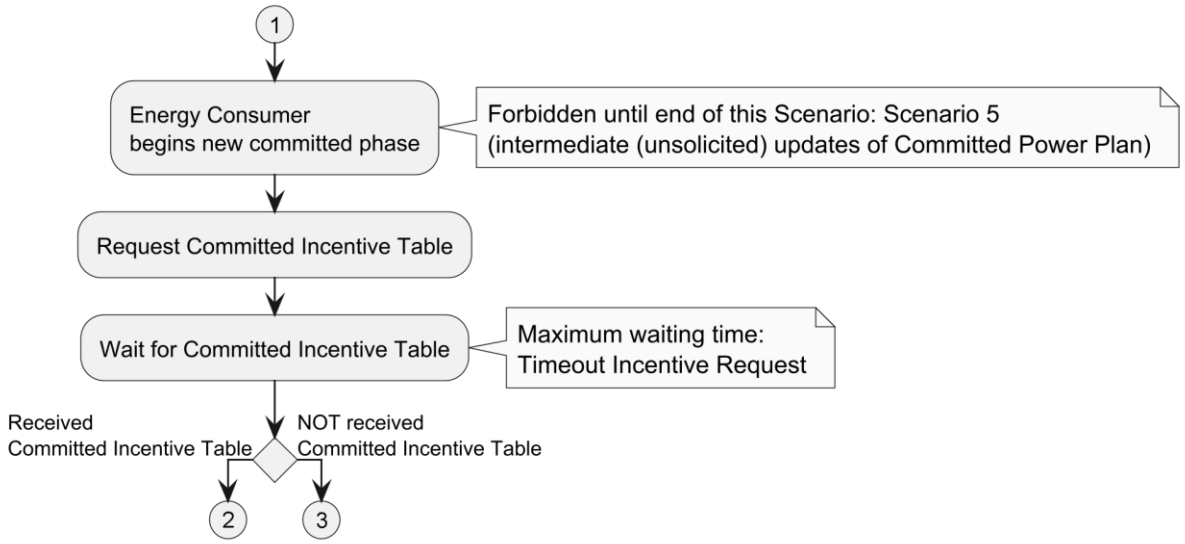


Figure 8: Activity begin diagram of Scenario 2

Step 1 (Energy Consumer begins new committed phase):
Until the end of Scenario 2, the Actor Energy Consumer SHALL NOT execute Scenario 5 ([ITPCM-001$^{\text{DISABLE_PLAN_UPDATE}}$]). During Scenario 2, updates of the Committed Power Plan MAY occur according to section 2.4.2.1.3 only.

Step 2 (Request Committed Incentive Table):
The Actor Energy Consumer SHALL allow Actor CEM to modify or create Actor Energy Consumer's Committed Incentive Table ([ITPCM-003$^{\text{IT_MODIFIABLE}}$]). The Actor Energy Consumer SHALL request towards Actor CEM to get a modified or new Committed Incentive Table ([ITPCM-004$^{\text{IT_REQUEST}}$]).

Step 3 (Wait for Committed Incentive Table or Timeout):
The Actor Energy Consumer SHALL wait for a maximum duration of Timeout Incentive Request ([ITPCM-005$^{\text{TIMEOUT_IT_REQUEST}}$]) to receive a valid update of the Committed Incentive Table. If the Actor Energy Consumer receives a valid update of the Committed Incentive Table in time, the Actor Energy Consumer SHALL continue as specified in section 2.4.2.1.3. Otherwise, the Actor Energy Consumer SHALL continue as specified in section 2.4.2.1.4.

2.4.2.1.3 Activity branch "Update of Committed Power Plan" ②

The Actor Energy Consumer is ready to update its Committed Power Plan as shown in Figure 9.

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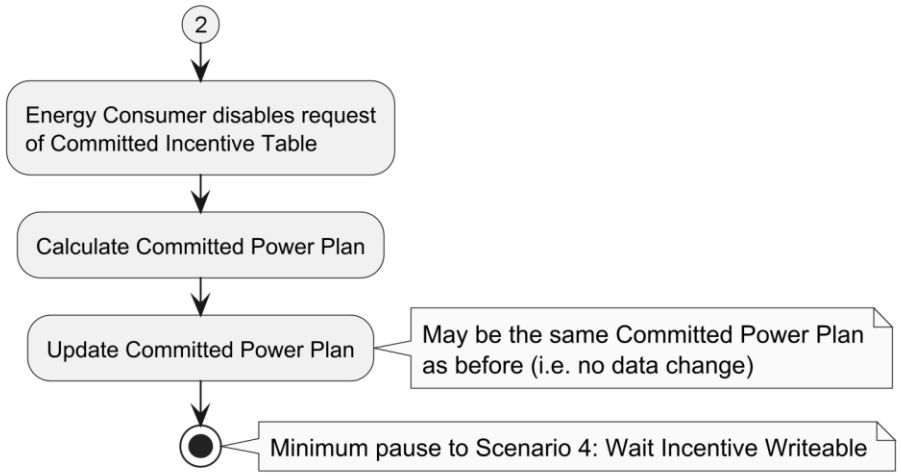


Figure 9: Activity branch "Update of Committed Power Plan" diagram of Scenario 2

Step 1 (Energy Consumer disables request of Committed Incentive Table):
The Actor Energy Consumer SHALL NOT allow Actor CEM to modify or create Actor Energy Consumer's Committed Incentive Table ([ITPCM-006^{IT_UNMODIFIABLE}]). The Actor Energy Consumer SHALL disable the request towards Actor CEM to get a modified or new Committed Incentive Table ([ITPCM-007^{IT_NO_REQUEST}]).

Step 2 (Calculate Committed Power Plan, Update Committed Power Plan):
The Actor Energy Consumer SHALL calculate and update its Committed Power Plan ([ITPCM-042^{NEGOTIATED_COM_PLAN_UPDATE}]).

Note (informative): Typically, the following (non-exhaustive) list should be used by the Actor Energy Consumer for the calculation:

1. The Committed Incentive Table (used for an optimization algorithm).
2. The current or anticipated energy demand (required energy incl. time frame to fulfil a task).
3. Any further constraints (imposed power limitation, e.g.).

Step 3 (End of Scenario 2):
The Actor Energy Consumer SHALL wait for a duration of Wait Incentive Writeable ([ITPCM-008^{WAIT_IT_MODIFIABLE}]) before it permits execution of Scenario 4.

2.4.2.1.4 Activity branch "Autonomous operation" ③

This is an error case where Actor CEM does not behave appropriately. The Actor Energy Consumer needs to continue its operation without update of the Committed Incentive Table as shown in Figure 10.

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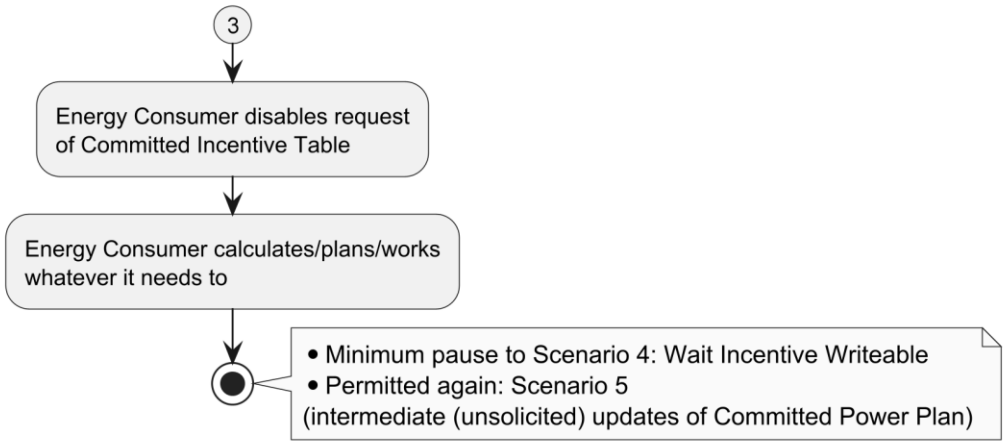


Figure 10: Activity branch "Autonomous operation" diagram of Scenario 2

Step 1 (Energy Consumer disables request of Committed Incentive Table):
The Actor Energy Consumer SHALL NOT allow Actor CEM to modify or create Actor Energy Consumer's Committed Incentive Table ([ITPCM-006^{IT_UNMODIFIABLE}]). The Actor Energy Consumer SHALL disable the request towards Actor CEM to get a modified or new Committed Incentive Table ([ITPCM-007^{IT_NO_REQUEST}]).

Step 2 (Consumer calculates/plans/works whatever it needs to):
The Actor Energy Consumer SHALL operate at its own discretion (with regards to this Use Case) to fulfil its task. This means the Actor Energy Consumer is not obliged to wait for any update of the Committed Incentive Table.

Note (informative): An Actor Energy Consumer may apply different optimization algorithms in comparison to the case with updated and valid Committed Incentive Table.

Step 3 (End of Scenario 2):
The Actor Energy Consumer SHALL wait for a duration of Wait Incentive Writeable ([ITPCM-008^{WAIT_IT_MODIFIABLE}]) before it permits execution of Scenario 4.

The Actor Energy Consumer MAY execute Scenario 5 ([ITPCM-002^{ENABLE_PLAN_UPDATE}]).

Note (informative): Even in this error case, the execution of Actor Energy Consumer's task will typically result in some kind of planning with proper power consumption. Therefore, it is recommended that the Actor Energy Consumer executes Scenario 5 to update its Committed Power Plan, if possible, as this may help Actor CEM (as soon as Actor CEM behaves appropriately again) for its overall energy management task.

The Actor Energy Consumer SHALL trigger the Scenario 2 again as soon as possible. I.e. the Actor Energy Consumer SHALL NOT remain in a mode of autonomous operation any longer than necessary.

2.4.2.2 Conditions

Triggering Event:
This Scenario is triggered by Actor Energy Consumer to schedule its current or imminent energy demand based upon valid committed incentives.

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699 **Pre-condition**:
700 The Actor CEM knows the negotiation options of Actor Energy Consumer from Scenario 1. The Actor
701 Energy Consumer has no or outdated incentives for the required period.

702 **Post-condition**:
703 In case of success, the Actor Energy Consumer received updated incentives for a certain period and
704 did provide an appropriate Committed Power Plan.

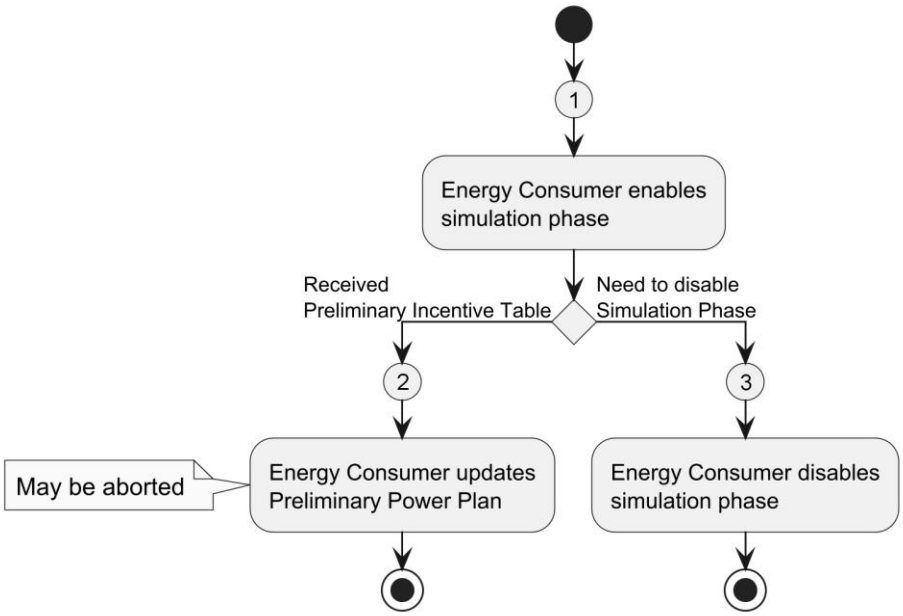
705 For error cases, please consider section 2.4.2.1.

706

707 ### 2.4.3 Scenario 3 - Negotiation of Preliminary Data

708 #### *2.4.3.1 Description*

709 *2.4.3.1.1 Overview*
710 In this Scenario, the Actor Energy Consumer permits Actor CEM to make use of a simulation phase.
711 The simulation phase may be aborted by Actor Energy Consumer any time. If the Actor Energy
712 Consumer receives a valid Preliminary Incentive Table within the simulation phase, it updates its
713 Preliminary Power Plan. An overview is shown in Figure 11.



714

715 *Figure 11: Overview of Scenario 3 activity*

716 The activity is split into several segments which begin at the circled numbers shown in Figure 11:

717 1. Activity begin ①
718 2. Activity branch "Update of Preliminary Power Plan" ②
719 3. Activity branch "Disable simulation phase" ③

720 Details are described in the subsequent sections.

721

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2.4.3.1.2 Activity begin ①

The Actor Energy Consumer is ready to permit a negotiation of preliminary data. It enables Actor CEM to provide a Preliminary Incentive Table and waits for the receipt. This is shown in Figure 12.

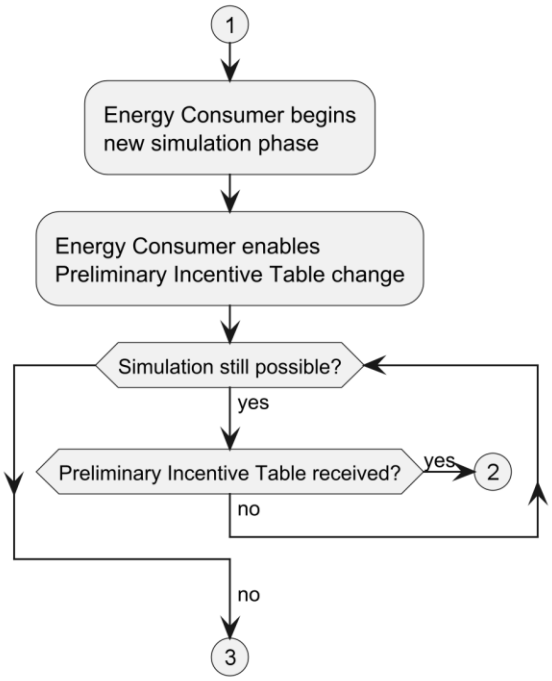


Figure 12: Activity begin diagram of Scenario 3

Step 1 (Energy Consumer begins new simulation phase):
Actor Energy Consumer is currently able to permit a negotiation of preliminary data.

Note: The Actor Energy Consumer's possibility to abort the simulation phase is given in subsequent steps.

Step 2 (Enable Preliminary Incentive Table change):
The Actor Energy Consumer SHALL allow Actor CEM to modify or create Actor Energy Consumer's Preliminary Incentive Table ([ITPCM-036^{PIT_MODIFIABLE}]).

Step 3 (Wait for Preliminary Incentive Table or Abort):
As long as Actor Energy Consumer does not need to abort the simulation phase, it waits to receive a valid update of the Preliminary Incentive Table.

If Actor CEM submits a Preliminary Incentive Table, it waits for a minimum period of TIMEOUT_INCENTIVE_PLAN = 60s to get a proper Preliminary Power Plan.

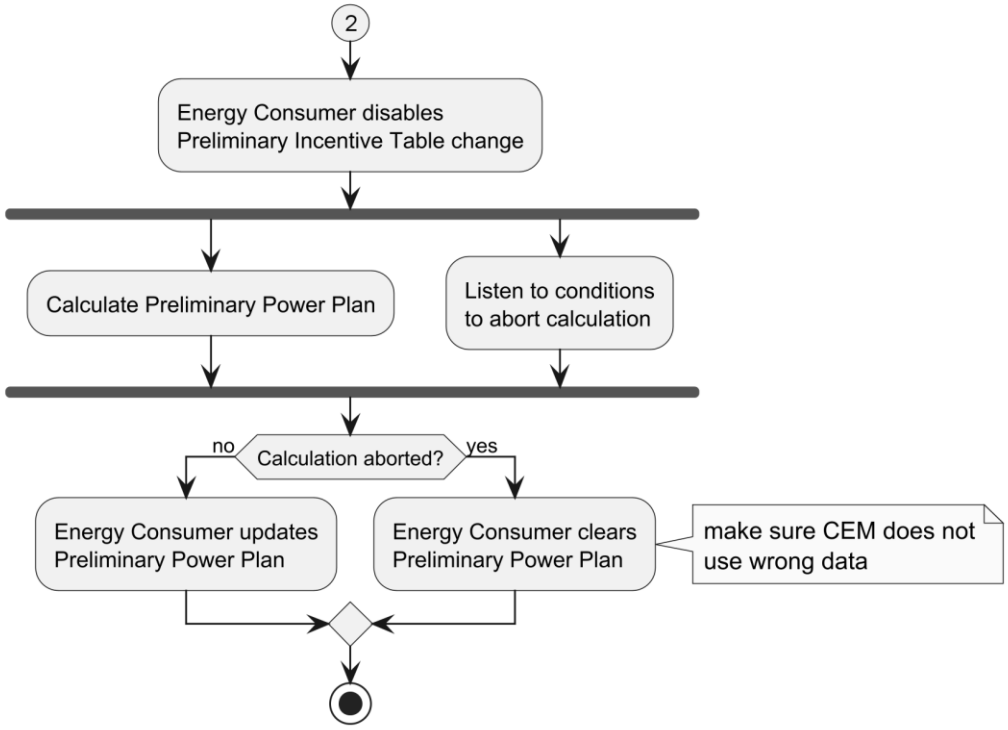
If the Actor Energy Consumer receives a valid update of the Preliminary Incentive Table, the Actor Energy Consumer SHALL continue as specified in section 2.4.3.1.3.

If the Actor Energy Consumer needs to abort the simulation phase (see section 2.4.3.1.5) before it received a Preliminary Incentive Table, it SHALL continue as specified in section 2.4.3.1.4.

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744 *2.4.3.1.3 Activity branch "Update of Preliminary Power Plan"* ②

745 The Actor Energy Consumer is ready to update its Preliminary Power Plan as shown in Figure 13.



746

747 *Figure 13: Activity branch "Update of Preliminary Power Plan" diagram of Scenario 3*

748 **Step 1 (Energy Consumer disables Preliminary Incentive Table change):**

749 The Actor Energy Consumer SHALL NOT allow Actor CEM to modify or create Actor Energy
750 Consumer's Preliminary Incentive Table ([ITPCM-037^{PIT_UNMODIFIABLE}]).

751 **Step 2 (Calculate Preliminary Power Plan, listen to conditions to abort calculation):**

752 The Actor Energy Consumer SHALL begin to calculate its Preliminary Power Plan. The Actor Energy
753 Consumer MAY abort the calculation if conditions require it to abort (see section 2.4.3.1.5).

754 This step ends as soon as the calculation completed successfully or the calculation was aborted.

755 Note (informative): Typically, the following (non-exhaustive) list should be used by the Actor Energy
756 Consumer for the calculation:

757 1. The Preliminary Incentive Table (used for an optimization algorithm).
758 2. The current or anticipated energy demand (required energy incl. time frame to fulfil a task).
759 3. Any further constraints (imposed power limitation, e.g.).

760 **Step 3 (Update or clear Preliminary Power Plan):**

761 If the calculation was aborted in step 2:

762 In this case, Actor Energy Consumer SHALL delete all power values from its Preliminary Power Plan
763 ([ITPCM-038^{PPLAN_DEL_POWER}]). This corresponds to an "empty" Preliminary Power Plan.

764 If the calculation completed successfully in step 2:

765 In this case, Actor Energy Consumer SHALL update its Preliminary Power Plan accordingly.

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Both cases end the Scenario 3. The Actor Energy Consumer MAY trigger the Scenario 3, again.

2.4.3.1.4 Activity branch "Disable simulation phase" ③

This is a case where Actor Energy Consumer needs to stop the simulation phase before it received a Preliminary Incentive Table. The process to stop the simulation phase is shown in Figure 14.

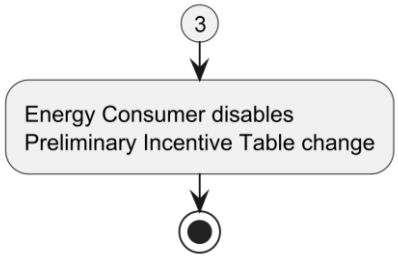


Figure 14: Activity branch "Disable simulation phase" diagram of Scenario 3

Step 1 (Energy Consumer disables Preliminary Incentive Table change):
The Actor Energy Consumer SHALL NOT allow Actor CEM to modify or create Actor Energy Consumer's Preliminary Incentive Table ([ITPCM-037^{PIT_UNMODIFIABLE}]).

This ends Scenario 3. The Actor Energy Consumer MAY trigger the Scenario 3, again.

2.4.3.1.5 Aborting the simulation phase

This is an informative section.

There are several situations where Actor Energy Consumer may need to abort a simulation phase. For example:

1. Impossibility of concurrent calculations:

   Actor Energy Consumer may enable Scenario 2 as well as 3 in parallel. However, this does not mean that Actor Energy Consumer is also capable of executing calculations of Scenario 3 in parallel to Scenario 2. If it is not capable of parallel calculations, Actor Energy Consumer has to set the data point "Concurrent Negotiation Capability" ([ITPCM-030^{CONCURRENT_SIMUL}]) accordingly, as described in Scenario 1. As Scenario 2 has a higher priority than Scenario 3, the receipt of a Committed Incentive Table would require aborting the simulation phase.

2. Scenario 2 invalidates the current calculation of Scenario 3:

   Actor Energy Consumer may enable Scenario 2 as well as 3 in parallel and may also be able of parallel calculations in general. However, the completion of a calculation for Scenario 2 (i.e. the Committed Power Plan) may alter the device's state in a way that assumptions applied for the parallel calculation of Scenario 3 (i.e. the Preliminary Power Plan) are not valid anymore and cannot be completed in a meaningful way.

3. Other changes of internal conditions:

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Actor Energy Consumer may be in a state where the execution of a simulation phase is not useful or even not possible.

2.4.3.2 Conditions

Triggering Event:
This Scenario is triggered by Actor Energy Consumer to permit Actor CEM making use of a simulation phase to find a potential schedule for Actor Energy Consumer's current or imminent energy demand based upon preliminary incentives.

Pre-condition:
The Actor CEM knows the negotiation options of Actor Energy Consumer from Scenario 1. The Actor Energy Consumer has no or outdated preliminary incentives for the required period.

Post-condition:
In case of success, the Actor Energy Consumer received updated preliminary incentives for a certain period and did provide an appropriate Preliminary Power Plan.

For error cases or the abortion of the simulation phase, please consider section 2.4.3.1.

2.4.4 Scenario 4 - Unsolicited update of Committed Incentive Table

2.4.4.1 Description

2.4.4.1.1 Overview

Unlike Scenario 2, Scenario 4 enables Actor Energy Consumer to receive unsolicited updates of the Committed Incentive Table. This means, Actor Energy Consumer signals the Actor CEM that the Committed Incentive Table is modifiable ([ITPCM-003^{IT_MODIFIABLE}]). Actor CEM can then decide on its own whether and when it will update the Actor Energy Consumer's Committed Incentive Table. Such updates can help Actor Energy Consumer to decide whether it should trigger Scenario 2 or 3 (by the according means defined in each Scenario).

2.4.4.1.2 Scenario details

If Actor Energy Consumer activates Scenario 4, it SHALL permit Actor CEM to update the Committed Incentive Table ([ITPCM-003^{IT_MODIFIABLE}]). But unlike Scenario 2, Actor Energy Consumer SHALL NOT explicitly request to receive a valid Committed Incentive Table ([ITPCM-007^{IT_NO_REQUEST}]).

Actor Energy Consumer MAY disable the possibility to receive updates of the Committed Incentive Table any time ([ITPCM-006^{IT_UNMODIFIABLE}]). This ends Scenario 4.

Unlike Scenario 2, the receipt of an unsolicited Committed Incentive Table SHALL NOT be used to update the Committed Power Plan. Instead, Actor Energy Consumer decides whether it wants to activate Scenario 2 or 3 or it ignores the updated Committed Incentive Table.

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2.4.4.2 Conditions

Triggering Event:
This Scenario is triggered by Actor Energy Consumer.

Pre-condition:
Scenario 2 is currently NOT executed.

Scenario 4 is currently NOT blocked by timer Wait Incentive Writeable ([ITPCM-008^{WAIT_IT_MODIFIABLE}]) from Scenario 2.

The Actor Energy Consumer has no or outdated incentives for the required period.

Post-condition:
In case of success, the Actor Energy Consumer received updated incentives for a certain period.

2.4.5 Scenario 5 - Unsolicited update of Committed Power Plan

2.4.5.1 Description

2.4.5.1.1 Overview
Unlike Scenario 2, Scenario 5 enables Actor CEM to receive unsolicited updates of Actor Energy Consumer's Committed Power Plan, but only if and only as long as the error case of Scenario 2 persists (see section 2.4.2.1.4) ([ITPCM-002^{ENABLE_PLAN_UPDATE}]). Such updates can help Actor CEM for its energy management task even if Scenario 2 can currently not be executed.

2.4.5.1.2 Scenario details
If this Scenario is supported and the pre-condition (see below) is fulfilled: Actor Energy Consumer SHALL update its Committed Power Plan if it anticipates substantial deviations from the already announced power values.

2.4.5.2 Conditions

Triggering Event:
This Scenario is triggered by Actor Energy Consumer.

Pre-condition:
The error case of Scenario 2 persists (see section 2.4.2.1.4) and Scenario 2 is currently NOT executed: This means that Actor Energy Consumer SHALL NOT execute Scenario 5 if it is currently blocked by Scenario 2 ([ITPCM-001^{DISABLE_PLAN_UPDATE}]), but it MAY execute Scenario 5 otherwise if the error case of Scenario 2 still persists ([ITPCM-002^{ENABLE_PLAN_UPDATE}]).

The Actor Energy Consumer has valid incentives for the required period (from an earlier, successful run of Scenario 2).

Post-condition:
In case of success, the Actor CEM received an updated Committed Power Plan for a certain period.

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2.5 Dependencies to other Use Cases

2.5.1 "Monitoring of Power Consumption"

The Use Case "Monitoring of Power Consumption" enables Actor CEM to monitor the actual power consumption of the Actor Energy Consumer. With regards to this Use Case, the data point "Total Active Power" of Use Case "Monitoring of Power Consumption" ([MPC-011]) is of special interest ([ITPCM-039^{P_TOTAL_ACTIVE}]).

The Actor CEM SHOULD support the Use Case "Monitoring of Power Consumption". In this case, the Actor CEM of this Use Case acts as Actor Monitoring Appliance within the Use Case "Monitoring of Power Consumption".

The Actor Energy Consumer SHOULD support the Use Case "Monitoring of Power Consumption". In this case, the Actor Energy Consumer of this Use Case acts as Actor Monitored Unit within the Use Case "Monitoring of Power Consumption".

2.6 Assumptions and Prerequisites

None.

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3 Technical SPINE solution

3.1 General rules and information

3.1.1 Underlying technology documents

This technical solution relies on the SPINE Resources Specification version 1.3.0 [ResourceSpecification].

For interoperable connectivity this technical solution relies on:

- SPINE Protocol Specification version 1.3.0 [ProtocolSpecification] as application protocol.
- SHIP Specification version 1.0.1 [SHIP] as transport protocol.

3.1.2 Use Case discovery rules

Use Case discovery SHALL be supported by each Actor and the following rules SHALL apply:

- The string content for the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseName" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be "incentiveTableBasedPowerConsumptionManagement". The string content SHALL only be defined by this Use Case (regardless of the Use Case version).
- The string content of the Element "nodeManagementUseCaseData. useCaseInformation. actor" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be set to the according value stated within the corresponding Actor's section.
- An Actor A that is implemented to support this Use Case specification SHALL set the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseVersion" within the Use Case discovery (please refer to [ProtocolSpecification]) to "1.0.0".
- If an Actor A supports multiple versions of this Use Case with the same major version number, only the highest one SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports multiple versions of this Use Case with the same major version number as supported by Actor A, the Actor A SHOULD evaluate from these versions of Actor B only the highest version number.
- If an Actor A supports multiple versions of this Use Case with different major version numbers, for each major version number only the highest version number SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports only versions with a major version number not implemented by Actor A, it still might be possible to run the Use Case or parts of the Use Case. Therefore, the Actor A should try to evaluate the Actor B as a valid partner for this Use Case.

3.1.3 Rules for "Content of Specialization..." tables and "Content of Function..." tables

3.1.3.1 General presence indication definitions

Abbreviations for the presence indication of Elements listed in the tables are defined as follows:

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Abbreviation	Meaning	Link to requirement keywords
M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 3: Presence indication description

An Actor MAY support Elements that are not listed in the tables. However, another Actor MAY ignore these Elements.

The presence indications "M", "R" and "O" are always meant relative to the respective parent Element. I.e. if a parent Element is optional ("O") and a child is mandatory ("M") the child Element can only be present if the parent Element is present as well.

Note: The indications and the aforementioned rules apply for "complete messages" (so-called "full function exchange", please refer to [ProtocolSpecification]). In contrast, the so-called "restricted function exchange" is designed to permit exchange of specific excerpts of data, i.e. fewer Elements than potentially available from the data owner (partially even not all "mandatory" Elements).

3.1.3.2 Presence indications for "Content of Specialization..." tables

This section only defines rules for the client side.

Elements that are marked with "M" SHALL be supported by the client in case of readable as well as writeable data. This Element may be optional on the server side.

The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- "R" means that the data SHOULD be supported by the client. In other words: If the server responds with the according Element, the client SHOULD be able to interpret the according Elements.
- "O" means that the data MAY be supported by the client. In other words: If the server responds with the according Element, the client MAY be able to interpret the according Elements.

The following applies for writeable data that is exchanged in a "write" operation:

- "R" means that the data SHOULD be written by the client.
- "O" means that the data MAY be written by the client.
- "F" means that the data SHALL NOT be written by the client.

The following applies for Elements that are not listed in the Actor section:

- In case of a received "reply" message: The client MAY ignore the Element.
- In case of a "write" operation to be created: The client MAY set the Element but SHALL consider that the server may ignore the Element.
- In case of a received "notify" message: The client MAY ignore the Element.

M, R or O may be combined with the suffix "(event)" to express that a supported Element or value only has to be supported during a certain event and hence does not need to be present at all times. If

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the event is not active the Element may be omitted or another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

In some cases, an Element may have a double presence indication, separated by a comma. Examples:

- M, O
- M \W, O \W

The description (rules) of the Element details in such a case the presence requirement. A typical example is a list-based Element where the "first" (or "last") Element has a different presence requirement than the other list Elements.

3.1.3.3 Presence indications for "Content of Function..." tables

This section only defines rules for the server side.

Elements that are marked with "M" SHALL be supported by the server in case of readable as well as writeable data. In case of writeable data (marked with "M \W") the server does not need to set the Element, because the Element is set only by the client.

The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- "R" means that the data SHOULD be provided by the server.
- "O" means that the data MAY be provided by the server.
- "F" means that the data SHALL NOT be provided by the server.

The following applies for writeable data that is exchanged in a "write" operation:

- "R" means that the data SHOULD be supported. In other words: If the client writes the Element, the server SHOULD accept those messages and the contained Elements.
- "O" means that the data MAY be supported. In other words: If the client writes the Element, the server MAY accept those messages and the contained Elements.

The following applies for Elements that are not listed in the Actor section:

- In case of a received "read" request: The according Element MAY be set in the reply.
- In case of a received "write" operation: The server MAY ignore the Element.
- In case of a "notify" operation to be created: The server MAY set the Element.

Note: The server will only accept write operations if the result fulfils the server Function requirements (permitted values, e.g.). Write operations on Elements that are not writeable MAY result in an error message.

M, R or O may be combined with the suffix "(event)" to express that a supported Element or value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active the Element may be omitted or another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

In some cases, an Element may have a double presence indication, separated by a comma. Examples:

- M, O

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- M \W, O \W

The description (rules) of the Element details in such a case the presence requirement. A typical example is a list-based Element where the "first" (or "last") Element has a different presence requirement than the other list Elements.

3.1.3.4 Cardinality indications on Elements and list items

A cardinality indication on an Element or list item expresses constraints on the number of occurrences of a given Element or data set. In this section we use "X" as representation for such an Element or data set. Furthermore, "a" and "b" represent constraints. The following rules apply for the occurrence of "X" and its content related to a specific Scenario (see note underneath the list):

1. X
   No cardinality indication.
2. X (a..b)
   This means "X" SHALL occur at least "a" times and at maximum "b" times.
3. X (a..)
   This means "X" SHALL occur at least "a" times and MAY occur more than "a" times.
4. X (..b)
   This means "X" SHALL occur at maximum "b" times and MAY occur less than "b" times (even zero occurrences are permissive).

Note: These rules apply only under consideration of presence indications and with regards to the given Scenario or Function definition for this Use Case.

The following table is an example to explain this for two different placements.

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
...	...		...
2: M \W	xFeatureType. xListData. xData. (1..3)		
2: M \W	xId	<*#(1..)>	PRIMARY IDENTIFIER
2: M \W	timePeriod		...
2: M \W	timePeriod. startTime	<xs:duration>	
2: M \W	xSlot. (1..)		
2: M \W	xSlot. xSlotId		...
2: M \W	xSlot. duration	<xs:duration>	...
...	...	...	...

Table 4: Example table for cardinality indications on Elements and list items

The field

xFeatureType. xListData. xData. (1..3)

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introduces a data pattern (required Elements and values) for "xData" instances used for Scenario 2. The field itself specifies that such an "xData" instance SHALL occur at least 1 time and at maximum 3 times within "xListData" of Feature Type "xFeatureType". However, this constraint holds only for Scenario 2 and only if such "xData" are required. In this case, they are required, as the left field

2: M \W

denotes that this data set is mandatory for Scenario 2.

The field

xSlot. (1..)

expresses that the Element "xSlot" SHALL occur at least one time within its "xData", but MAY occur more than one time.

For the expression "<*#(1..)>" of Element "xId" please see section 3.1.3.6.

The remaining fields do not have an explicit cardinality indication.

Note: Cardinality expressions are also used within placeholder expressions as defined in section 3.1.3.6. In many cases such placeholder expressions define the number of required or permitted Elements or list items as they explicitly define how many different values for a given Identifier are required or permitted for a given Scenario.

3.1.3.5 Writability and changeability indication

In the same column where the presence indications are denoted, a mark is used to distinguish between writeable, changeable or readable Elements:

- Elements that are marked with "\W" are written by a client and SHALL be writeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\C" are changed by a client and SHALL be changeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\RW" are read and written by a client and SHALL be writeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are marked with "\RC" are read and changed by a client and SHALL be changeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are not marked are only read by a client and SHALL be provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.

"Writeable" means that the Element and its value may be written by a client. This includes the possibility to modify (if the Element is already present), create (if the Element is not present yet), and

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delete the Element. The server SHALL adjust its Function according to the received "write" operation (unless the server cannot accept the "write" operation according to section 3.1.3.3).

"Changeable" means that the Element's value may be changed by a client. If the Element is not present at the resource before, it probably **cannot** be created by the client via the "write" operation. In this case the server MAY decline such a message.

Note: "\W" includes "\C" already.

Note: Depending on the resource a client might need to request a proper binding before the server accepts a "write" operation.

3.1.3.6 "Value" placeholders

3.1.3.6.1 Introduction

Specializations may use placeholders to model relations between different Elements or even list items of different Functions. The main purpose is to declare which Identifier values relate to each other. As a Use Case does not prescribe specific values to be used for a given Identifier, a placeholder like "<x1>" can be used in "Value" columns to express the intended relations.

There are two styles to identify a placeholder that can be referenced:

1. <xM>
2. <xM#S>

where

1. "x" is any alphabetical prefix like "m", "t", "ec", "cc", etc.
2. "M" is a (major) number like "1", "2", "15", etc.
3. "S" is a sub-number like "1", "7", "10", etc.

Examples for the first style are "<ec1>", "<z12>". Examples for the second style are "<p4#2>", "<m22#3>". For a given placeholder, only one of the styles can be used.

In addition, there are also styles for placeholders that do not need to be referenced:

1. <*>
2. <*#S>

The second style is only used with so-called cardinality expressions.

3.1.3.6.2 Uniqueness of placeholders

A given placeholder <xM> or <xM#S> represents the same value throughout a given Use Case specification for a given set of its parent Identifier values. This shall be explained in a brief example:

We assume a list item with PRIMARY IDENTIFIER "pId". It also has a SUB IDENTIFIER "sId" with placeholder "<s1>". Then, each occurrence of "<s1>" represents the same value for a given value of

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pId. This means that "<s1>" of a list item with pId=1 can differ from "<s1>" of a list item with pId=2. But it also means that "<s1>" represents the same value in all list items with pId=1.

Note: Typically, parent Identifiers like "pId" will also be expressed with a placeholder like "<p5>", e.g. In this case, the uniqueness rule applies for "<p5>" likewise.

Note: The uniqueness also applies for placeholders used as FOREIGN IDENTIFIER.

3.1.3.6.3 Placeholder expressions with cardinalities

For some Identifiers, more than one placeholder is needed. Several notations are used for this purpose, which make use of cardinality expressions. The general notation is as follows:

1. <xM#(a..b)>

This is equivalent to this explicit definition:

> At least a and at maximum b placeholders of this list: <xM#1> <xM#2> ... <xM#b>

This means that the implementation of a given Use Case (or Scenario) requires a minimum of "a" distinct values of the respective Identifier. In total, there can be up to "b" distinct values of the respective Identifier.

Additionally, the following notations may occur:

2. <xM#(a..)>
   This is equivalent to "<xM#(a..b)>" with "b" equal to infinity.
3. <xM#(..b)>
   This is equivalent to "<xM#(a..b)>" with "a" equal to zero.

"<xM#(a..)>" has only a lower bound of "a" distinct values, but no upper bound. "<xM#(..b)>", on the other hand, expresses that the Identifier may not be required at all, but it is permitted to have up to "b" distinct values.

Similarly, the cardinality can be used for placeholders that are not referenced, i.e. <*#(a..b)> etc.

Note: The cardinality does NOT express which of the sub-numbers have to be used! I.e., it does NOT mean that the Identifiers <xM#1> ... <xM#a> are always used and just those with larger sub-numbers (<xM#a+1> ... <xM#b>) are optional. If, for instance, a placeholder expression "<xM#(3..5)>" is given, it may well happen that an implementation makes use of <xM#2>, <xM#4>, and <xM#5> (i.e., it does NOT use <xM#1>, <xM#3>). Which sub-numbers are used usually depends on other parts of a Specialization and their references to required placeholders, which is explained in section 3.1.3.6.4.

3.1.3.6.4 References to placeholders and relations

According to the styles for placeholders that can be referenced, an enumeration value "e" can refer to a particular placeholder:

1. e(-><xM>)
2. e(-><xM#S>)

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This denotes that "e" is to be used with "<xM>" or "<xM#S>", resp.

Example: A Specialization contains the Elements "mId" and "phase". "mId" is an Identifier with placeholder definition <m2#(1..3)>. "phase" is a string that permits the values "a", "b", and "c" using this expression:

"a"(-><m2#1>)
"b"(-><m2#2>)
"c"(-><m2#3>)

This expresses that the enumeration value "a" is to be used with the placeholder <m2#1>, "b" is to be used with <m2#2> and "c" with <m2#3>.

Similarly, a placeholder "yN" can refer to a particular placeholder:

3. <yN->xM>
4. <yN->xM#S>
5. <yN#T->xM>
6. <yN#T->xM#S>

where "T" is a sub-number of "yN".

It is also feasible to associate placeholders with cardinalities:

7. <yN#(a..b)->xM#(a..b)>

denotes that <yN#1> is to be used with <xM#1>, <yN#2> is to be used with <xM#2>, etc.

Note: In this case, the placeholder expressions of yN and xM must have the same cardinality.

In some cases, there is a need to express that multiple list items with similar values are feasible or required, but only particular combinations of these different data are permitted. The following example shows that several "fData" list items with different "phase" value are required, but that these list items may only express either the "phase" value combination { "a", "b", "c" } or the "phase" value combination { "a", "abc", "neutral" }. The permitted combinations are defined in a note below a table:

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	F. fListData. fData.		
2: M	zId	<z3#(3..5)>	
2: M	phase	"a"(-><z3#1>)	
		"b"(-><z3#2>)	
		"c"(-><z3#3>)	
		"abc"(-><z3#4>)	
		"neutral"(-><z3#5>)	

Table 5: Content of an example Specialization

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Note: One of the following combinations SHALL be used at least: {<z3#1>, <z3#2>, <z3#3>} or {<z3#1>, <z3#4>, <z3#5>}.

3.1.3.7 Rules for content of "Value" column

For a given Scenario, the "Value" column may restrict the permitted content of a Function's Element to one or more particular values. This means that Elements with values deviating from the restriction (i.e. from the permitted values) do not belong to the respective Scenario and need to be considered as if the Element is not set. If more than one particular value is permitted for an Element, the values are in a single line each.

If a presence indication is set for the value (in an additional column before the value), the following rules SHALL be applied:

- "M" means that the value SHALL be supported. This means the value needs to be set at a certain point in time (depending on the value rules) or for a certain Element within a list entry.
- "R" means that the value SHOULD be supported.
- "O" means that the value MAY be supported.

If all possible values of a given mandatory Element are optional or recommended and this Element is used for the purpose of the respective Scenario, one of the values SHALL be set. If all possible values of a given optional or recommended Element are optional or recommended, this Element MAY contain also other values, but then this Element SHALL NOT be considered as part of the respective Scenario.

M, R or O may be combined with the suffix "(event)" to express that a supported value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

If no presence indication is set for the value, the following rules SHALL be applied:

- In case of Elements where the server may set or change an Element on its own (see section 3.1.3.5):
  - within the tables in the "Server data - Resources" sections:
    - the server SHALL support at least one of the listed values.
  - within the tables in the "Client data - Specializations" sections:
    - the client SHALL support all listed values.
- In case of Elements that are writeable or changeable (see section 3.1.3.5):
  - within the tables in the "Server data - Resources" sections:
    - the server SHALL support all listed values.
  - within the tables in the "Client data - Specializations" sections:
    - the client SHALL support at least one of the listed values.

Depending on the Element, different values may be used during runtime. If this is the case, those rules are described within the value rules.

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If a value is placed in parenthesis, the corresponding value is a recommendation. The actual value MAY deviate from this, e.g. "(1024)".

3.1.3.8 General information on how to interpret the "Content of Function..." and "Content of Specialization..." tables

Within the "Client data - Specializations" sections each Specialization is described in an own sub-section with the name "Specialization "<name of the Specialization>"" (e.g. "Specialization "Measurement_GridFeedInEnergy""). It contains only one table that includes all Elements needed for this Specialization. The different Functions are mentioned in a continuous row, highlighted with grey background colour. This row contains the following parts:

<Feature Type>. <Function>.[<list entry instance name>.]

The <list entry instance name> is only included if the <Function> is a list-based Function. An example could be:

DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData.
deviceConfigurationKeyValueDescriptionData.

In the following rows, only the names of the Elements are stated, without the prefix described above.

Within the "Server data - Resources" sections each Feature Type is described in an own sub-section with the name "Feature Type "<name of the Feature Type>"" (e.g. "Feature Type "Measurement""). It contains sub-sections for each Function named "Function "<name of the Function>"" (e.g. "Function "measurementListData""). These sections contain one table with all Elements needed for this resource. The list entries are mentioned in a continuous row, highlighted with grey background colour. This row contains the following parts:

<Feature Type>. <Function>.[<list entry instance name>.]

The <list entry instance name> is only included if the <Function> is a list-based Function. An example could be:

Measurement. measurementDescriptionListData. measurementDescriptionData.

In the following rows, only the names of the Elements are stated, without the prefix described above.

For both kinds of tables, the following applies:

- Parent Elements are marked with a dot at the end of the name:
  <parent Element>.
  E.g.:
  value.
- If there are sub-Elements, they are described in own rows with the name of the parent Element as prefix, separated by a dot and a blank space:

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<parent Element>. <sub-Element>

E.g.:

value. number

3.1.4 Rules for "Feature Types and Functions..." tables

3.1.4.1 Presence indications for "Feature Types and Functions..." tables

The following presence indications are used:

Abbreviation	Meaning	Link to requirement keywords
M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 6: Presence indication of Feature Types and Functions support

If at least one Function of a Feature has the presence indication "M", it is mandatory to support the Feature.

3.1.4.2 Rules for "Possible operations" column

Within the "Feature Types and Functions..." tables the column "Possible operations" state whether the Function is read- or writeable (as defined in the detailed discovery mechanism, see [ProtocolSpecification]).

If the "partial" concept (also called "restricted function exchange") SHALL be supported, the following notation is used (separated for read and write access):

read (M). partial (M)
write (M). partial (M)

If the "partial" concept SHOULD be supported, the following notation is used:

read (M). partial (R)
write (M). partial (R)

If the "partial" concept MAY be supported, the following notation is used:

read (M). partial (O)
write (M). partial (O)

The server can decide whether a notification is submitted complete or partial (as described in [ProtocolSpecification]) if not defined differently within this Use Case Specification.

3.1.5 "Actor ... overview" diagram rules

Within the "Actor [...] overview" diagrams in the "Actors" sub-sections the complete functionality of this Use Case is provided, including optional Scenarios. Which Scenarios are optional can be found in Table 1. The Actor MAY have more functionality implemented than needed for this Use Case.

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For the following Actor overview example, a brief description of the graphical symbols will be described.

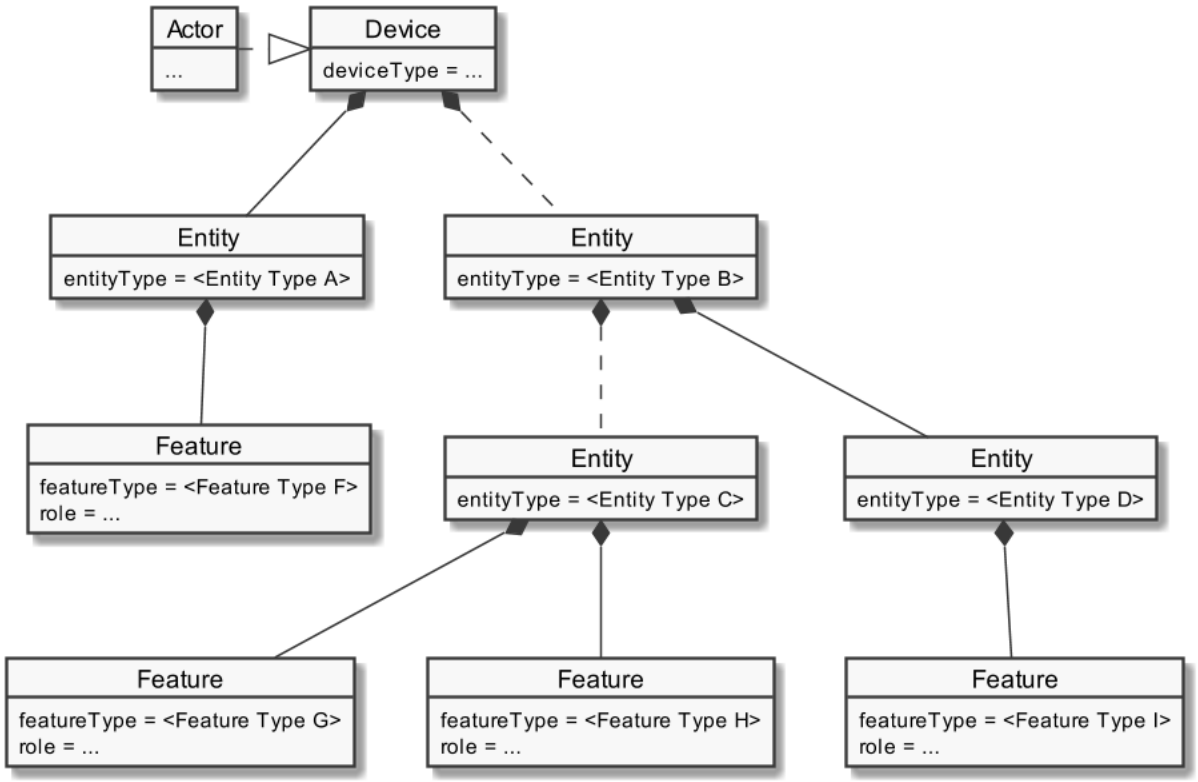


Figure 15: Actor overview example

The solid lines in the figure represent an immediate parent-childhood relation: The Entity with "<Entity Type A>" is a direct child of "Device". The Entity with "<Entity Type D>" is a direct child of the Entity with "<Entity Type B>". All Features are immediate child of the respective Entity.

The dashed lines in the figure express that there MAY be additional Entities between the shown Entities: A vendor's implementation MAY have one or more Entities between "Device" and the Entity with "<Entity Type B>". Likewise, a vendor's implementation MAY have one or more Entities between the Entity with "<Entity Type B>" and the Entity with "<Entity Type C>".

3.1.6 Specializations

Within the "Actors" sub-sections, Specializations are referenced. A Specialization describes a dataset necessary to fulfil the specific requirements of a High-Level Use Case and its Scenarios. Often, data from multiple different Features and Functions are needed to fulfil the requirements. Therefore, a Specialization defines a dataset that may encompass multiple related Functions from one or more different Features.

As different Use Cases sometimes share similar requirements, Specializations are also important from a re-usability perspective. This approach is used to improve consistency across Use Cases and avoid multiple variances of basically the same dataset. This is especially important in the case when an implementation supports multiple Use Cases. E.g. if a power measurement is necessary in two different Use Cases, both Use Cases could define slightly different datasets. In this case, the server as

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well as the client functionality would have to implement both variances if both Use Cases are supported. This means, depending on the number of Use Cases, two or more datasets need to be generated, transmitted and stored instead of one. Therefore, common Specializations are defined to enable consolidated implementations. Specializations are detailed per Actor, as distinct Actors may use different parts of a Specialization or even have different permissions to change data. In general, this may also depend on the respective Scenario.

If a Feature server can provide the data of a Specialization, the data does not necessarily always need to be available at the Feature server. There might be situations where the user deactivates a Use Case. There may also be other reasons why Use Case data cannot be provided currently. Therefore, a client always needs to be subscribed (as described in section 3.3.4) on the corresponding dataset to stay updated.

The SPINE resource descriptions given in the "SPINE resources of the Actor" sections are derived from the Specializations given in the Actor's overview diagram.

3.1.7 Order of messages within the sequence diagrams

There are several sequence diagrams in this document describing message flows. The order of the messages SHOULD be kept by the communications partners, but there might be cases where a different order makes sense. The communications partners SHALL be able to handle the Scenario functionalities even if the messages are transmitted in a different order by the other Actor(s). The sequence diagrams can be seen as examples.

3.1.8 Further information and rules

3.1.8.1 Frequently used Element rules for the Resource and Specialization tables

This section serves as a collection of rules frequently used by Resource and Specialization tables of the subsequent sections. Each rule applies only where referenced explicitly in the tables.

Note: The purpose of this collection is just to reduce the size of the tables. As such, no rule has a meaning without a reference indicating the required rule. A reference looks like "See [Scaled number rules]", e.g.

[String-length rules]:

The string-length SHOULD NOT be longer than 256 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 256 characters.

[Scaled number rules]:

The sub-Elements "number" and "scale" represent a value according to the following formula:
number * 10^{scale}

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3.1.8.2 Rules regarding the usage of time-related information

Times SHALL be presented as absolute times in UTC ([ITPCM-041^{UTC}]). This means relative times SHALL NOT be used, unless the data point only supports relative times (e.g. for a "duration").

Note: This means the Actors need to have a common understanding of these absolute times. It is strongly recommended that each Actor is "sufficiently synchronised" with the official "Coordinated Universal Time" (UTC).

Note: A UTC start time plus a duration yield a UTC end time of the duration ([ITPCM-041^{UTC}]).

3.2 Actors

3.2.1 Energy Consumer

3.2.1.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "EnergyConsumer" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor Energy Consumer's resource hierarchy.

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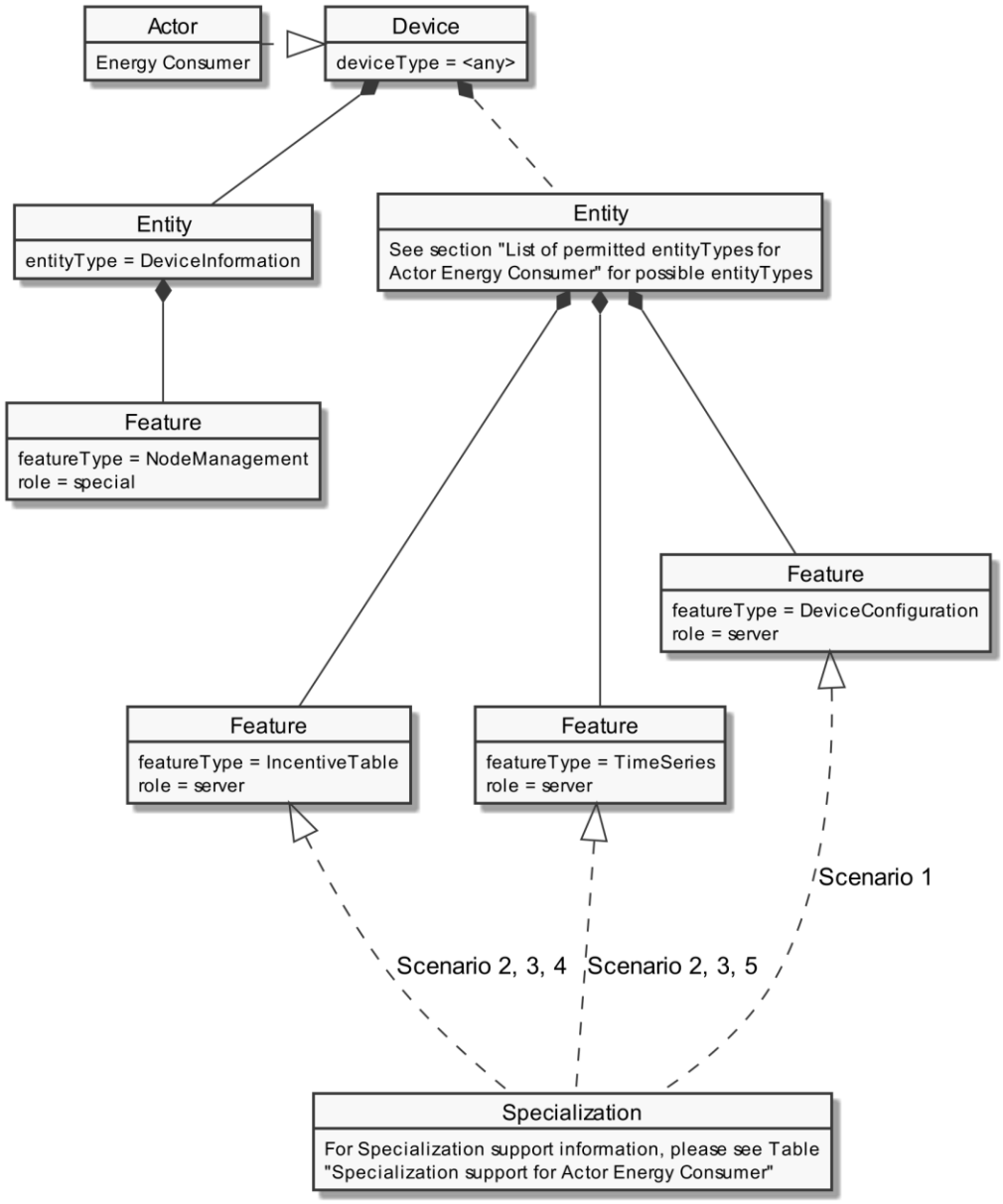


Figure 16: Actor "Energy Consumer" overview

The ""Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind one of the entityTypes listed in section 3.2.1.1.1. The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a

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corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Used Feature...	...in tables
DeviceConfiguration_IncentivesSimulationCyclesMax	1	DeviceConfiguration	Table 14 Table 15
DeviceConfiguration_IncentivesSimulationConcurrent	1	DeviceConfiguration	Table 14 Table 15
DeviceConfiguration_IncentivesTimeoutIncentiveRequest	1	DeviceConfiguration	Table 14 Table 15
DeviceConfiguration_IncentivesWaitIncentiveWriteable	1	DeviceConfiguration	Table 14 Table 15
IncentiveTable_SimpleCommittedIncentiveTable	2, 4	IncentiveTable	Table 9 Table 10 Table 11
IncentiveTable_SimplePreliminaryIncentiveTable	3	IncentiveTable	Table 9 Table 10 Table 11
TimeSeries_CommittedPowerPlan	2, 5	TimeSeries	Table 12 Table 13
TimeSeries_PreliminaryPowerPlan	3	TimeSeries	Table 12 Table 13

Table 7: Specialization support for Actor Energy Consumer

3.2.1.1.1 List of permitted entityTypes for Actor Energy Consumer

In this version of this Use Case, only the following entityTypes are permitted for the Actor Energy Consumer:

- HeatPumpAppliance

This may change at a later version and be extended to further entityTypes.

3.2.1.2 Server data - Resources

Behind one of the entityTypes listed in section 3.2.1.1.1, the Actor Energy Consumer SHALL offer the Feature Types and Functions given in the table below.

Feature Type	Scenario: M/R/O	Function	Possible operations
DeviceConfiguration	1: M	deviceConfigurationKeyValueDescriptionListData	read (M). partial (R)
	1: M	deviceConfigurationKeyValueListData	read (M). partial (R)

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IncentiveTable	2: M 3: M 4: M	incentiveTableDescriptionData	read (M). partial (R) write (M). partial (M)
	2: M 3: M 4: M	incentiveTableConstraintsData	read (M). partial (M)
	2: M 3: M 4: M	incentiveTableData	read (M). partial (M) write (M). partial (M)
TimeSeries	2: M 3: M 5: M	timeSeriesDescriptionListData	read (M). partial (R)
	2: M 3: M 5: M	timeSeriesListData	read (M). partial (M)

Table 8: Feature Types and Functions used within this Use Case by the Actor Energy Consumer

For each of these Feature Types, the following rule applies: There SHALL be at maximum one Feature with the Feature Type in the Entity.

Note: As a consequence of the previous rule, an implementation may need to have Feature data from different Scenarios/Specializations or even Use Cases in a given Feature.

The Scenario number shows in which Scenarios the Energy Consumer acts as server and which Feature Types and Functions are relevant in each Scenario.

A detailed definition of the Elements and values that shall be supported in each Function is given in the following sub-sections.

Note: If in the table above "partial" read is not mentioned or is only optional, it still might be mandatory to support partial notifications. The details of "partial" support are described within the Scenario sections.

Note: The presence indications stated above are meant relative to the ones of the according Scenario stated in Table 1. I.e., if a Scenario is optional ("O") and a Feature Type is mandatory ("M"), the Feature Type need only be supported if the Scenario is supported, too.

Note: Further Features MAY be implemented on the same Entities; also, further Functions MAY be implemented in the used Entities.

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3.2.1.2.1 Feature Type "IncentiveTable"

3.2.1.2.1.1 Function "incentiveTableDescriptionData"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M 4: M	IncentiveTable. incentiveTableDescriptionData. incentiveTableDescription.		
2: M 4: M	tariffDescription.		
2: M 4: M	tariffDescription. tariffId	<tf1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER of an incentiveTable.
2: M 4: M	tariffDescription. tariffWriteable	"true"	[ITPCM-003^{IT_MODIFIABLE}]
		"false"	[ITPCM-006^{IT_UNMODIFIABLE}]
2: M 4: M	tariffDescription. updateRequired	"true" [ITPCM-004^{IT_REQUEST}]	If updateRequired is "true", the server expects the responsible client to update the writeable or changeable data related to the same PRIMARY IDENTIFIER. When the client updates the server the client SHALL ignore updateRequired until the writeable and changeable data related to the same PRIMARY IDENTIFIER was updated successfully. The server SHALL set updateRequired back to "false", as soon as the update of the writeable or changeable data related to the same PRIMARY IDENTIFIER was successful. The server SHALL ensure that only one responsible client is able to update the related data. However, also if updateRequired is set to "false" a server SHOULD in general allow updates of the data from the responsible client. The server MAY choose to withdraw the update request at any time by setting updateRequired back to "false".
		"false" [ITPCM-007^{IT_NO_REQUEST}]	

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2: M 4: M	tariffDescription. scopeType	"simpleCommittedIncentiveTable"		[ITPCM-045IT_COMMITTED]
2: M 4: M	tariffDescription. slotIdSupport	"true"		
2: M \W 4: M \W	tier.			[ITPCM-013IT_TIER]
2: M \W 4: M \W	tier. tierDescription.			
2: M \W 4: M \W	tier. tierDescription. tierId	<tr1#(1..5)>		SHALL be set as SUB IDENTIFIER of tariffId.
2: M \W 4: M \W	tier. tierDescription. tierType	"dynamicCost"		[ITPCM-010IT_GEN] The tier has a cost incentive that MAY vary over time.
2: R \W 4: R \W	tier. tierDescription. label			[ITPCM-029TIER_DESCR] [String-length rules]
2: O \W 4: O \W	tier. tierDescription. description			[ITPCM-029TIER_DESCR] [String-length rules]
2: M \W 4: M \W	tier. boundaryDescription.			
2: M \W 4: M \W	tier. boundaryDescription. boundaryId	<by1#(1..1)>		SHALL be set as SUB IDENTIFIER of tierId.
2: M \W 4: M \W	tier. boundaryDescription. boundaryType	"powerBoundary"		[ITPCM-012IT_POWER] The boundary SHALL be an electrical power. E.g. if the surplus power is consumed, more expensive power has to be used. Positive powerBoundary values describe which power can be consumed with certain incentives, while negative powerBoundary values describe which power can be produced with certain incentives.
2: M \W 4: M \W	tier. boundaryDescription. boundaryUnit	"W"		If set, the unit SHALL be applied to the lowerBoundaryValue and upperBoundaryValue.
2: M \W 4: M \W	tier. incentiveDescription.			[ITPCM-019UNIQUE_I_TYPES]
2: M \W 4: M \W	tier. incentiveDescription. incentiveId	<ie1#(1..3)>		SHALL be set as SUB IDENTIFIER of tierId.
2: M \W 4: M \W	tier. incentiveDescription. incentiveType Note: At least one of the values SHALL be used.	2: M \W 4: M \W	"absoluteCost"	[ITPCM-016ABSOLUTE_PRICE] Each related value SHALL be an absolute cost. Positive values SHALL be interpreted as costs. Negative values SHALL be interpreted as profits. The element "currency" SHALL be set to the respective currency. The element "unit" SHALL be set to "Wh". The cost refers to "currency" per "unit" (EUR/Wh, e.g.).

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		2: O \W 4: O \W	"renewableEnergyPercentage"	[ITPCM-017[RENEWABLE_PCT]] The "unit" Element of the incentive SHALL be set to "pct" (percentage). The value SHALL be in the range from 0% to 100%.
		2: O \W 4: O \W	"co2Emission"	[ITPCM-018[CO2]] The "unit" Element of the incentive SHALL be set to "kg/Wh". Only positive values SHALL be used.
2: O \W 4: O \W	tier. incentiveDescription. currency			If set, the currency SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
2: O \W 4: O \W	tier. incentiveDescription. unit			If set, the unit SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
3: M	IncentiveTable. incentiveTableDescriptionData. incentiveTableDescription.			
3: M	tariffDescription.			
3: M	tariffDescription. tariffId	<tf2#(1..1)>		SHALL be set as PRIMARY IDENTIFIER of an incentiveTable.
3: M	tariffDescription. tariffWriteable	"true"		[ITPCM-036[PIT_MODIFIABLE]]
		"false"		[ITPCM-037[PIT_UNMODIFIABLE]]
3: F	tariffDescription. updateRequired			
3: M	tariffDescription. scopeType	"simplePreliminaryIncentiveTable"		[ITPCM-046[IT_PRELIMINARY]]
3: M	tariffDescription. slotIdSupport	"true"		
3: M \W	tier.			[ITPCM-013[IT_TIER]]
3: M \W	tier. tierDescription.			
3: M \W	tier. tierDescription. tierId	<tr2#(1..5)>		SHALL be set as SUB IDENTIFIER of tariffId.
3: M \W	tier. tierDescription. tierType	"dynamicCost"		[ITPCM-010[IT_GEN]] The tier has a cost incentive that MAY vary over time.
3: R \W	tier. tierDescription. label			[ITPCM-029[TIER_DESCR]] [String-length rules]
3: O \W	tier. tierDescription. description			[ITPCM-029[TIER_DESCR]] [String-length rules]
3: M \W	tier. boundaryDescription.			
3: M \W	tier. boundaryDescription. boundaryId	<by2#(1..1)>		SHALL be set as SUB IDENTIFIER of tierId.
3: M \W	tier. boundaryDescription. boundaryType	"powerBoundary"		[ITPCM-012[IT_POWER]] The boundary SHALL be an electrical power. E.g. if the surplus power is consumed, more expensive power has to be used. Positive powerBoundary values describe which power can be consumed with certain incentives, while negative

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				powerBoundary values describe which power can be produced with certain incentives.
3: M \W	tier. boundaryDescription. boundaryUnit	"W"		If set, the unit SHALL be applied to the lowerBoundaryValue and upperBoundaryValue.
3: M \W	tier. incentiveDescription.			[ITPCM-019^{UNIQUE_I_TYPES}]
3: M \W	tier. incentiveDescription. incentiveId	<ie2#(1..3)>		SHALL be set as SUB IDENTIFIER of tierId.
3: M \W	tier. incentiveDescription. incentiveType Note: At least one of the values SHALL be used.	3: M \W	"absoluteCost"	[ITPCM-016^{ABSOLUTE_PRICE}] Each related value SHALL be an absolute cost. Positive values SHALL be interpreted as charge. Negative values SHALL be interpreted as profits. The element "currency" SHALL be set to the respective currency. The element "unit" SHALL be set to "Wh". The cost refers to "currency" per "unit" (EUR/Wh, e.g.).
		3: O \W	"renewableEnergyPercentage"	[ITPCM-017^{RENEWABLE_PCT}] The "unit" Element of the incentive SHALL be set to "pct" (percentage). The value SHALL be in the range from 0% to 100%.
		3: O \W	"co2Emission"	[ITPCM-018^{CO2}] The "unit" Element of the incentive SHALL be set to "kg/Wh". Only positive values SHALL be used.
3: O \W	tier. incentiveDescription. currency			If set, the currency SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
3: O \W	tier. incentiveDescription. unit			If set, the unit SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.

Table 9: Content of Function "incentiveTableDescriptionData" at Actor Energy Consumer

3.2.1.2.1.2 Function "incentiveTableConstraintsData"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
2: M 4: M	IncentiveTable. incentiveTableConstraintsData. incentiveTableConstraints.		
2: M 4: M	tariff.		

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2: M 4: M	tariff. tariffId	<tf1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER of an incentiveTable.
2: M 4: M	tariffConstraints.		
2: M 4: M	tariffConstraints. maxTiersPerTariff	M: ≥ 3 R: ≥ 5	[ITPCM-009^{MAX_TIERS}] If set, the incentiveTable SHALL NOT include more tiers than given in maxTiersPerTariff.
2: M 4: M	tariffConstraints. maxBoundariesPerTier	≥ 1	[ITPCM-043^{MAX_BOUNDARIES}] If set, the related "incentiveTableData. incentiveTable. incentiveSlot. tier" instance SHALL NOT include more "boundary" instances than given in maxBoundariesPerTier.
2: M 4: M	tariffConstraints. maxIncentivesPerTier	≥ 1	[ITPCM-014^{MAX_INCENTIVES}] If set, the tier within the incentiveTable SHALL NOT include more incentives than given in maxIncentivesPerTier.
2: M 4: M	incentiveSlotConstrain ts.		
2: M 4: M	incentiveSlotConstrain ts. slotCountMax		[ITPCM-044^{MAX_SLOTS}] If set, the incentiveTable SHALL NOT include more slots than given in slotCountMax.
3: M	IncentiveTable. incentiveTableConstraintsData. incentiveTableConstraints.		
3: M	tariff.		
3: M	tariff. tariffId	<tf2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER of an incentiveTable.
3: M	tariffConstraints.		
3: M	tariffConstraints. maxTiersPerTariff	M: ≥ 3 R: ≥ 5	[ITPCM-009^{MAX_TIERS}] If set, the incentiveTable SHALL NOT include more tiers than given in maxTiersPerTariff.
3: M	tariffConstraints. maxBoundariesPerTier	≥ 1	[ITPCM-043^{MAX_BOUNDARIES}] If set, the related "incentiveTableData. incentiveTable. incentiveSlot. tier" instance SHALL NOT include more "boundary" instances than given in maxBoundariesPerTier.
3: M	tariffConstraints. maxIncentivesPerTier	≥ 1	[ITPCM-014^{MAX_INCENTIVES}] If set, the tier within the incentiveTable SHALL NOT include more incentives than given in maxIncentivesPerTier.
3: M	incentiveSlotConstrain ts.		
3: M	incentiveSlotConstrain ts. slotCountMax		[ITPCM-044^{MAX_SLOTS}]

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			If set, the incentiveTable SHALL NOT include more slots than given in slotCountMax.

1381 *Table 10: Content of Function "incentiveTableConstraintsData" at Actor Energy Consumer*

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1383 *3.2.1.2.1.3 Function "incentiveTableData"*

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M \W 4: M \W	IncentiveTable. incentiveTableData. incentiveTable.		
2: M \W 4: M \W	tariff.		
2: M \W 4: M \W	tariff. tariffId	<tf1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M \W 4: M \W	incentiveSlot.		[ITPCM-011^{IT_SLOT}]
2: M \W 4: M \W	incentiveSlot. timeInterval.		[ITPCM-021^{SLOT_PERIOD}] SHALL be set. The timeInterval of different incentiveSlots within an incentiveTable SHALL NOT overlap in time.
2: M \W 4: M \W	incentiveSlot. timeInterval. timeSlotId	<ti1#(1..)>	SUB IDENTIFIER of tariffId.
2: M \W 4: M \W	incentiveSlot. timeInterval. startTime.		
2: M \W 4: M \W	incentiveSlot. timeInterval. startTime. dateTime		
2: M \W, O \W 4: M \W, O \W	incentiveSlot. timeInterval. endTime.		[ITPCM-033^{IT_GAPLESS}] SHALL be set for the last slot. MAY be set for all other slots.
2: M \W 4: M \W	incentiveSlot. timeInterval. endTime. dateTime		
2: M \W 4: M \W	incentiveSlot. tier.		[ITPCM-013^{IT_TIER}]
2: M \W 4: M \W	incentiveSlot. tier. tier.		
2: M \W 4: M \W	incentiveSlot. tier. tier. tierId	<tr1#(1..5)>	SUB IDENTIFIER of tariffId to identify a tier of a tariff. SHALL be set.

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2: M \W 4: M \W	incentiveSlot. tier. boundary.		The boundary list defines the boundaries of a tier. The boundary range of different tiers within an incentiveSlot defined by lowerBoundaryValue and upperBoundaryValue SHALL NOT overlap for boundaries with the same boundaryType. The boundary range of a tier includes all values that are equal or greater than lowerBoundaryValue and lower than upperBoundaryValue.
2: M \W 4: M \W	incentiveSlot. tier. boundary. boundaryId	<by1#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set.
2: M \W 4: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue.		[ITPCM-035^{IT_CONSUMPTION}], [ITPCM-040^{IT_P_GAPLESS}] Within a boundary lowerBoundaryValue SHALL be smaller than upperBoundaryValue. **Lower and upper boundaries of adjacent tiers**: If lowerBoundaryValue is omitted and upperBoundaryValue is set, the lowerBoundaryValue SHALL be interpreted as if it is equal (without a gap) to the upperBoundaryValue of the next lower tier within the incentiveSlot. **Minus infinity lowerBoundary rule**: If lowerBoundaryValue is omitted and there exists no lower tier with a lowerBoundaryValue or upperBoundaryValue within the incentiveSlot, the omitted lowerBoundaryValue SHALL be interpreted as minus infinity. See [Scaled number rules].
2: M \W 4: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. number		
2: M \W 4: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. scale		
2: M \W 4: M \W	incentiveSlot. tier. incentive.		[ITPCM-022^{INCENTIVE}]
2: M \W 4: M \W	incentiveSlot. tier. incentive. incentiveId	<ie1#(1..3)>	SHALL be used as SUB IDENTIFIER of tierId.

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2: M \W 4: M \W	incentiveSlot. tier. incentive. value.		[ITPCM-023^{INCENTIVE_VALUE}] See [Scaled number rules].
2: M \W 4: M \W	incentiveSlot. tier. incentive. value. number		
2: M \W 4: M \W	incentiveSlot. tier. incentive. value. scale		
3: M \W	IncentiveTable. incentiveTableData. incentiveTable.		
3: M \W	tariff.		
3: M \W	tariff. tariffId	<tf2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M \W	incentiveSlot.		[ITPCM-011^{IT_SLOT}]
3: M \W	incentiveSlot. timeInterval.		[ITPCM-021^{SLOT_PERIOD}] SHALL be set. The timeInterval of different incentiveSlots within an incentiveTable SHALL NOT overlap in time.
3: M \W	incentiveSlot. timeInterval. timeSlotId	<ti2#(1..)>	SUB IDENTIFIER of tariffId.
3: M \W	incentiveSlot. timeInterval. startTime.		
3: M \W	incentiveSlot. timeInterval. startTime. dateTime		
3: M \W, O \W	incentiveSlot. timeInterval. endTime.		[ITPCM-033^{IT_GAPLESS}] SHALL be set for the last slot. MAY be set for all other slots.
3: M \W	incentiveSlot. timeInterval. endTime. dateTime		
3: M \W	incentiveSlot. tier.		[ITPCM-013^{IT_TIER}]
3: M \W	incentiveSlot. tier. tier.		
3: M \W	incentiveSlot. tier. tier. tierId	<tr2#(1..5)>	SUB IDENTIFIER of tariffId to identify a tier of a tariff. SHALL be set.
3: M \W	incentiveSlot. tier. boundary.		The boundary list defines the boundaries of a tier. The boundary range of different tiers within an incentiveSlot defined by lowerBoundaryValue and upperBoundaryValue SHALL NOT overlap for boundaries with the same boundaryType. The boundary range of a tier includes all values that are equal or greater than lowerBoundaryValue and lower than upperBoundaryValue.
3: M \W	incentiveSlot. tier. boundary. boundaryId	<by2#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set.

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3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue.		[ITPCM-035^{IT_CONSUMPTION}], [ITPCM-040^{IT_P_GAPLESS}] Within a boundary lowerBoundaryValue SHALL be smaller than upperBoundaryValue. **Lower and upper boundaries of adjacent tiers**: If lowerBoundaryValue is omitted and upperBoundaryValue is set, the lowerBoundaryValue SHALL be interpreted as if it is equal (without a gap) to the upperBoundaryValue of the next lower tier within the incentiveSlot. **Minus infinity lowerBoundary rule**: If lowerBoundaryValue is omitted and there exists no lower tier with a lowerBoundaryValue or upperBoundaryValue within the incentiveSlot, the omitted lowerBoundaryValue SHALL be interpreted as minus infinity. See [Scaled number rules].
3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. number		
3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. scale		
3: M \W	incentiveSlot. tier. incentive.		[ITPCM-022^{INCENTIVE}]
3: M \W	incentiveSlot. tier. incentive. incentiveId	<ie2#(1..3)>	SHALL be used as SUB IDENTIFIER of tierId.
3: M \W	incentiveSlot. tier. incentive. value.		[ITPCM-023^{INCENTIVE_VALUE}] See [Scaled number rules].
3: M \W	incentiveSlot. tier. incentive. value. number		
3: M \W	incentiveSlot. tier. incentive. value. scale		

1384 *Table 11: Content of Function "incentiveTableData" at Actor Energy Consumer*

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1386 *3.2.1.2.2 Feature Type "TimeSeries"*

1387 *3.2.1.2.2.1 Function "timeSeriesDescriptionListData"*

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M 5: M	TimeSeries. timeSeriesDescriptionListData. timeSeriesDescriptionData.		
2: M 5: M	timeSeriesId	<ts1#(1..1)>	PRIMARY IDENTIFIER of timeSeries.
2: M 5: M	timeSeriesType	"plan"	"plan" SHALL be used to model how a certain value will change over time.
2: M 5: M	timeSeriesWriteable	"false"	If set to false or omitted, this timeSeriesData SHALL NOT be writeable.
2: R 5: R	measurementId	<m1#(1..1)>	[ITPCM-039^{P_TOTAL_ACTIVE}] FOREIGN IDENTIFIER. If measurementId is used, it SHALL link to a measurement value of the total active power ([ITPCM-039^{P_TOTAL_ACTIVE}]).
2: M 5: M	unit	"W"	The unit SHALL be applied to the value in timeSeriesData
2: M 5: M	scopeType	"committedPowerPlan"	[ITPCM-047^{PLAN_COMMITTED}]
3: M	TimeSeries. timeSeriesDescriptionListData. timeSeriesDescriptionData.		
3: M	timeSeriesId	<ts2#(1..1)>	PRIMARY IDENTIFIER of timeSeries.
3: M	timeSeriesType	"plan"	"plan" SHALL be used to model how a certain value will change over time.
3: M	timeSeriesWriteable	"false"	If set to false or omitted, this timeSeriesData SHALL NOT be writeable.
3: R	measurementId	<m1#(1..1)>	[ITPCM-039^{P_TOTAL_ACTIVE}] FOREIGN IDENTIFIER. If measurementId is used, it SHALL link to a measurement value of the total active power ([ITPCM-039^{P_TOTAL_ACTIVE}]).
3: M	unit	"W"	The unit SHALL be applied to the value in timeSeriesData
3: M	scopeType	"preliminaryPowerPlan"	[ITPCM-048^{PLAN_PRELIMINARY}]

1388 *Table 12: Content of Function "timeSeriesDescriptionListData" at Actor Energy Consumer*

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1390 *3.2.1.2.2.2 Function "timeSeriesListData"*

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M 5: M	TimeSeries. timeSeriesListData. timeSeriesData.		
2: M 5: M	timeSeriesId	<ts1#(1..1)>	PRIMARY IDENTIFIER of timeSeries
2: F 5: F	timePeriod.		
2: M 5: M	timeSeriesSlot.		[ITPCM-025^{PLAN_SLOT}]
2: M 5: M	timeSeriesSlot. timeSeriesSlotId		If timeSeriesSlotId is omitted it SHALL be interpreted as zero.
2: M 5: M	timeSeriesSlot. timePeriod.		
2: M 5: M	timeSeriesSlot. timePeriod. startTime	<xs:dateTime>	SHALL be set.
2: M 5: M	timeSeriesSlot. duration		[ITPCM-034^{PLAN_GAPLESS}] SHALL be set and express the duration of the slot. SHALL only contain values greater than zero seconds. The duration SHALL last to the start time of the next slot, if any.
2: M 5: M	timeSeriesSlot. value.		[ITPCM-027^{PLAN_PEXP}] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If minValue is set the content of value SHALL be greater or equal. If maxValue is set the content of value SHALL be smaller or equal. See [Scaled number rules].
2: M 5: M	timeSeriesSlot. value. number	≥0	
2: O 5: O	timeSeriesSlot. value. scale		
2: O 5: O	timeSeriesSlot. minValue.		[ITPCM-026^{PLAN_PMIN}] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If value is set the content of

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			minValue SHALL be smaller or equal. If value is not set but maxValue is set the content of minValue SHALL be smaller or equal. See [Scaled number rules].
2: M 5: M	timeSeriesSlot. minValue. number	≥0	
2: O 5: O	timeSeriesSlot. minValue. scale		
2: O 5: O	timeSeriesSlot. maxValue.		[ITPCM-028^{PLAN_PMAX}] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If value is set the content of maxValue SHALL be greater or equal. If value is not set but minValue is set the content of maxValue SHALL be greater or equal. See [Scaled number rules].
2: M 5: M	timeSeriesSlot. maxValue. number	≥0	
2: O 5: O	timeSeriesSlot. maxValue. scale		
3: M	TimeSeries. timeSeriesListData. timeSeriesData.		
3: M	timeSeriesId	<ts2#(1..1)>	PRIMARY IDENTIFIER of timeSeries
3: F	timePeriod.		
3: M	timeSeriesSlot.		[ITPCM-025^{PLAN_SLOT}]
3: M	timeSeriesSlot. timeSeriesSlotId		If timeSeriesSlotId is omitted it SHALL be interpreted as zero.
3: M	timeSeriesSlot. timePeriod.		
3: M	timeSeriesSlot. timePeriod. startTime	<xs:dateTime>	SHALL be set.
3: M	timeSeriesSlot. duration		[ITPCM-034^{PLAN_GAPLESS}] SHALL be set and express the duration of the slot. SHALL only contain values greater than zero seconds. The duration SHALL last to the start time of the next slot, if any.
3: M	timeSeriesSlot. value.		[ITPCM-027^{PLAN_PEXP}] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If minValue is set the content of value SHALL be greater or equal. If maxValue is set the content of value SHALL be smaller or equal. See [Scaled number rules].
3: M	timeSeriesSlot. value. number	≥0	

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3: O	timeSeriesSlot. value. scale		
3: O	timeSeriesSlot. minValue.		[ITPCM-026^{PLAN_PMIN}] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If value is set the content of minValue SHALL be smaller or equal. If value is not set but maxValue is set the content of minValue SHALL be smaller or equal. See [Scaled number rules].
3: M	timeSeriesSlot. minValue. number	≥0	
3: O	timeSeriesSlot. minValue. scale		
3: O	timeSeriesSlot. maxValue.		[ITPCM-028^{PLAN_PMAX}] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If value is set the content of maxValue SHALL be greater or equal. If value is not set but minValue is set the content of maxValue SHALL be greater or equal. See [Scaled number rules].
3: M	timeSeriesSlot. maxValue. number	≥0	
3: O	timeSeriesSlot. maxValue. scale		

Table 13: Content of Function "timeSeriesListData" at Actor Energy Consumer

3.2.1.2.3 Feature Type "DeviceConfiguration"

3.2.1.2.3.1 Function "deviceConfigurationKeyValueDescriptionListData"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
1: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
1: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	keyName	"incentivesSimulationCyclesMax"	
1: M	valueType	"integer"	

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1: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
1: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	keyName	"incentivesSimulationConcurrent"	
1: M	valueType	"boolean"	
1: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
1: M	keyId	<k3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	keyName	"incentivesTimeoutIncentiveRequest"	
1: M	valueType	"duration"	
1: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
1: M	keyId	<k4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	keyName	"incentivesWaitIncentiveWriteable"	
1: M	valueType	"duration"	

Table 14: Content of Function "deviceConfigurationKeyValueDescriptionListData" at Actor Energy Consumer

3.2.1.2.3.2 Function "deviceConfigurationKeyValueListData"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
1: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
1: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	value.		[ITPCM-031^{MAX_SIMUL_CYCLES}] Exactly one of the child Elements SHALL be set. This SHALL match with the content of *valueType* Element within the key value description part (see above).
1: M	value. integer		SHALL be used.
1: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
1: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	value.		[ITPCM-030^{CONCURRENT_SIMUL}] Exactly one of the child Elements SHALL be set. This SHALL match with the content of *valueType* Element within the key value description part (see above).
1: M	value. boolean		SHALL be used.
1: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
1: M	keyId	<k3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	value.		[ITPCM-005^{TIMEOUT_IT_REQUEST}]

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			Exactly one of the child Elements SHALL be set. This SHALL match with the content of *valueType* Element within the key value description part (see above).
1: M	value. duration	*1	SHALL be used.
1: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
1: M	keyId	<k4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	value.		[ITPCM-008WAIT_IT_MODIFIABLE] Exactly one of the child Elements SHALL be set. This SHALL match with the content of *valueType* Element within the key value description part (see above).
1: M	value. duration	*1	SHALL be used.

Table 15: Content of Function "deviceConfigurationKeyValueListData" at Actor Energy Consumer

*1: Values within the duration Element can be represented in different ways. E.g.: "2 hours" can be represented as "PT2H" or "PT120M".

3.2.1.3 Client Data - Specializations

As this Actor has only server functionality, no Specializations are described within this section.

3.2.2 CEM

3.2.2.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "CEM" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor CEM's resource hierarchy.

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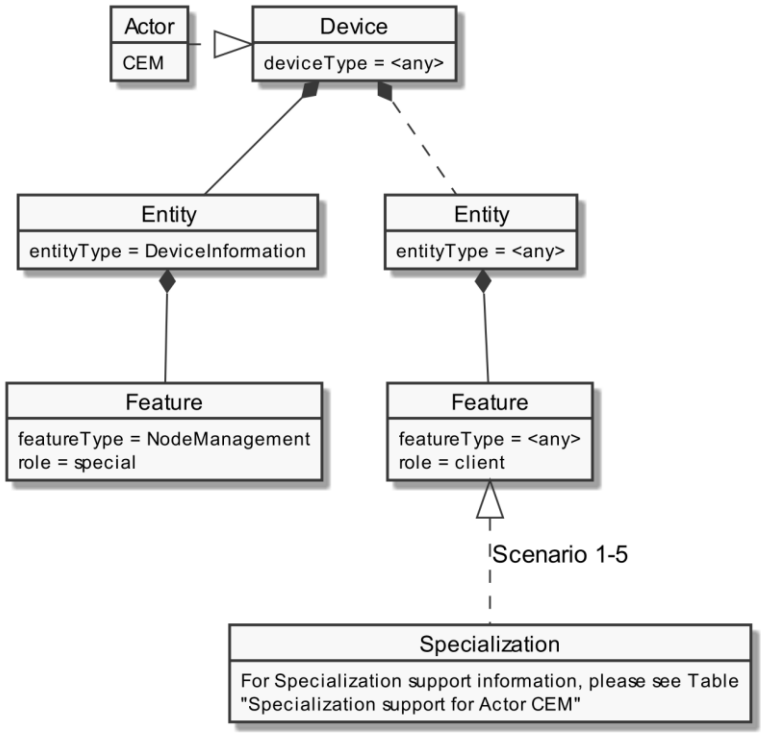


Figure 17: Actor "CEM" overview

The ""Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind any entityType. The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Described in table
DeviceConfiguration_IncentivesSimulationCyclesMax	1	Table 21
DeviceConfiguration_IncentivesSimulationConcurrent	1	Table 22
DeviceConfiguration_IncentivesTimeoutIncentiveRequest	1	Table 23
DeviceConfiguration_IncentivesWaitIncentiveWriteable	1	Table 24
IncentiveTable_SimpleCommittedIncentiveTable	2, 4	Table 17
IncentiveTable_SimplePreliminaryIncentiveTable	3	Table 18

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TimeSeries_CommittedPowerPlan	2, 5	Table 19
TimeSeries_PreliminaryPowerPlan	3	Table 20

Table 16: Specialization support for Actor CEM

3.2.2.2 Server Data - Resources

As this Actor has only client functionality, no resources are described within this section.

3.2.2.3 Client data - Specializations

3.2.2.3.1 Topic "IncentiveTable"

3.2.2.3.1.1 Specialization "IncentiveTable_SimpleCommittedIncentiveTable"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M 4: M	IncentiveTable. incentiveTableDescriptionData. incentiveTableDescription.		
2: M 4: M	tariffDescription.		
2: M 4: M	tariffDescription. tariffId	<tf1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER of an incentiveTable.
2: M 4: M	tariffDescription. tariffWriteable	"true"	[ITPCM-003^{IT_MODIFIABLE}]
		"false"	[ITPCM-006^{IT_UNMODIFIABLE}]
2: M 4: M	tariffDescription. updateRequired	"true" [ITPCM-004^{IT_REQUEST}]	If updateRequired is "true", the server expects the responsible client to update the writeable or changeable data related to the same PRIMARY IDENTIFIER. When the client updates the server the client SHALL ignore updateRequired until the writeable and changeable data related to the same PRIMARY IDENTIFIER was updated successfully. The server SHALL set updateRequired back to "false", as soon as the update of the writeable or changeable data related to the same PRIMARY IDENTIFIER was successful.

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		"false" [ITPCM-007^{IT_NO_REQUEST}]	The server SHALL ensure that only one responsible client is able to update the related data. However, also if updateRequired is set to "false" a server SHOULD in general allow updates of the data from the responsible client. The server MAY choose to withdraw the update request at any time by setting updateRequired back to "false".
2: M 4: M	tariffDescription. scopeType	"simpleCommittedIncentiveTable"	[ITPCM-045^{IT_COMMITTED}]
2: M 4: M	tariffDescription. slotIdSupport	"true"	
2: M \W 4: M \W	tier.		[ITPCM-013^{IT_TIER}]
2: M \W 4: M \W	tier. tierDescription.		
2: M \W 4: M \W	tier. tierDescription. tierId	<tr1#(1..5)>	SHALL be set as SUB IDENTIFIER of tariffId.
2: M \W 4: M \W	tier. tierDescription. tierType	"dynamicCost"	[ITPCM-010^{IT_GEN}] The tier has a cost incentive that MAY vary over time.
2: R \W 4: R \W	tier. tierDescription. label		[ITPCM-029^{TIER_DESCR}] [String-length rules]
2: O \W 4: O \W	tier. tierDescription. description		[ITPCM-029^{TIER_DESCR}] [String-length rules]
2: M \W 4: M \W	tier. boundaryDescription.		
2: M \W 4: M \W	tier. boundaryDescription. boundaryId	<by1#(1..1)>	SHALL be set as SUB IDENTIFIER of tierId.
2: M \W 4: M \W	tier. boundaryDescription. boundaryType	"powerBoundary"	[ITPCM-012^{IT_POWER}] The boundary SHALL be an electrical power. E.g. if the surplus power is consumed, more expensive power has to be used. Positive powerBoundary values describe which power can be consumed with certain incentives, while negative powerBoundary values describe which power can be produced with certain incentives.

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2: M \W 4: M \W	tier. boundaryDescription. boundaryUnit	"W"		If set, the unit SHALL be applied to the lowerBoundaryValue and upperBoundaryValue.
2: M \W 4: M \W	tier. incentiveDescription.			[ITPCM-019^{UNIQUE_I_TYPES}]
2: M \W 4: M \W	tier. incentiveDescription. incentiveId	<ie1#(1..3)>		SHALL be set as SUB IDENTIFIER of tierId.
2: M \W 4: M \W	tier. incentiveDescription. incentiveType Note: At least one of the values SHALL be used.	2: M \W 4: M \W	"absoluteCost"	[ITPCM-016^{ABSOLUTE_PRICE}] Each related value SHALL be an absolute cost. Positive values SHALL be interpreted as charge. Negative values SHALL be interpreted as profits. The element "currency" SHALL be set to the respective currency. The element "unit" SHALL be set to "Wh". The cost refers to "currency" per "unit" (EUR/Wh, e.g.).
		2: O \W 4: O \W	"renewableEnergyPercentage"	[ITPCM-017^{RENEWABLE_PCT}] The "unit" Element of the incentive SHALL be set to "pct" (percentage). The value SHALL be in the range from 0% to 100%.
		2: O \W 4: O \W	"co2Emission"	[ITPCM-018^{CO2}] The "unit" Element of the incentive SHALL be set to "kg/Wh". Only positive values SHALL be used.
2: O \W 4: O \W	tier. incentiveDescription. currency			If set, the currency SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
2: O \W 4: O \W	tier. incentiveDescription. unit			If set, the unit SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
2: M 4: M	IncentiveTable. incentiveTableConstraintsData. incentiveTableConstraints.			
2: M 4: M	tariff.			
2: M 4: M	tariff. tariffId	<tf1#(1..1)>		SHALL be set as PRIMARY IDENTIFIER of an incentiveTable.
2: M 4: M	tariffConstraints.			
2: M 4: M	tariffConstraints. maxTiersPerTariff	M: ≥ 3 R: ≥ 5		[ITPCM-009^{MAX_TIERS}] If set, the incentiveTable SHALL NOT include more tiers than given in maxTiersPerTariff.
2: M 4: M	tariffConstraints. maxBoundariesPerTier	≥1		[ITPCM-043^{MAX_BOUNDARIES}] If set, the related "incentiveTableData. incentiveTable. incentiveSlot. tier" instance SHALL NOT include more

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			"boundary" instances than given in maxBoundariesPerTier.
2: M 4: M	tariffConstraints. maxIncentivesPerTier	≥1	[ITPCM-014[MAX_INCENTIVES]] If set, the tier within the incentiveTable SHALL NOT include more incentives than given in maxIncentivesPerTier.
2: M 4: M	incentiveSlotConstraints.		
2: M 4: M	incentiveSlotConstraints. slotCountMax		[ITPCM-044[MAX_SLOTS]] If set, the incentiveTable SHALL NOT include more slots than given in slotCountMax.
2: M \W 4: M \W	IncentiveTable. incentiveTableData. incentiveTable.		
2: M \W 4: M \W	tariff.		
2: M \W 4: M \W	tariff. tariffId	<tf1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M \W 4: M \W	incentiveSlot.		[ITPCM-011[IT_SLOT]]
2: M \W 4: M \W	incentiveSlot. timeInterval.		[ITPCM-021[SLOT_PERIOD]] SHALL be set. The timeInterval of different incentiveSlots within an incentiveTable SHALL NOT overlap in time.
2: M \W 4: M \W	incentiveSlot. timeInterval. timeSlotId	<ti1#(1..)>	SUB IDENTIFIER of tariffId.
2: M \W 4: M \W	incentiveSlot. timeInterval. startTime.		
2: M \W 4: M \W	incentiveSlot. timeInterval. startTime. dateTime		
2: M \W, O \W 4: M \W, O \W	incentiveSlot. timeInterval. endTime.		[ITPCM-033[IT_GAPLESS]] SHALL be set for the last slot. MAY be set for all other slots.
2: M \W 4: M \W	incentiveSlot. timeInterval. endTime. dateTime		
2: M \W 4: M \W	incentiveSlot. tier.		[ITPCM-013[IT_TIER]]
2: M \W 4: M \W	incentiveSlot. tier. tier.		
2: M \W 4: M \W	incentiveSlot. tier. tier. tierId	<tr1#(1..5)>	SUB IDENTIFIER of tariffId to identify a tier of a tariff. SHALL be set.
2: M \W 4: M \W	incentiveSlot. tier. boundary.		The boundary list defines the boundaries of a tier. The boundary

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			range of different tiers within an incentiveSlot defined by lowerBoundaryValue and upperBoundaryValue SHALL NOT overlap for boundaries with the same boundaryType. The boundary range of a tier includes all values that are equal or greater than lowerBoundaryValue and lower than upperBoundaryValue.
2: M \W 4: M \W	incentiveSlot. tier. boundary. boundaryId	<by1#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set.
2: M \W 4: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue.		[ITPCM-035^{IT_CONSUMPTION}], [ITPCM-040^{IT_P_GAPLESS}] Within a boundary lowerBoundaryValue SHALL be smaller than upperBoundaryValue. **Lower and upper boundaries of adjacent tiers**: If lowerBoundaryValue is omitted and upperBoundaryValue is set, the lowerBoundaryValue SHALL be interpreted as if it is equal (without a gap) to the upperBoundaryValue of the next lower tier within the incentiveSlot. **Minus infinity lowerBoundary rule**: If lowerBoundaryValue is omitted and there exists no lower tier with a lowerBoundaryValue or upperBoundaryValue within the incentiveSlot, the omitted lowerBoundaryValue SHALL be interpreted as minus infinity. See [Scaled number rules].
2: M \W 4: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. number		
2: M \W 4: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. scale		
2: M \W 4: M \W	incentiveSlot. tier. incentive.		[ITPCM-022^{INCENTIVE}]
2: M \W 4: M \W	incentiveSlot. tier. incentive. incentiveId	<ie1#(1..3)>	SHALL be used as SUB IDENTIFIER of tierId.
2: M \W 4: M \W	incentiveSlot. tier. incentive. value.		[ITPCM-023^{INCENTIVE_VALUE}] See [Scaled number rules].

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2: M \W 4: M \W	incentiveSlot. tier. incentive. value. number		
2: M \W 4: M \W	incentiveSlot. tier. incentive. value. scale		

1436 *Table 17: Content of Specialization "IncentiveTable_SimpleCommittedIncentiveTable" at Actor CEM*

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1438 *3.2.2.3.1.2 Specialization "IncentiveTable_SimplePreliminaryIncentiveTable"*

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	IncentiveTable. incentiveTableDescriptionData. incentiveTableDescription.		
3: M	tariffDescription.		
3: M	tariffDescription. tariffId	<tf2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER of an incentiveTable.
3: M	tariffDescription. tariffWriteable	"true"	[ITPCM-036^{PIT_MODIFIABLE}]
		"false"	[ITPCM-037^{PIT_UNMODIFIABLE}]
3: F	tariffDescription. updateRequired		
3: M	tariffDescription. scopeType	"simplePreliminaryIncentiveTable"	[ITPCM-046^{IT_PRELIMINARY}]
3: M	tariffDescription. slotIdSupport	"true"	
3: M \W	tier.		[ITPCM-013^{IT_TIER}]
3: M \W	tier. tierDescription.		
3: M \W	tier. tierDescription. tierId	<tr2#(1..5)>	SHALL be set as SUB IDENTIFIER of tariffId.
3: M \W	tier. tierDescription. tierType	"dynamicCost"	[ITPCM-010^{IT_GEN}] The tier has a cost incentive that MAY vary over time.
3: R \W	tier. tierDescription. label		[ITPCM-029^{TIER_DESCR}] [String-length rules]
3: O \W	tier. tierDescription. description		[ITPCM-029^{TIER_DESCR}] [String-length rules]
3: M \W	tier. boundaryDescription.		
3: M \W	tier. boundaryDescription. boundaryId	<by2#(1..1)>	SHALL be set as SUB IDENTIFIER of tierId.
3: M \W	tier. boundaryDescription. boundaryType	"powerBoundary"	[ITPCM-012^{IT_POWER}] The boundary SHALL be an electrical power. E.g. if the surplus power is consumed, more expensive power has to be used. Positive powerBoundary values describe

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				which power can be consumed with certain incentives, while negative powerBoundary values describe which power can be produced with certain incentives.
3: M \W	tier. boundaryDescription. boundaryUnit	"W"		If set, the unit SHALL be applied to the lowerBoundaryValue and upperBoundaryValue.
3: M \W	tier. incentiveDescription.			[ITPCM-019^{UNIQUE_I_TYPES}]
3: M \W	tier. incentiveDescription. incentiveId	<ie2#(1..3)>		SHALL be set as SUB IDENTIFIER of tierId.
3: M \W	tier. incentiveDescription. incentiveType Note: At least one of the values SHALL be used.	3: M \W	"absoluteCost"	[ITPCM-016^{ABSOLUTE_PRICE}] Each related value SHALL be an absolute cost. Positive values SHALL be interpreted as charge. Negative values SHALL be interpreted as profits. The element "currency" SHALL be set to the respective currency. The element "unit" SHALL be set to "Wh". The cost refers to "currency" per "unit" (EUR/Wh, e.g.).
		3: O \W	"renewableEnergyPercentage"	[ITPCM-017^{RENEWABLE_PCT}] The "unit" Element of the incentive SHALL be set to "pct" (percentage). The value SHALL be in the range from 0% to 100%.
		3: O \W	"co2Emission"	[ITPCM-018^{CO2}] The "unit" Element of the incentive SHALL be set to "kg/Wh". Only positive values SHALL be used.
3: O \W	tier. incentiveDescription. currency			If set, the currency SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
3: O \W	tier. incentiveDescription. unit			If set, the unit SHALL be applied to the value of the incentive. See Element "incentiveType" for rules.
3: M	IncentiveTable. incentiveTableConstraintsData. incentiveTableConstraints.			
3: M	tariff.			
3: M	tariff. tariffId	<tf2#(1..1)>		SHALL be set as PRIMARY IDENTIFIER of an incentiveTable.
3: M	tariffConstraints.			
3: M	tariffConstraints. maxTiersPerTariff	M: ≥ 3 R: ≥ 5		[ITPCM-009^{MAX_TIERS}] If set, the incentiveTable SHALL NOT include more tiers than given in maxTiersPerTariff.
3: M	tariffConstraints. maxBoundariesPerTier	≥ 1		[ITPCM-043^{MAX_BOUNDARIES}] If set, the related "incentiveTableData. incentiveTable. incentiveSlot. tier"

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			instance SHALL NOT include more "boundary" instances than given in maxBoundariesPerTier.
3: M	tariffConstraints. maxIncentivesPerTier	≥ 1	[ITPCM-014^{MAX_INCENTIVES}] If set, the tier within the incentiveTable SHALL NOT include more incentives than given in maxIncentivesPerTier.
3: M	incentiveSlotConstrai nts.		
3: M	incentiveSlotConstrai nts. slotCountMax		[ITPCM-044^{MAX_SLOTS}] If set, the incentiveTable SHALL NOT include more slots than given in slotCountMax.
3: M \W	IncentiveTable. incentiveTableData. incentiveTable.		
3: M \W	tariff.		
3: M \W	tariff. tariffId	<tf2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M \W	incentiveSlot.		[ITPCM-011^{IT_SLOT}]
3: M \W	incentiveSlot. timeInterval.		[ITPCM-021^{SLOT_PERIOD}] SHALL be set. The timeInterval of different incentiveSlots within an incentiveTable SHALL NOT overlap in time.
3: M \W	incentiveSlot. timeInterval. timeSlotId	<ti2#(1..)>	SUB IDENTIFIER of tariffId.
3: M \W	incentiveSlot. timeInterval. startTime.		
3: M \W	incentiveSlot. timeInterval. startTime. dateTime		
3: M \W, O \W	incentiveSlot. timeInterval. endTime.		[ITPCM-033^{IT_GAPLESS}] SHALL be set for the last slot. MAY be set for all other slots.
3: M \W	incentiveSlot. timeInterval. endTime. dateTime		
3: M \W	incentiveSlot. tier.		[ITPCM-013^{IT_TIER}]
3: M \W	incentiveSlot. tier. tier.		
3: M \W	incentiveSlot. tier. tier. tierId	<tr2#(1..5)>	SUB IDENTIFIER of tariffId to identify a tier of a tariff. SHALL be set.
3: M \W	incentiveSlot. tier. boundary.		The boundary list defines the boundaries of a tier. The boundary range of different tiers within an incentiveSlot defined by lowerBoundaryValue and upperBoundaryValue SHALL NOT overlap for boundaries with the same boundaryType. The boundary

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			range of a tier includes all values that are equal or greater than lowerBoundaryValue and lower than upperBoundaryValue.
3: M \W	incentiveSlot. tier. boundary. boundaryId	<by2#(1..1)>	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set.
3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue.		[ITPCM-035^{IT_CONSUMPTION}], [ITPCM-040^{IT_P_GAPLESS}] Within a boundary lowerBoundaryValue SHALL be smaller than upperBoundaryValue. **Lower and upper boundaries of adjacent tiers**: If lowerBoundaryValue is omitted and upperBoundaryValue is set, the lowerBoundaryValue SHALL be interpreted as if it is equal (without a gap) to the upperBoundaryValue of the next lower tier within the incentiveSlot. **Minus infinity lowerBoundary rule**: If lowerBoundaryValue is omitted and there exists no lower tier with a lowerBoundaryValue or upperBoundaryValue within the incentiveSlot, the omitted lowerBoundaryValue SHALL be interpreted as minus infinity. See [Scaled number rules].
3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. number		
3: M \W	incentiveSlot. tier. boundary. lowerBoundaryValue. scale		
3: M \W	incentiveSlot. tier. incentive.		[ITPCM-022^{INCENTIVE}]
3: M \W	incentiveSlot. tier. incentive. incentiveId	<ie2#(1..3)>	SHALL be used as SUB IDENTIFIER of tierId.
3: M \W	incentiveSlot. tier. incentive. value.		[ITPCM-023^{INCENTIVE_VALUE}] See [Scaled number rules].
3: M \W	incentiveSlot. tier. incentive. value. number		
3: M \W	incentiveSlot. tier. incentive. value. scale		

1439 *Table 18: Content of Specialization "IncentiveTable_SimplePreliminaryIncentiveTable" at Actor CEM*

1440

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1441 *3.2.2.3.2 Topic "TimeSeries"*

1442 *3.2.2.3.2.1 Specialization "TimeSeries_CommittedPowerPlan"*

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M 5: M	TimeSeries. timeSeriesDescriptionListData. timeSeriesDescriptionData.		
2: M 5: M	timeSeriesId	<ts1#(1..1)>	PRIMARY IDENTIFIER of timeSeries.
2: M 5: M	timeSeriesType	"plan"	"plan" SHALL be used to model how a certain value will change over time.
2: M 5: M	timeSeriesWriteable	"false"	If set to false or omitted, this timeSeriesData SHALL NOT be writeable.
2: M 5: M	measurementId	<m1#(1..1)>	[ITPCM-039^{P_TOTAL_ACTIVE}] FOREIGN IDENTIFIER. If measurementId is used, it SHALL link to a measurement value of the total active power ([ITPCM-039^{P_TOTAL_ACTIVE}]).
2: M 5: M	unit	"W"	The unit SHALL be applied to the value in timeSeriesData
2: M 5: M	scopeType	"committedPowerPlan"	[ITPCM-047^{PLAN_COMMITTED}]
2: M 5: M	TimeSeries. timeSeriesListData. timeSeriesData.		
2: M 5: M	timeSeriesId	<ts1#(1..1)>	PRIMARY IDENTIFIER of timeSeries
2: F 5: F	timePeriod.		
2: M 5: M	timeSeriesSlot.		[ITPCM-025^{PLAN_SLOT}]
2: M 5: M	timeSeriesSlot. timeSeriesSlotId		If timeSeriesSlotId is omitted it SHALL be interpreted as zero.
2: M 5: M	timeSeriesSlot. timePeriod.		
2: M 5: M	timeSeriesSlot. timePeriod. startTime	<xs:dateTime>	
2: M 5: M	timeSeriesSlot. duration		[ITPCM-034^{PLAN_GAPLESS}] SHALL be set and express the duration of the slot. SHALL only contain values greater than zero seconds. The duration SHALL last to the start time of the next slot, if any.
2: M	timeSeriesSlot. value.		[ITPCM-027^{PLAN_PEXP}]

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5: M			A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If minValue is set the content of value SHALL be greater or equal. If maxValue is set the content of value SHALL be smaller or equal. See [Scaled number rules].
2: M 5: M	timeSeriesSlot. value. number	≥0	
2: M 5: M	timeSeriesSlot. value. scale		
2: M 5: M	timeSeriesSlot. minValue.		[ITPCM-026^{PLAN_PMIN}] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If value is set the content of minValue SHALL be smaller or equal. If value is not set but maxValue is set the content of minValue SHALL be smaller or equal. See [Scaled number rules].
2: M 5: M	timeSeriesSlot. minValue. number	≥0	
2: M 5: M	timeSeriesSlot. minValue. scale		
2: M 5: M	timeSeriesSlot. maxValue.		[ITPCM-028^{PLAN_PMAX}] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If value is set the content of maxValue SHALL be greater or equal. If value is not set but minValue is set the content of maxValue SHALL be greater or equal. See [Scaled number rules].
2: M 5: M	timeSeriesSlot. maxValue. number	≥0	
2: M 5: M	timeSeriesSlot. maxValue. scale		

1443 *Table 19: Content of Specialization "TimeSeries_CommittedPowerPlan" at Actor CEM*

1444

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1445 *3.2.2.3.2.2 Specialization "TimeSeries_PreliminaryPowerPlan"*

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	TimeSeries. timeSeriesDescriptionListData. timeSeriesDescriptionData.		
3: M	timeSeriesId	<ts2#(1..1)>	PRIMARY IDENTIFIER of timeSeries.
3: M	timeSeriesType	"plan"	"plan" SHALL be used to model how a certain value will change over time.
3: M	timeSeriesWriteable	"false"	If set to false or omitted, this timeSeriesData SHALL NOT be writeable.
3: M	measurementId	<m1#(1..1)>	[ITPCM-039^{P_TOTAL_ACTIVE}] FOREIGN IDENTIFIER. If measurementId is used, it SHALL link to a measurement value of the total active power ([ITPCM-039^{P_TOTAL_ACTIVE}]).
3: M	unit	"W"	The unit SHALL be applied to the value in timeSeriesData
3: M	scopeType	"preliminaryPowerPlan"	[ITPCM-048^{PLAN_PRELIMINARY}]
3: M	TimeSeries. timeSeriesListData. timeSeriesData.		
3: M	timeSeriesId	<ts2#(1..1)>	PRIMARY IDENTIFIER of timeSeries
3: F	timePeriod.		
3: M	timeSeriesSlot.		[ITPCM-025^{PLAN_SLOT}]
3: M	timeSeriesSlot. timeSeriesSlotId		If timeSeriesSlotId is omitted it SHALL be interpreted as zero.
3: M	timeSeriesSlot. timePeriod.		
3: M	timeSeriesSlot. timePeriod. startTime	<xs:dateTime>	
3: M	timeSeriesSlot. duration		[ITPCM-034^{PLAN_GAPLESS}] SHALL be set and express the duration of the slot. SHALL only contain values greater than zero seconds. The duration SHALL last to the start time of the next slot, if any.
3: M	timeSeriesSlot. value.		[ITPCM-027^{PLAN_PEXP}] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If minValue is set the content of value SHALL be greater or equal. If maxValue is set the content of value SHALL be smaller or equal.

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			See [Scaled number rules].
3: M	timeSeriesSlot. value. number	≥0	
3: M	timeSeriesSlot. value. scale		
3: M	timeSeriesSlot. minValue.		[ITPCM-026^{PLAN_PMIN}] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If value is set the content of minValue SHALL be smaller or equal. If value is not set but maxValue is set the content of minValue SHALL be smaller or equal. See [Scaled number rules].
3: M	timeSeriesSlot. minValue. number	≥0	
3: M	timeSeriesSlot. minValue. scale		
3: M	timeSeriesSlot. maxValue.		[ITPCM-028^{PLAN_PMAX}] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If value is set the content of maxValue SHALL be greater or equal. If value is not set but minValue is set the content of maxValue SHALL be greater or equal. See [Scaled number rules].
3: M	timeSeriesSlot. maxValue. number	≥0	
3: M	timeSeriesSlot. maxValue. scale		

Table 20: Content of Specialization "TimeSeries_PreliminaryPowerPlan" at Actor CEM

3.2.2.3.3 Topic "DeviceConfiguration"

3.2.2.3.3.1 Specialization "DeviceConfiguration_IncentivesSimulationCyclesMax"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
1: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
1: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	keyName	"incentivesSimulationCyclesMax"	

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1: M	valueType	"integer"	
1: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
1: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	value.		[ITPCM-031^{MAX_SIMUL_CYCLES}] Exactly one of the child Elements SHALL be set. This SHALL match with the content of *valueType* Element within the key value description part (see above).
1: M	value. integer		SHALL be used.

Table 21: Content of Specialization "DeviceConfiguration_IncentivesSimulationCyclesMax" at Actor CEM

3.2.2.3.3.2 Specialization "DeviceConfiguration_IncentivesSimulationConcurrent"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
1: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
1: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	keyName	"incentivesSimulationConcurrent"	
1: M	valueType	"boolean"	
1: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
1: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	value.		[ITPCM-030^{CONCURRENT_SIMUL}] Exactly one of the child Elements SHALL be set. This SHALL match with the content of *valueType* Element within the key value description part (see above).
1: M	value. boolean		SHALL be used.

Table 22: Content of Specialization "DeviceConfiguration_IncentivesSimulationConcurrent" at Actor CEM

3.2.2.3.3.3 Specialization "DeviceConfiguration_IncentivesTimeoutIncentiveRequest"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
1: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
1: M	keyId	<k3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.

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1: M	keyName	"incentivesTimeoutIncentiveRequest"	
1: M	valueType	"duration"	
1: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
1: M	keyId	<k3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	value.		[ITPCM-005^{TIMEOUT_IT_REQUEST}] Exactly one of the child Elements SHALL be set. This SHALL match with the content of *valueType* Element within the key value description part (see above).
1: M	value. duration	*1	SHALL be used.

Table 23: Content of Specialization "DeviceConfiguration_IncentivesTimeoutIncentiveRequest" at Actor CEM

*1: Values within the duration Element can be represented in different ways. E.g.: "2 hours" can be represented as "PT2H" or "PT120M".

3.2.2.3.3.4 Specialization "DeviceConfiguration_IncentivesWaitIncentiveWriteable"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
1: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
1: M	keyId	<k4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	keyName	"incentivesWaitIncentiveWriteable"	
1: M	valueType	"duration"	
1: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
1: M	keyId	<k4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	value.		[ITPCM-008^{WAIT_IT_MODIFIABLE}] Exactly one of the child Elements SHALL be set. This SHALL match with the content of *valueType* Element within the key value description part (see above).
1: M	value. duration	*1	SHALL be used.

Table 24: Content of Specialization "DeviceConfiguration_ IncentivesWaitIncentiveWriteable" at Actor CEM

*1: Values within the duration Element can be represented in different ways. E.g.: "2 hours" can be represented as "PT2H" or "PT120M".

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3.3 Pre-Scenario communication

3.3.1 General information

The Pre-Scenario communication is needed if a client does not know the corresponding addresses on the server or if the required subscriptions or bindings are not active. In this case certain general communication mechanisms SHALL be used within SPINE:

a) Detailed discovery: allows to discover resource addresses.
b) Binding: allows to bind to resource address, which is frequently necessary to obtain write privileges.
c) Subscription: allows to subscribe to resource addresses, which is necessary to receive unsolicited notifications if a resource changes during runtime.

It is possible to combine those steps for multiple Scenarios or also multiple Use Cases:

- E.g. if multiple Scenarios in multiple Use Cases use the same Feature, only one subscription needs to occur.
- E.g. a complete detailed discovery or a subscription to the NodeManagement Feature needs to occur only once for all Use Cases.

Depending on which Entity, Feature and Functions are used within a Scenario the payload of the corresponding messages may slightly differ, but the basic principles and messages used stay the same.

The subsequent messages SHALL be exchanged for those parts that have not already been performed since the current connection is established or if those parts are outdated for another reason (e.g. if the detailed discovery is needed, but the bindings and subscriptions are still active from a previous connection only the detailed discovery messages need to be exchanged). If all Pre-Scenario communication parts are up to date, this section MAY be skipped, and the implementation can proceed as described in the corresponding "Scenario communication" sections.

After the connection is re-established (e.g. due to a power loss or a firmware update) the Pre-Scenario communication SHALL be performed as well. There may be circumstances where messages from the Pre-Scenario communication may be exchanged again.

Often the necessary messages of different Scenarios can be combined, so that only one single message is needed instead of multiple messages for the different Scenarios. This also is the case for the Pre-Scenario communication. In most cases only one "read" operation on the detailed discovery is necessary, as well as only one subscription request or binding request is needed for each Feature. Often multiple Scenarios within a Use Case access the same Feature, so only one subscription or binding is necessary.

3.3.2 Detailed discovery

For the functionality where a client already has current detailed discovery information (i.e. independent of this Use Case or any Scenario of it) the remainder of this section SHOULD be skipped.

Otherwise, the following procedure SHALL be performed in the given order:

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1. If a client is not subscribed to the primary NodeManagement instance, the client SHALL acquire a subscription according to the figure provided within this sub-section.
2. A client SHALL read the detailed discovery information according to the figure provided within this sub-section. It SHALL keep the received information as far as it concerns mandatory and supported optional Entity Types, Feature Types and Functions of this Use Case that are needed by the client. This means that a client may choose how to store the necessary information. E.g. a client Actor can store the information how to address the necessary Features of the implemented Scenarios but may discard the Entity information.
3. If and as long as a client has a subscription to the detailed discovery information of an Actor and receives proper notifications, it SHALL consider (i.e. integrate into the kept detailed discovery information) the received information as far as it concerns mandatory and supported optional Entity Types, Feature Types and Functions of this Use Case.

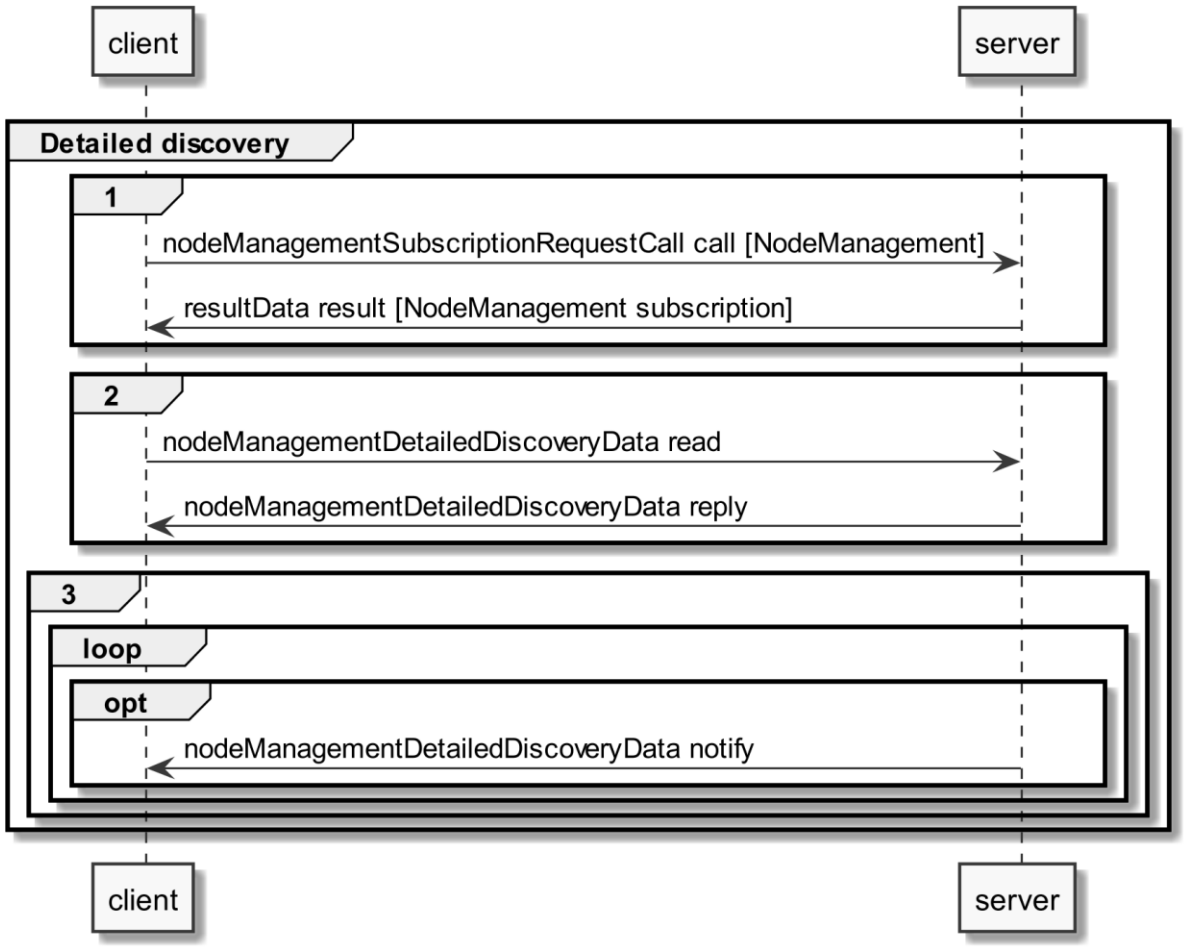


Figure 18: Pre-Scenario communication - Detailed discovery sequence diagram

Example XMLs are provided within Annex A.

If the "nodeManagementDetailedDiscoveryData read" fails, the client SHOULD retry to read the detailed discovery information until the "nodeManagementDetailedDiscoveryData reply" message was received successfully.

If all functionality is present at all times: The "nodeManagementDetailedDiscoveryData reply" message contains at least the mandatory Entities and Features given in the "Actor [...] overview"

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diagrams as well as the used Functions and their "possible operations" described in section 3.2 and its sub-sections.

If functionality is added or removed dynamically: The "nodeManagementDetailedDiscoveryData reply" message does not need to contain all mandatory Entities and Features given in the "Actor [...] overview" diagrams as well as all needed Functions and their "possible operations" described in section 3.2 and its sub-sections. However, as soon as the functionality is available it will be announced via a "nodeManagementDetailedDiscoveryData notify" message.

For the nodeManagementDetailedDiscoveryData read Function it is recommended to use a partial read with separated Selectors that may use one of the following Elements:

- entityType
- featureType

Note: Even with the usage of Selectors Features and Entities that are not relevant for this Use Case may be discovered. However, only Features and Entities that fulfil the hierarchical order as described within the Actors' sections shall be considered for this Use Case.

A "partial" notify SHALL be supported without using Selectors and Elements. Partial "delete" notify SHOULD also be supported with separated Selectors that may use one of the following Elements:

- entityAddress
- featureAddress

3.3.3 Binding

If binding is required by a Scenario that uses Features with writeable or changeable data, the server SHALL support binding for the respective Features. Before a write on a Function of a Feature occurs, the client SHALL create a binding to the corresponding Feature. For this the nodeManagementBindingRequestCall Function is used as shown in the following sequence diagram:

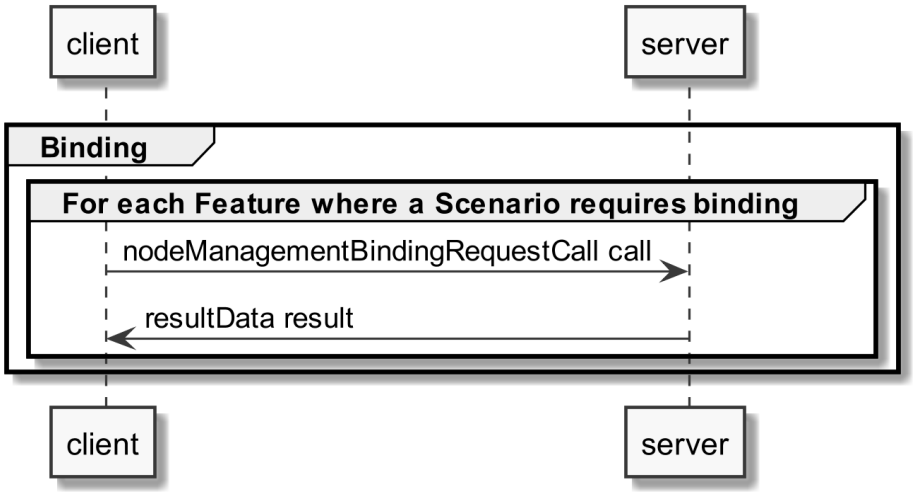


Figure 19: Pre-Scenario communication - Binding sequence diagram

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If functionality is added or removed dynamically, binding may not be possible at all times on the required Functions. A client SHALL retry to create a binding again when receiving according updated detailed discovery information.

3.3.4 Subscription

A server SHALL support subscription for all Features that contain readable data that may change during runtime. The client SHALL create a subscription for all Features that the client wants to read. For this the nodeManagementSubscriptionRequestCall Function is used as shown in the following sequence diagram:

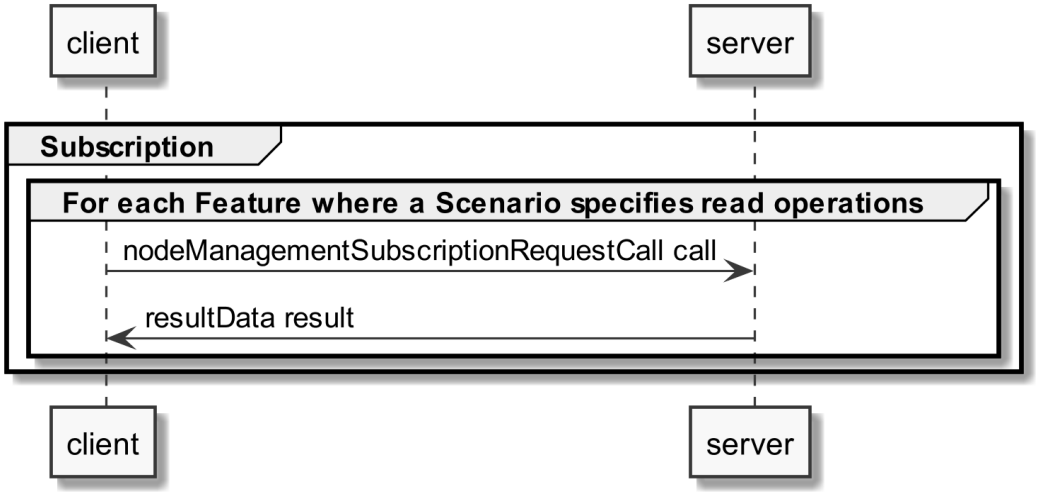


Figure 20: Pre-Scenario communication - Subscription sequence diagram

If the subscription request fails (e.g. because it is not supported by the server or the maximum number of possible subscriptions is reached), the client SHOULD read the data periodically (so-called "polling").

If functionality is added or removed dynamically, subscription may not be possible at all times on the required Functions. A client SHALL retry its subscription procedure again when receiving according updated detailed discovery information.

3.3.5 Dynamic behaviour

In case Entities or Features are removed, a nodeManagementDetailedDiscoveryData "notify" is transmitted that informs about the deleted Entities and Features. All existing binding or subscription entries on the deleted Features SHALL be deleted by each device.

In case Entities or Features are added the Pre-Scenario communication starts with transmitting a nodeManagementDetailedDiscoveryData "notify" that contains the added Entities and Features.

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3.4 Scenarios

3.4.1 Scenario 1 - Announcement of negotiation options

3.4.1.1 Pre-Scenario communication

1. **Detailed discovery**: Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding**: Binding SHOULD NOT be used for this Scenario.
3. **Subscription**: Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.1.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

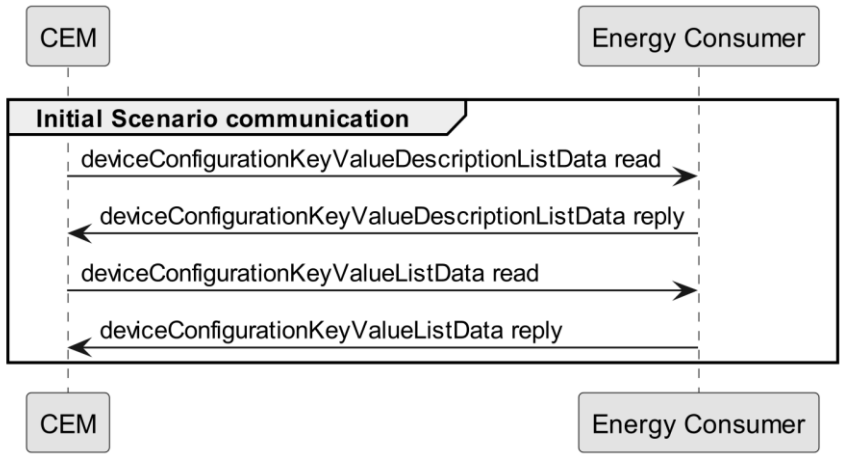


Figure 21: Scenario 1 - Initial Scenario communication sequence diagram

Partial data information regarding [ITPCM-031^{MAX_SIMUL_CYCLES}]:

The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- keyName = "incentivesSimulationCyclesMax"

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The deviceConfigurationKeyValueListData read SHOULD be "partial" read operation with the following Selector:

- keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

Partial data information regarding [ITPCM-030^{CONCURRENT_SIMUL}]:

The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- keyName = "incentivesSimulationConcurrent"

The deviceConfigurationKeyValueListData read SHOULD be "partial" read operation with the following Selector:

- keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

Partial data information regarding [ITPCM-005^{TIMEOUT_IT_REQUEST}]:

The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- keyName = "incentivesTimeoutIncentiveRequest"

The deviceConfigurationKeyValueListData read SHOULD be "partial" read operation with the following Selector:

- keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

Partial data information regarding [ITPCM-008^{WAIT_IT_MODIFIABLE}]:

The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- keyName = "incentivesWaitIncentiveWriteable"

The deviceConfigurationKeyValueListData read SHOULD be "partial" read operation with the following Selector:

- keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

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The following table shows where the required content of the reply messages from the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
deviceConfigurationKeyValueDescriptionListData reply	Table 14	1
deviceConfigurationKeyValueListData reply	Table 15	1

Table 25: Initial Scenario communication content references for reply messages of Scenario 1

XML examples can be found in Annex A.

Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be provided completely, but later during Runtime Scenario communication.

3.4.1.3 Runtime Scenario communication

Based on the Initial Scenario communication, the Runtime Scenario communication provides updates during runtime.

If one of the referenced server Functions' data change, the server SHALL submit the change as shown in the following figure:

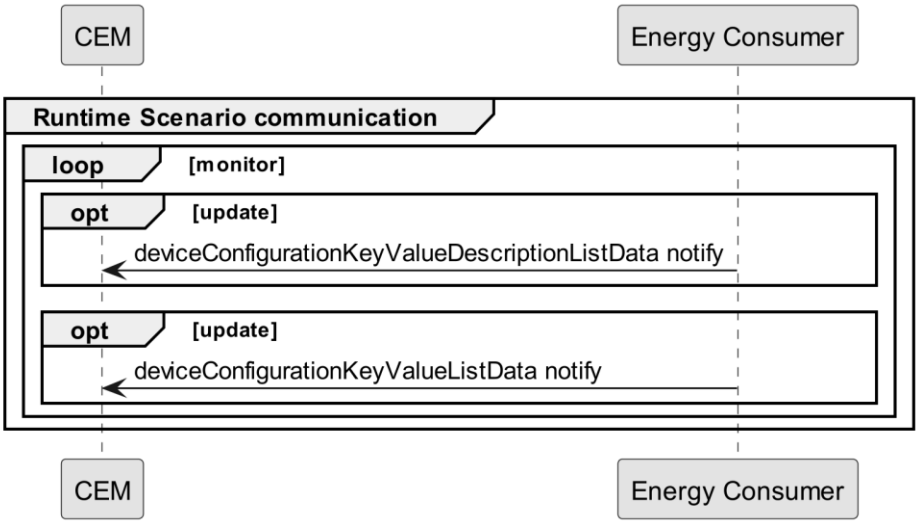


Figure 22: Scenario 1 - Runtime Scenario notifications

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For deviceConfigurationKeyValueDescriptionListData notify and deviceConfigurationKeyValueListData notify "partial" delete notifications SHOULD be supported with the Selector:

- keyId

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Note: To interpret partial notification messages correctly the information obtained during the Initial Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the notify messages of the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
deviceConfigurationKeyValueDescriptionListData notify	Table 14	1
deviceConfigurationKeyValueListData notify	Table 15	1

Table 26: Runtime Scenario communication content references for notify messages of Scenario 1

XML examples can be found in Annex A.

3.4.1.4 Additional information

None.

3.4.2 Scenario 2 - Negotiation of Committed Data

3.4.2.1 Pre-Scenario communication

1. **Detailed discovery**: Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding**: Actors that write parts of a Feature within this Scenario, need to create a binding, as described in section 3.3.3. Only one binding partner is allowed to write the data specified in this Scenario.
3. **Subscription**: Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

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3.4.2.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

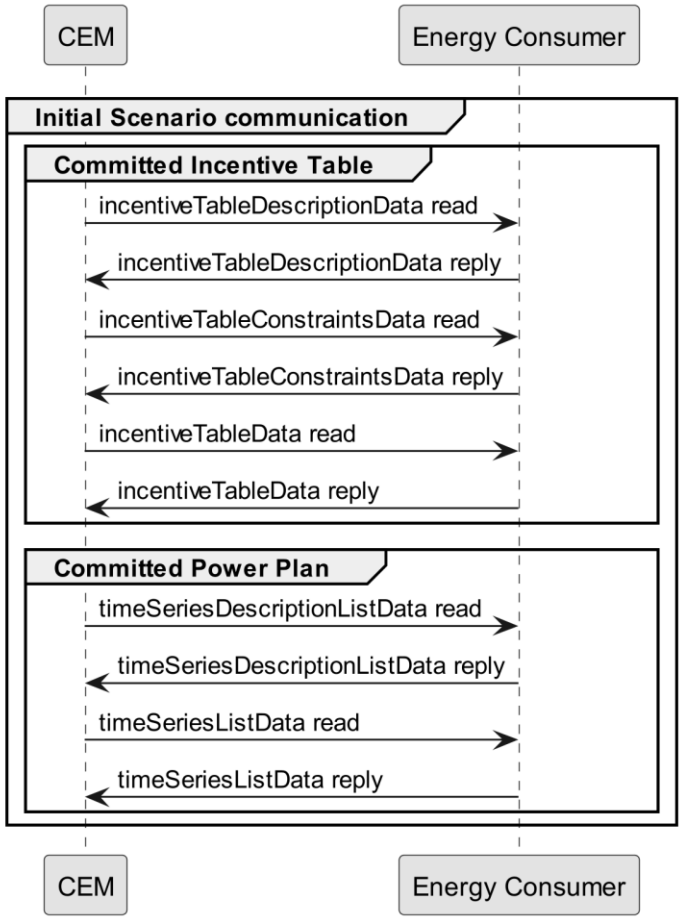


Figure 23: Scenario 2 - Initial Scenario communication sequence diagram

Partial data information regarding [ITPCM-045^{IT_COMMITTED}]:

The incentiveTableDescriptionData read SHOULD be a "partial" read operation with the following Selector:

- tariffDescription. scopeType = "simpleCommittedIncentiveTable"

The incentiveTableConstraintsData and incentiveTableData read SHOULD be "partial" read operations with the following Selector:

- tariff. tariffId (with tariffId derived from the incentiveTableDescriptionData reply)

Partial data information regarding [ITPCM-047^{PLAN_COMMITTED}]:

The timeSeriesDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

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- scopeType = "committedPowerPlan"

The timeSeriesListData read SHOULD be a "partial" read operation with the following Selector:

- timeSeriesId (derived from the timeSeriesDescriptionListData reply)

Note: If partial read is not supported a full read SHALL occur.

The following table shows where the required content of the reply messages from the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
incentiveTableDescriptionData reply	Table 9	2
incentiveTableConstraintsData reply	Table 10	2
incentiveTableData reply	Table 11	2
timeSeriesDescriptionListData reply	Table 12	2
timeSeriesListData reply	Table 13	2

Table 27: Initial Scenario communication content references for reply messages of Scenario 2

XML examples can be found in Annex A.

Note: The Scenario comprises several steps that need to be processed in a particular order (see section 2.4.2). In practice it may happen that Actors are switched on at different times, e.g. from a reboot. This means the Initial Scenario communication may occur before the first step, after the last step, or in between. Dependent on the time of execution, the Initial Scenario communication provides respective data. Subsequently we assume that the Initial Scenario communication has been executed before the first step. Data of the first and subsequent steps will be explained in the Runtime Scenario communication.

3.4.2.3 Runtime Scenario communication

Based on the Initial Scenario communication, the Runtime Scenario communication provides updates during runtime. These can be either notifications (i.e. data changes reported by a server) or write operations (i.e. data change request submitted to a server).

Order of communication:

This section just lists the required messages in general, i.e. without consideration of any order or dependency. Please consider section 2.4.2 for rules on the occurrence and content of the messages.

Notifications:

If one of the referenced server Functions' data change, the server SHALL submit the change as shown in the following figure:

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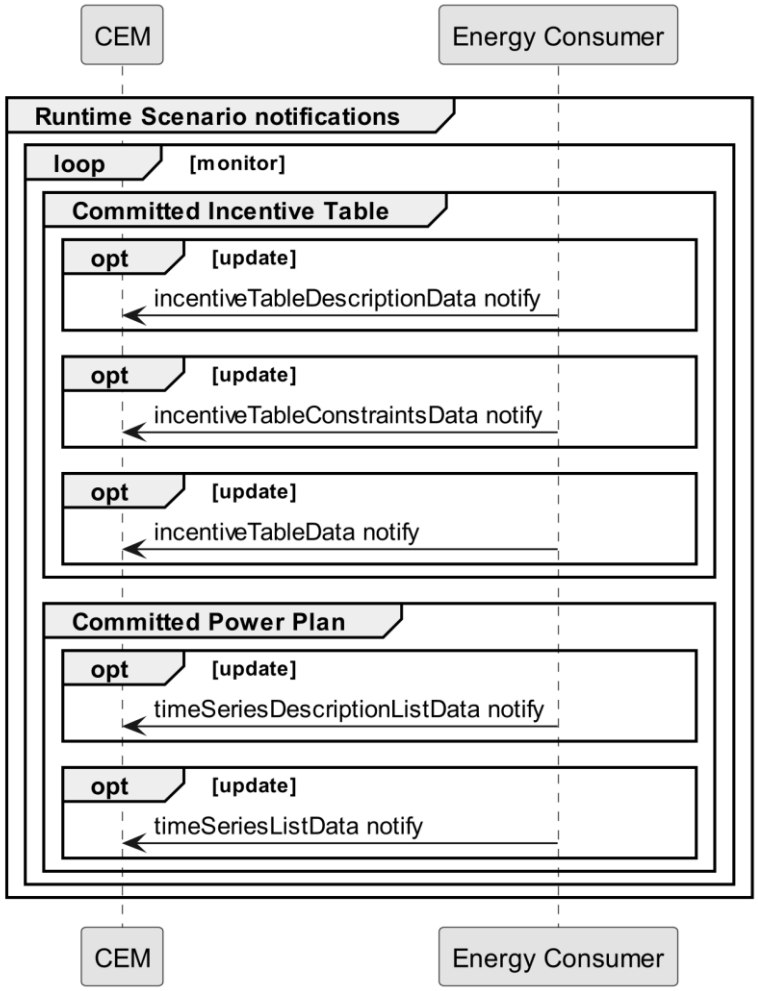


Figure 24: Scenario 2 - Runtime Scenario notifications

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

Partial delete notifications for incentiveTableDescriptionData SHOULD be supported with the Selector:

- tariffDescription. tariffId

Partial delete notifications for incentiveTableConstraintsData and incentiveTableData SHOULD be supported with the Selector:

- tariff. tariffId

Partial delete notifications for timeSeriesDescriptionListData and timeSeriesListData SHOULD be supported with the Selector:

- timeSeriesId

Note: To interpret partial notification messages correctly the information obtained during the Initial Scenario communication phase is required.

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Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the notify messages of the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
incentiveTableDescriptionData notify	Table 9	2
incentiveTableConstraintsData notify	Table 10	2
incentiveTableData notify	Table 11	2
timeSeriesDescriptionListData notify	Table 12	2
timeSeriesListData notify	Table 13	2

Table 28: Runtime Scenario communication content references for notify messages of Scenario 2

Write operations:

Figure 25 shows possible write operations of this Scenario:

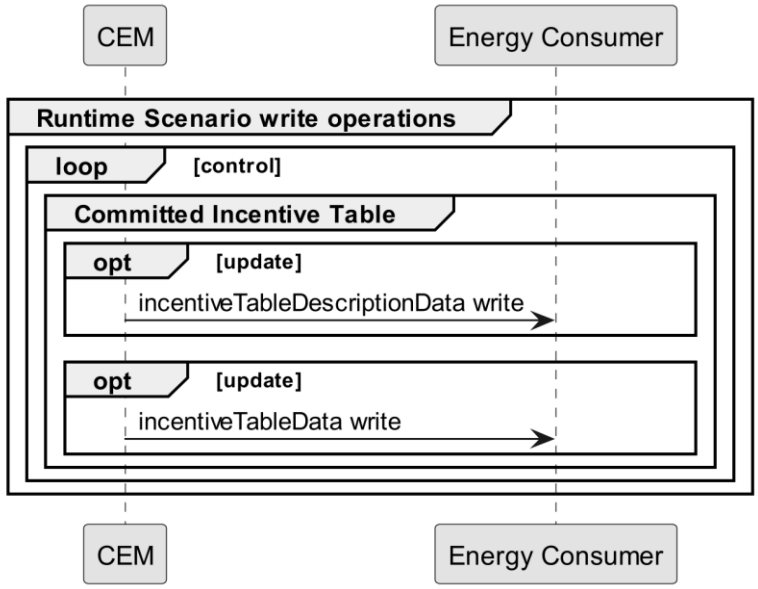


Figure 25: Scenario 2 - Runtime Scenario write operations

For partial delete write operations the same rules apply as for notifications.

For required content of write messages please consider the respective Specializations.

XML examples can be found in Annex A.

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3.4.2.4 Additional information

3.4.2.4.1 Committed Incentive Table

3.4.2.4.1.1 Recommendations on "tariffId"

This Use Case does not prescribe a particular value for the Element "tariffId". But once a Committed Incentive Table instance has been created by Actor Energy Consumer, the respective value of "tariffId" SHOULD be preserved. This means that updates of the Committed Incentive Table SHOULD reuse the value of "tariffId" (in contrast to delete data with the last value of "tariffId" and write the new Committed Incentive Table with a new value in "tariffId").

Note: The value of "tariffId" of the Committed Incentive Table differs from the one of a Preliminary Incentive Table.

3.4.2.4.1.2 Recommendations on "timeSlotId"

This Use Case does not prescribe particular values for the Element "timeSlotId". But once a slot (with a dedicated time period) has been added to a Committed Incentive Table, the respective value of "timeSlotId" SHOULD be preserved. This recommendation applies only as long as an update of a Committed Incentive Table still uses the same time periods for its slots as before.

3.4.2.4.1.3 Recommendations on "partial operations" on slots

This section applies to the Function "incentiveTableData" only. In general, an update of a Committed Incentive Table's slots may comprise of the removal of (past) slots, the addition of new slots, the change of existing slots, or even a replacement of existing slots if a given period needs to be split into a different number of slots with altered durations. These changes may be expressed with "partial operations" in different ways. As a proper modelling and evaluation of partial changes can be prone to errors, some definitions and recommendations will be given. For details and technical background information please consider [ProtocolSpecification].

"New slots":

1. The Actor CEM MAY append one or more future slots to an existing Committed Incentive Table (without time gaps).

"Past slots":

The following rules aim at cleaning up the Incentive Table to focus on current and subsequent activities:

1. The Actor CEM SHOULD NOT write slots whose end time is in the past.
2. The Actor Energy Consumer MAY remove slots whose end time is in the past. This removal SHALL be notified properly.

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Partial write operations by Actor CEM to adjust slots:

If the Actor CEM just wants to add "new slots" as described above, the Actor CEM can send a simple "write" operation with a single "cmdControl" Element containing the Element "partial" together with the new slots. This kind of message is also applicable if an already existing slot just need to get a value change in one of its Elements.

In every other case of slot adjustment (remove "past slots", replace existing slots with new slots, etc.) it is recommended to send the "write" operation using a so-called "delete-add" pattern: For a given value of "tariffId" the "write" operation contains a "filter" Element with "delete" as well as a "filter" Element with "partial". This is shown in the following excerpt of a "datagram":

```
...
<payload>
    <cmd>
        <function>incentiveTableData</function>
        <filter>
            <cmdControl><delete/></cmdControl>
            <incentiveTableDataSelectors>
                <tariff>
                    <tariffId>(*1)</tariffId>
                </tariff>
            </incentiveTableDataSelectors>
        </filter>
        <filter>
            <cmdControl><partial/></cmdControl>
        </filter>
        <incentiveTableData>
            (*2)
        </incentiveTableData>
    </cmd>
</payload>
```

where "(*1)" must be replaced by the proper value of "tariffId" and "(*2)" must contain a proper "incentiveTable" instance with the same value of "tariffId" and slots to be stored by Actor Energy Consumer. This pattern replaces the previous "incentiveTable" instance (for the given value of "tariffId") with the new one ("(*2)").

Partial notifications by Actor Energy Consumer on adjusted slots:

If the Actor Energy Consumer just wants to report "new slots" as described above, the Actor Energy Consumer can send a simple "notify" operation with a single "cmdControl" Element containing the Element "partial" together with the new slots. This kind of message is also applicable if an already existing slot just got a value change in one of its Elements.

In every other case of slot adjustment (removal of "past slots", replacement of existing slots with new slots, etc.) it is recommended to send the "notify" operation using the same "delete-add" pattern as described above.

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3.4.2.4.1.4 Recommendations on "partial operations" on "incentiveTableDescriptionData"

This section applies to the Function "incentiveTableDescriptionData" only. Similar to section 3.4.2.4.1.3, an update of a Committed Incentive Table's incentiveTableDescriptionData may comprise of the removal or addition or change of "tier" instances or other changes. Esp. the Actor CEM needs to consider that Actor Energy Consumer may contain "unexpected" "tier" instances: This can happen if the CEM is a replacement of a broken CEM that already submitted an Incentive Table before it broke. It may also happen if the Actor CEM has been reset to factory default and does not remember previously written tiers. A CEM may also need to remove "tier" instances it previously wrote, but that are not valid anymore due to a switch to a different electricity provider with a new tariff structure, e.g.

In every case, the Actor CEM needs to ensure that no "unknown" or "old" "tier" information is present in Actor Energy Consumer's Committed Incentive Table after Actor CEM's write operation.

Partial write operations by Actor CEM to remove "unknown" or "old" "tier" information:

Whenever the CEM needs to remove "unknown" or "old" "tier" information it is recommended to send the "write" operation using a so-called "delete-add" pattern: For a given value of "tariffId" the "write" operation contains a "filter" Element with "delete" as well as a "filter" Element with "partial". This is shown in the following excerpt of a "datagram":

```
...
<payload>
    <cmd>
        <function>incentiveTableDescriptionData</function>
        <filter>
            <cmdControl><delete/></cmdControl>
            <incentiveTableDescriptionDataSelectors>
                <tariffDescription>
                    <tariffId>(*1)</tariffId>
                </tariffDescription>
            </incentiveTableDescriptionDataSelectors>
        </filter>
        <filter>
            <cmdControl><partial/></cmdControl>
        </filter>
        <incentiveTableDescriptionData>
            (*2)
        </incentiveTableDescriptionData>
    </cmd>
</payload>
```

where "(*1)" must be replaced by the proper value of "tariffId" and "(*2)" must contain a proper "incentiveTableDescription" instance with the same value of "tariffId" and tier and further data to be stored by Actor Energy Consumer. This pattern replaces the previous "incentiveTableDescription" instance (for the given value of "tariffId") with the new one ("(*2)").

Note: This "delete-add" pattern removes all writeable data of the related "incentiveTableDescription" instance, i.e. not just contained "tier" instances. This means the write operation may need to set such Elements as well with current values.

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Partial notifications by Actor Energy Consumer:

If the Actor Energy Consumer just wants to report "new tiers", the Actor Energy Consumer can send a simple "notify" operation with a single "cmdControl" Element containing the Element "partial" together with the new "tier" instances. This kind of message is also applicable if an already existing "tier" just got a value change in one of its Elements.

In case of the removal of "tier" instances, it is recommended to send the "notify" operation using the same "delete-add" pattern as described above.

3.4.2.4.2 Committed Power Plan

3.4.2.4.2.1 Recommendations on "timeSeriesId"

This Use Case does not prescribe a particular value for the Element "timeSeriesId". But once a Committed Power Plan instance has been created by Actor Energy Consumer, the respective value of "timeSeriesId" SHOULD be preserved. This means that updates of the Committed Power Plan SHOULD reuse the value of "timeSeriesId" (in contrast to delete data with the last value of "timeSeriesId" and write the new Committed Power Plan with a new value in "timeSeriesId").

Note: The value of "timeSeriesId" of the Committed Power Plan differs from the one of a Preliminary Power Plan.

3.4.2.4.2.2 Recommendations on "timeSeriesSlotId"

Section 3.4.2.4.1.2 specifies recommendations on "timeSlotId" of a Committed Incentive Table. These recommendations apply to "timeSeriesSlotId" of a Committed Power Plan accordingly.

Note: "timeSlotId" and "timeSeriesSlotId" are not related to each other.

3.4.2.4.2.3 Recommendations on "partial operations" on slots

This section applies to the Function "timeSeriesListData" only. In general, an update of a Committed Power Plan's slots may comprise of the removal of (past) slots, the addition of new slots, the change of existing slots, or even a replacement of existing slots if a given period needs to be split into a different number of slots with altered durations. These changes may be expressed with "partial operations" in different ways. As a proper modelling and evaluation of partial changes can be prone to errors, some definitions and recommendations will be given. For details and technical background information please consider [ProtocolSpecification].

"New slots":

1. The Actor Energy Consumer MAY append one or more future slots to an existing Committed Power Plan (without time gaps).

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"Past slots":

The following rules aim at cleaning up the Power Plan to focus on current and subsequent activities:

1. The Actor Energy Consumer SHOULD NOT announce slots whose end time is in the past.
2. The Actor Energy Consumer MAY remove slots whose end time is in the past. This removal SHALL be notified properly.

Partial notifications by Actor Energy Consumer to adjust slots:

If the Actor Energy Consumer just wants to add "new slots" as described above, the Actor Energy Consumer can send a simple "notify" operation with a single "cmdControl" Element containing the Element "partial" together with the new slots. This kind of message is also applicable if an already existing slot just needs to get a value change in one of its Elements.

In every other case of slot adjustment (remove "past slots", replace existing slots with new slots, etc.) it is recommended to send the "notify" operation using a so-called "delete-add" pattern: For a given value of "timeSeriesId" the "notify" operation contains a "filter" Element with "delete" as well as a "filter" Element with "partial". This is shown in the following excerpt of a "datagram":

```
...
<payload>
    <cmd>
        <function>timeSeriesListData</function>
        <filter>
            <cmdControl><delete/></cmdControl>
            <timeSeriesListDataSelectors>
                <timeSeriesId>(*1)</timeSeriesId>
            </timeSeriesListDataSelectors>
        </filter>
        <filter>
            <cmdControl><partial/></cmdControl>
        </filter>
        <timeSeriesListData>
            (*2)
        </timeSeriesListData>
    </cmd>
</payload>
```

where "(*1)" must be replaced by the proper value of "timeSeriesId" and "(*2)" must contain a proper "timeSeriesData" instance with the same value of "timeSeriesId" and all slots of Actor Energy Consumer. This pattern replaces the previous "timeSeriesData" instance (for the given value of "timeSeriesId") with the new one ("(*2)").

3.4.3 Scenario 3 - Negotiation of Preliminary Data

3.4.3.1 Pre-Scenario communication

1. **Detailed discovery**: Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.

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2. **Binding**: Actors that write parts of a Feature within this Scenario, need to create a binding, as described in section 3.3.3. Only one binding partner is allowed to write the data specified in this Scenario.
3. **Subscription**: Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.3.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

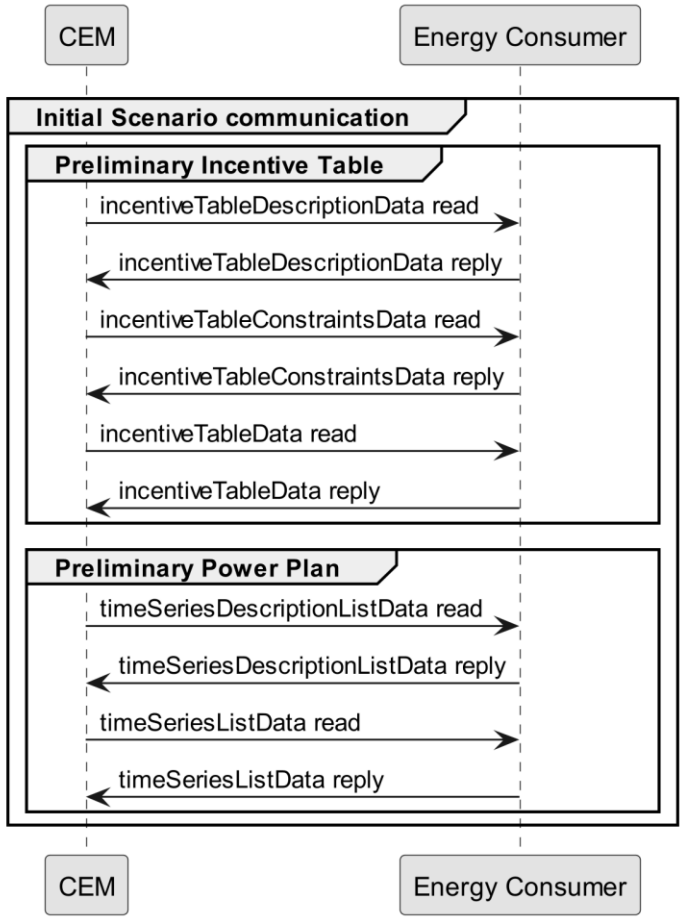


Figure 26: Scenario 3 - Initial Scenario communication sequence diagram

Partial data information regarding [ITPCM-046^{IT_PRELIMINARY}]:

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The incentiveTableDescriptionData read SHOULD be a "partial" read operation with the following Selector:

- tariffDescription. scopeType = "simplePreliminaryIncentiveTable"

The incentiveTableConstraintsData and incentiveTableData read SHOULD be "partial" read operations with the following Selector:

- tariff. tariffId (with tariffId derived from the incentiveTableDescriptionData reply)

Partial data information regarding [ITPCM-048^{PLAN_PRELIMINARY}]:

The timeSeriesDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- scopeType = "preliminaryPowerPlan"

The timeSeriesListData read SHOULD be a "partial" read operation with the following Selector:

- timeSeriesId (derived from the timeSeriesDescriptionListData reply)

Note: If partial read is not supported a full read SHALL occur.

The following table shows where the required content of the reply messages from the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
incentiveTableDescriptionData reply	Table 9	3
incentiveTableConstraintsData reply	Table 10	3
incentiveTableData reply	Table 11	3
timeSeriesDescriptionListData reply	Table 12	3
timeSeriesListData reply	Table 13	3

Table 29: Initial Scenario communication content references for reply messages of Scenario 3

XML examples can be found in Annex A.

Note: The Scenario comprises several steps that need to be processed in a particular order (see section 2.4.3). In practice it may happen that Actors are switched on at different times, e.g. from a reboot. This means the Initial Scenario communication may occur before the first step, after the last step, or in between. Dependent on the time of execution, the Initial Scenario communication provides respective data. Subsequently we assume that the Initial Scenario communication has been executed before the first step. Data of the first and subsequent steps will be explained in the Runtime Scenario communication.

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3.4.3.3 Runtime Scenario communication

Based on the Initial Scenario communication, the Runtime Scenario communication provides updates during runtime. These can be either notifications (i.e. data changes reported by a server) or write operations (i.e. data change request submitted to a server).

Order of communication:

This section just lists the required messages in general, i.e. without consideration of any order or dependency. Please consider section 2.4.3 for rules on the occurrence and content of the messages.

Notifications:

If one of the referenced server Functions' data change, the server SHALL submit the change as shown in the following figure:

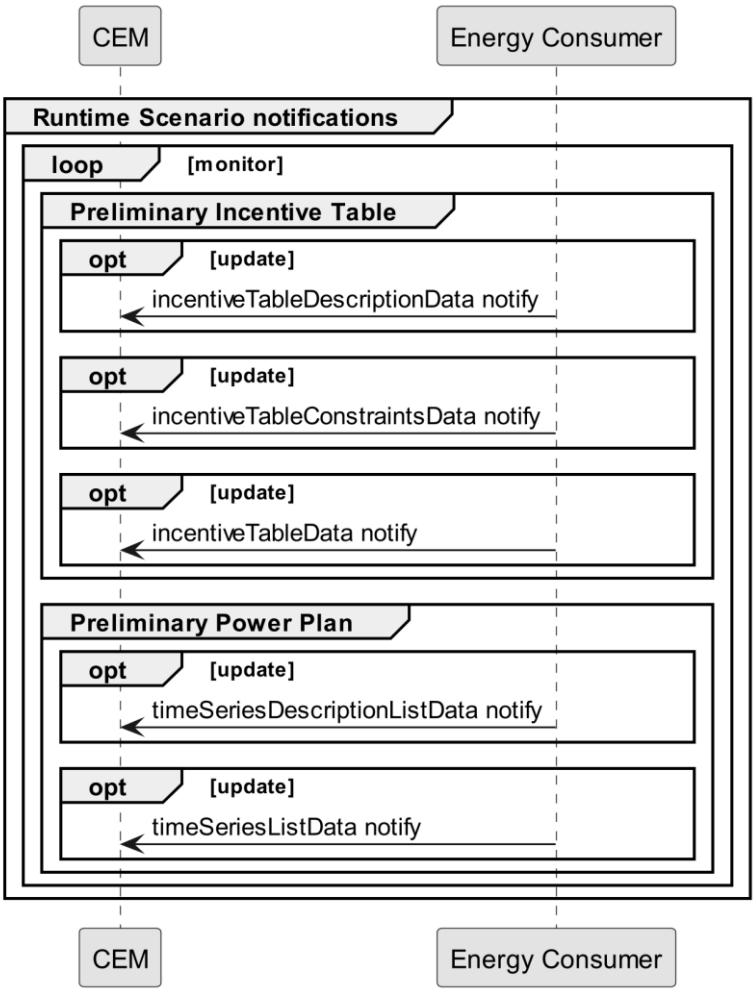


Figure 27: Scenario 3 - Runtime Scenario notifications

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

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Partial delete notifications for incentiveTableDescriptionData SHOULD be supported with the Selector:

- tariffDescription. tariffId

Partial delete notifications for incentiveTableConstraintsData and incentiveTableData SHOULD be supported with the Selector:

- tariff. tariffId

Partial delete notifications for timeSeriesDescriptionListData and timeSeriesListData SHOULD be supported with the Selector:

- timeSeriesId

Note: To interpret partial notification messages correctly the information obtained during the Initial Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the notify messages of the sequence diagram is described:

Message name from sequence diagram	**Content description in table**	**Scenario number in table**
incentiveTableDescriptionData notify	Table 9	3
incentiveTableConstraintsData notify	Table 10	3
incentiveTableData notify	Table 11	3
timeSeriesDescriptionListData notify	Table 12	3
timeSeriesListData notify	Table 13	3

Table 30: Runtime Scenario communication content references for notify messages of Scenario 3

Write operations:

Figure 28 shows possible write operations of this Scenario:

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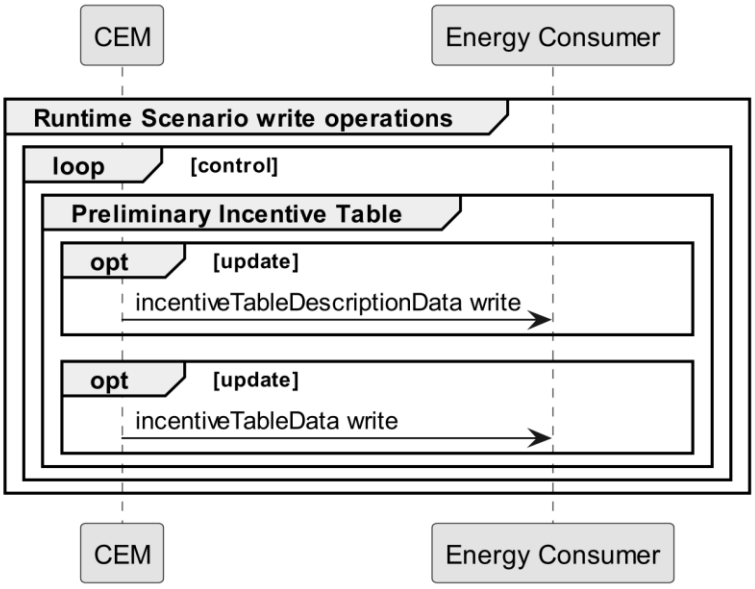


Figure 28: Scenario 3 - Runtime Scenario write operations

For partial delete write operations the same rules apply as for notifications.

For required content of write messages please consider the respective Specializations.

XML examples can be found in Annex A.

3.4.3.4 Additional information

3.4.3.4.1 Preliminary Incentive Table

3.4.3.4.1.1 Recommendations on "tariffId"

This Use Case does not prescribe a particular value for the Element "tariffId". But once a Preliminary Incentive Table instance has been created by Actor Energy Consumer, the respective value of "tariffId" SHOULD be preserved. This means that updates of a Preliminary Incentive Table SHOULD reuse the value of "tariffId" (in contrast to delete data with the last value of "tariffId" and write the new Preliminary Incentive Table with a new value in "tariffId").

Note: The value of "tariffId" of the Preliminary Incentive Table differs from the one of a Committed Incentive Table.

3.4.3.4.1.2 Recommendations on "partial operations" on slots

This section applies to the Function "incentiveTableData" only. In general, an update of a Preliminary Incentive Table's slots may change the number of slots and may even alter some or all timeSlotId values. These changes may be expressed with "partial operations" in different ways. As a proper modelling and evaluation of partial changes can be prone to errors, some recommendations will be given. For details and technical background information please consider [ProtocolSpecification].

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Partial write operations by Actor CEM to adjust slots:

It is recommended to send the "write" operation using a so-called "delete-add" pattern: For a given value of "tariffId" the "write" operation contains a "filter" Element with "delete" as well as a "filter" Element with "partial". This is shown in the following excerpt of a "datagram":

```
...
<payload>
    <cmd>
        <function>incentiveTableData</function>
        <filter>
            <cmdControl><delete/></cmdControl>
            <incentiveTableDataSelectors>
                <tariff>
                    <tariffId>(*1)</tariffId>
                </tariff>
            </incentiveTableDataSelectors>
        </filter>
        <filter>
            <cmdControl><partial/></cmdControl>
        </filter>
        <incentiveTableData>
            (*2)
        </incentiveTableData>
    </cmd>
</payload>
```

where "(*1)" must be replaced by the proper value of "tariffId" and "(*2)" must contain a proper "incentiveTable" instance with the same value of "tariffId" and slots to be stored by Actor Energy Consumer. This pattern replaces the previous "incentiveTable" instance (for the given value of "tariffId") with the new one ("(*2)").

Partial notifications by Actor Energy Consumer on adjusted slots:

If the Actor Energy Consumer just wants to report "new slots" as described above, the Actor Energy Consumer can send a simple "notify" operation with a single "cmdControl" Element containing the Element "partial" together with the new slots. This kind of message is also applicable if an already existing slot just got a value change in one of its Elements.

In every other case of slot adjustment (removal of "past slots", replacement of existing slots with new slots, etc.) it is recommended to send the "notify" operation using the same "delete-add" pattern as described above.

3.4.3.4.1.3 Recommendations on "partial operations" on "incentiveTableDescriptionData"

Section 3.4.2.4.1.4 describes some recommendations for the Committed Incentive Table. These recommendations apply for the Preliminary Incentive Table likewise.

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3.4.3.4.2 Preliminary Power Plan

3.4.3.4.2.1 Recommendations on "timeSeriesId"

This Use Case does not prescribe a particular value for the Element "timeSeriesId". But once a Preliminary Power Plan instance has been created by Actor Energy Consumer, the respective value of "timeSeriesId" SHOULD be preserved. This means that updates of a Preliminary Power Plan SHOULD reuse the value of "timeSeriesId" (in contrast to delete data with the last value of "timeSeriesId" and write the new Preliminary Power Plan with a new value in "timeSeriesId").

Note: The value of "timeSeriesId" of the Preliminary Power Plan differs from the one of a Committed Power Plan.

3.4.3.4.2.2 Recommendations on "partial operations" on slots

This section applies to the Function "timeSeriesListData" only. In general, an update of a Preliminary Power Plan's slots may change the number of slots and may even alter some or all timeSlotId values. These changes may be expressed with "partial operations" in different ways. As a proper modelling and evaluation of partial changes can be prone to errors, some recommendations will be given. For details and technical background information please consider [ProtocolSpecification].

Partial notifications by Actor Energy Consumer to adjust slots:

It is recommended to send the "notify" operation using a so-called "delete-add" pattern: For a given value of "timeSeriesId" the "notify" operation contains a "filter" Element with "delete" as well as a "filter" Element with "partial". This is shown in the following excerpt of a "datagram":

```
...
<payload>
    <cmd>
        <function>timeSeriesListData</function>
        <filter>
            <cmdControl><delete/></cmdControl>
            <timeSeriesListDataSelectors>
                <timeSeriesId>(*1)</timeSeriesId>
            </timeSeriesListDataSelectors>
        </filter>
        <filter>
            <cmdControl><partial/></cmdControl>
        </filter>
        <timeSeriesListData>
            (*2)
        </timeSeriesListData>
    </cmd>
</payload>
```

where "(*1)" must be replaced by the proper value of "timeSeriesId" and "(*2)" must contain a proper "timeSeriesData" instance with the same value of "timeSeriesId" and all slots of Actor Energy Consumer. This pattern replaces the previous "timeSeriesData" instance (for the given value of "timeSeriesId") with the new one ("(*2)").

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If Actor Energy Consumer has no slots ("empty" Preliminary Power Plan ([ITPCM-038[PPLAN_DEL_POWER]])), the "delete-add" pattern applies anyhow: In this case, "(*2)" just contains a "timeSeriesData" instance with the same value of "timeSeriesId":

```
...
<timeSeriesListData>
    <timeSeriesData>
        <timeSeriesId>(*1)</timeSeriesId>
    </timeSeriesData>
</timeSeriesListData>
...
```

3.4.4 Scenario 4 - Unsolicited Update of Committed Incentive Table

3.4.4.1 Pre-Scenario communication

See section 3.4.2, as Scenario 2 is the basis for Scenario 4.

3.4.4.2 Initial Scenario communication

See section 3.4.2, as Scenario 2 is the basis for Scenario 4.

3.4.4.3 Runtime Scenario communication

See section 3.4.2, as Scenario 2 is the basis for Scenario 4 and provides updates (notifications) of the Actor Energy Consumer during runtime.

The Actor CEM executes Scenario 4 by sending the required changes to Actor Energy Consumer as specified in "Write operations" of section 3.4.2.3.

3.4.4.4 Additional information

3.4.4.4.1 Committed Incentive Table

See section 3.4.2.4.1.

3.4.5 Scenario 5 - Unsolicited Update of Committed Power Plan

3.4.5.1 Pre-Scenario communication

See section 3.4.2, as Scenario 2 is the basis for Scenario 5.

3.4.5.2 Initial Scenario communication

See section 3.4.2, as Scenario 2 is the basis for Scenario 5.

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3.4.5.3 Runtime Scenario communication

See section 3.4.2, as Scenario 2 is the basis for Scenario 5 and provides updates (notifications) of the Actor Energy Consumer during runtime.

The Actor Energy Consumer executes Scenario 5 by sending the required changes to Actor CEM as specified in "Notifications" of section 3.4.2.3 for the Functions in this table:

Message name from sequence diagram	**Content description in table**	**Scenario number in table**
timeSeriesDescriptionListData notify	Table 12	5
timeSeriesListData notify	Table 13	5

Table 31: Runtime Scenario communication content references for notify messages of Scenario 5

3.4.5.4 Additional information

3.4.5.4.1 Committed Power Plan

See section 3.4.2.4.2.

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Annex A - SPINE best practice

A.1 Sequence diagrams

A.1.1 Pre-Scenario communication

These steps need to be done only once.

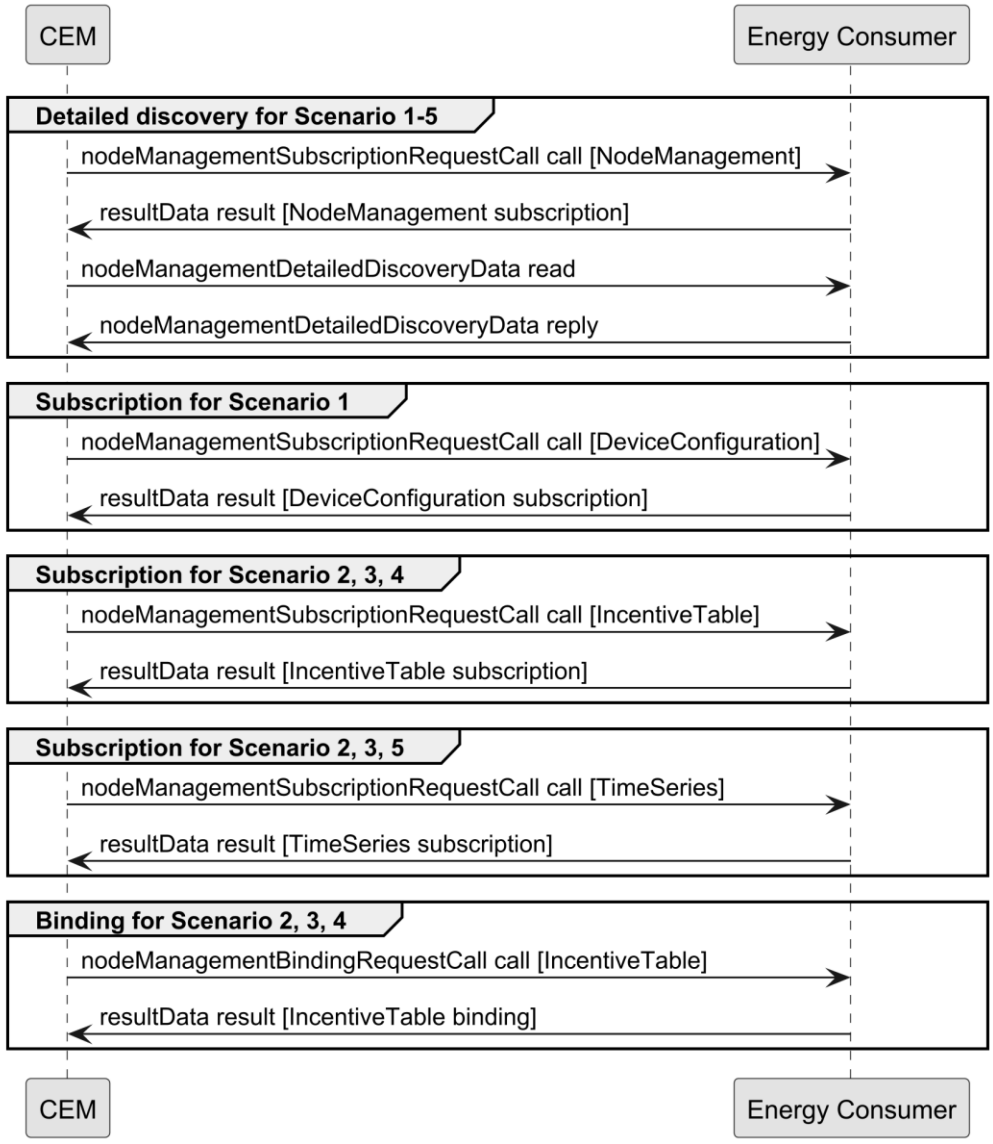


Figure 29: Best practice - Pre-Scenario communication

The message content is provided in the "XML examples" section.

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A.1.2 Initial Scenario communication

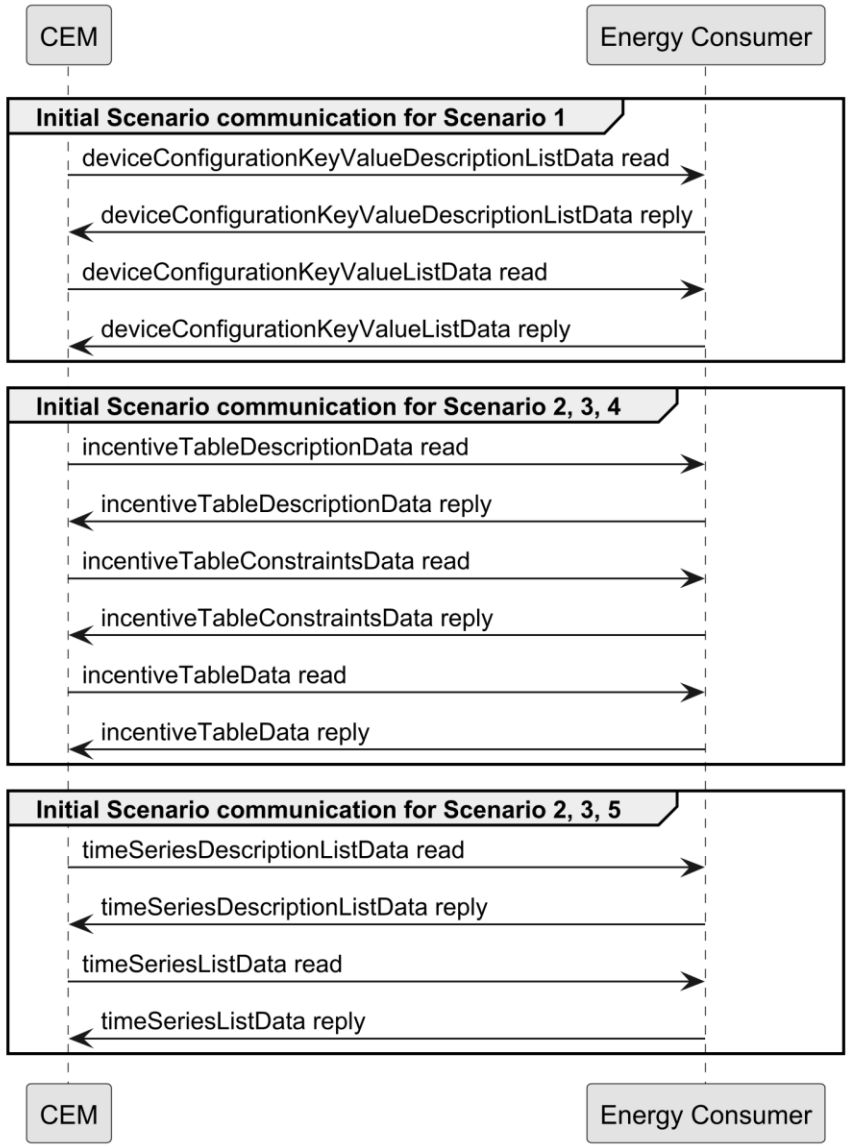


Figure 30: Best practice - Initial Scenario communication

The message content is provided in the "XML examples" section.

All read operations should be partial messages as described in the corresponding Scenario sections.

Note: It is considered best practice to read xDescriptionListData before xConstraintsListData and xConstraintsListData before xListData.

A.1.3 Runtime Scenario communication

After the Initial Scenario communication, the following messages can occur whenever the described processes permit and require the respective data exchange.

Notifications:

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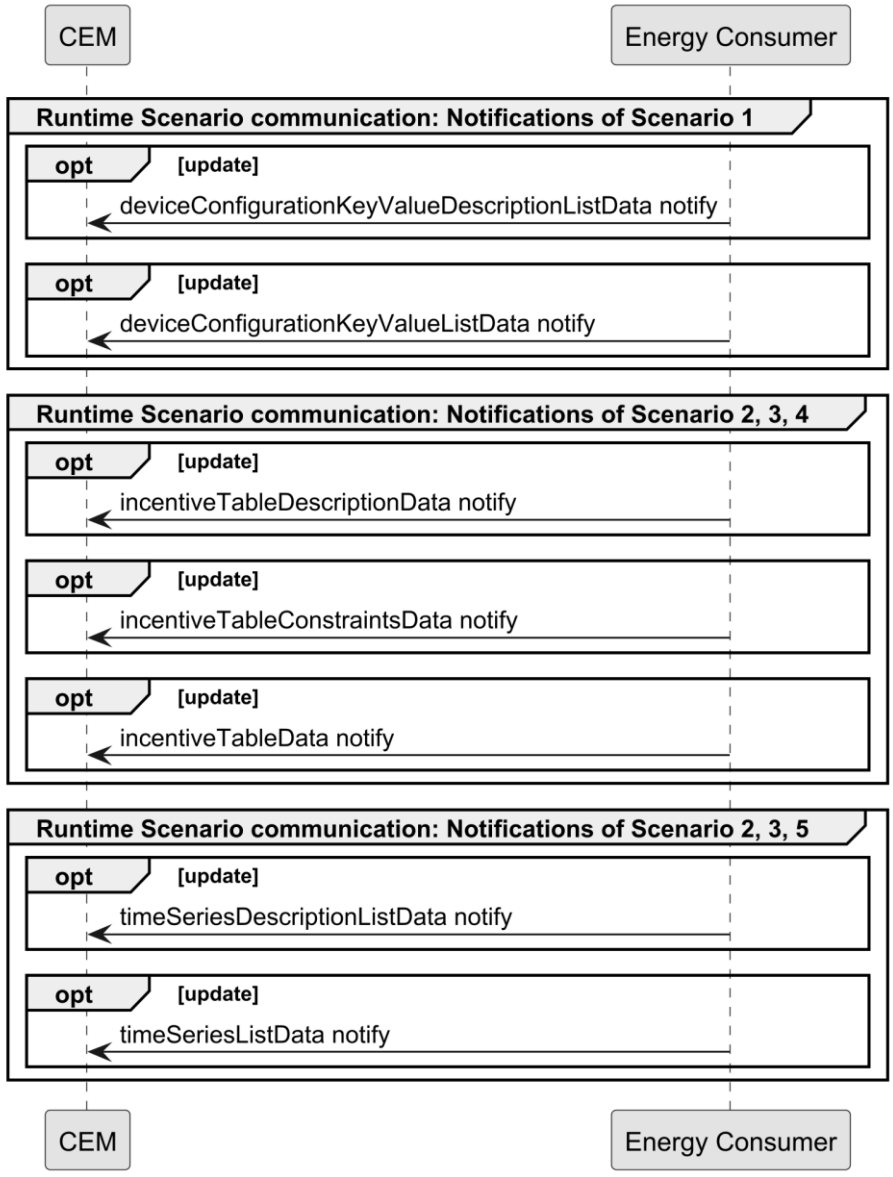


Figure 31: Best practice - Runtime Scenario communication - Notifications

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Write operations:

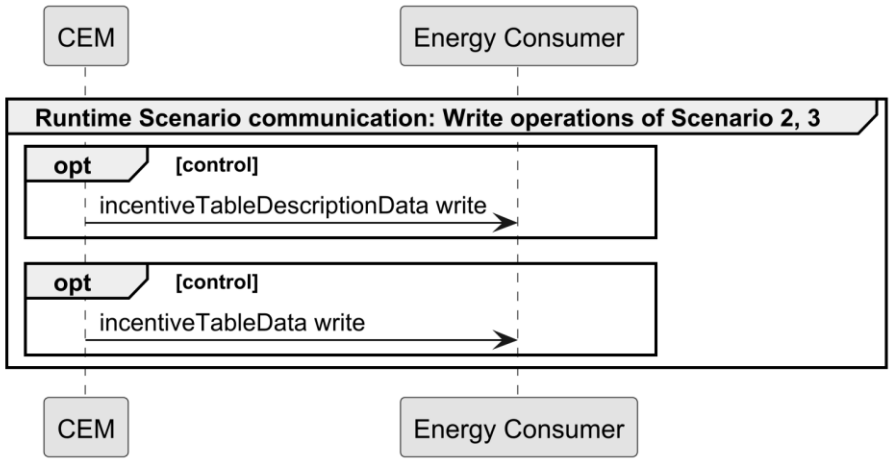


Figure 32: Best practice - Runtime Scenario communication - Write operations

The message content is provided in the "XML examples" section.

All notify operations should be partial messages as described in the corresponding Scenario sections.

A.2 XML examples

Within the XMLs two devices are used as example: A CEM with the device address "d:_i:46925_CEM" and a Heat Pump Compressor (as Energy Consumer) with device address "d:_i:46925_HeatPump".

It is assumed that the device addresses are already known (from a previous connection, e.g.) and do not need to be discovered with the XML examples given in this best practice.

A.2.1 Pre-Scenario communication

A.2.1.1 Example content of "nodeManagementSubscriptionRequestCall call [NodeManagement]"

Detailed discovery subscription for all Scenarios.

As a device may have implemented dynamic Features or Entities that are added during runtime, a subscription on the NodeManagement Feature itself enables notifications of added or removed Features or Entities.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_CEM</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_HeatPump</device>
      <entity>0</entity>
```

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```
        <feature>0</feature>
      </addressDestination>
      <msgCounter>1</msgCounter>
      <cmdClassifier>call</cmdClassifier>
      <ackRequest>true</ackRequest>
      <timestamp>2026-07-15T16:00:01.0Z</timestamp>
    </header>
    <payload>
      <cmd>
        <nodeManagementSubscriptionRequestCall>
          <subscriptionRequest>
            <clientAddress>
              <entity>0</entity>
              <feature>0</feature>
            </clientAddress>
            <serverAddress>
              <entity>0</entity>
              <feature>0</feature>
            </serverAddress>
          </subscriptionRequest>
        </nodeManagementSubscriptionRequestCall>
      </cmd>
    </payload>
  </datagram>
```

A.2.1.2 Example content of "resultData result [NodeManagement subscription]"

Detailed discovery subscription result for all Scenarios.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressDestination>
    <msgCounter>1</msgCounter>
    <msgCounterReference>1</msgCounterReference>
    <cmdClassifier>result</cmdClassifier>
    <timestamp>2026-07-15T16:00:02.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <resultData>
        <errorNumber>0</errorNumber>
      </resultData>
    </cmd>
  </payload>
</datagram>
```

A.2.1.3 Example content of "nodeManagementDetailedDiscoveryData read"

Detailed discovery read for all Scenarios (it is assumed that the full functionality of this Use Case is realized in both devices).

Note: It is also possible to add further Selectors to search for Entity Types and Feature Types used by other supported Use Cases. Alternatively, a full read may be used.

```
<?xml version="1.0" encoding="UTF-8"?>
```

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```
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
      xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_CEM</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_HeatPump</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressDestination>
    <msgCounter>2</msgCounter>
    <cmdClassifier>read</cmdClassifier>
    <timestamp>2026-07-15T16:00:03.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>nodeManagementDetailedDiscoveryData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
        <nodeManagementDetailedDiscoveryDataSelectors>
          <entityInformation>
            <entityType>HeatPumpAppliance</entityType>
          </entityInformation>
        </nodeManagementDetailedDiscoveryDataSelectors>
        <nodeManagementDetailedDiscoveryDataSelectors>
          <featureInformation>
            <featureType>DeviceConfiguration</featureType>
          </featureInformation>
        </nodeManagementDetailedDiscoveryDataSelectors>
        <nodeManagementDetailedDiscoveryDataSelectors>
          <featureInformation>
            <featureType>TimeSeries</featureType>
          </featureInformation>
        </nodeManagementDetailedDiscoveryDataSelectors>
        <nodeManagementDetailedDiscoveryDataSelectors>
          <featureInformation>
            <featureType>IncentiveTable</featureType>
          </featureInformation>
        </nodeManagementDetailedDiscoveryDataSelectors>
      </filter>
      <nodeManagementDetailedDiscoveryData/>
    </cmd>
  </payload>
</datagram>
```

A.2.1.4 Example content of "nodeManagementDetailedDiscoveryData reply"

Detailed discovery reply for all Scenarios (it is assumed that the full functionality of this Use Case is realized in both devices).

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
      xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
      file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>0</entity>
      <feature>0</feature>
```

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```
      </addressDestination>
      <msgCounter>2</msgCounter>
      <msgCounterReference>2</msgCounterReference>
      <cmdClassifier>reply</cmdClassifier>
      <timestamp>2026-07-15T16:00:04.0Z</timestamp>
    </header>
    <payload>
      <cmd>
        <function>nodeManagementDetailedDiscoveryData</function>
        <filter>
          <cmdControl>
            <partial/>
          </cmdControl>
        </filter>
        <nodeManagementDetailedDiscoveryData>
          <entityInformation>
            <description>
              <entityAddress>
                <entity>1</entity>
              </entityAddress>
              <entityType>HeatPumpAppliance</entityType>
            </description>
          </entityInformation>
          <featureInformation>
            <description>
              <featureAddress>
                <entity>1</entity>
                <feature>1</feature>
              </featureAddress>
              <featureType>TimeSeries</featureType>
              <role>server</role>
              <supportedFunction>
                <function>timeSeriesDescriptionListData</function>
                <possibleOperations>
                  <read>
                    <partial/>
                  </read>
                </possibleOperations>
              </supportedFunction>
              <supportedFunction>
                <function>timeSeriesListData</function>
                <possibleOperations>
                  <read>
                    <partial/>
                  </read>
                </possibleOperations>
              </supportedFunction>
            </description>
          </featureInformation>
          <featureInformation>
            <description>
              <featureAddress>
                <entity>1</entity>
                <feature>2</feature>
              </featureAddress>
              <featureType>IncentiveTable</featureType>
              <role>server</role>
              <supportedFunction>
                <function>incentiveTableDescriptionData</function>
                <possibleOperations>
                  <read>
                    <partial/>
                  </read>
                  <write>
                    <partial/>
                  </write>
                </possibleOperations>
              </supportedFunction>
              <supportedFunction>
                <function>incentiveTableConstraintsData</function>
                <possibleOperations>
                  <read>
                    <partial/>
                  </read>
                </possibleOperations>
              </supportedFunction>
              <supportedFunction>
                <function>incentiveTableData</function>
```

```
              <possibleOperations>
                <read>
                  <partial/>
                </read>
                <write>
                  <partial/>
                </write>
              </possibleOperations>
            </supportedFunction>
          </description>
        </featureInformation>
        <featureInformation>
          <description>
            <featureAddress>
              <entity>1</entity>
              <feature>3</feature>
            </featureAddress>
            <featureType>DeviceConfiguration</featureType>
            <role>server</role>
            <supportedFunction>
              <function>deviceConfigurationKeyValueDescriptionListData</function>
              <possibleOperations>
                <read>
                  <partial/>
                </read>
              </possibleOperations>
            </supportedFunction>
            <supportedFunction>
              <function>deviceConfigurationKeyValueListData</function>
              <possibleOperations>
                <read>
                  <partial/>
                </read>
              </possibleOperations>
            </supportedFunction>
          </description>
        </featureInformation>
      </nodeManagementDetailedDiscoveryData>
    </cmd>
  </payload>
</datagram>
```

A.2.1.5 Example content of "nodeManagementSubscriptionRequestCall call [DeviceConfiguration]"

DeviceConfiguration subscription for Scenario 1.

Note: Currently, it is not possible to subscribe to parts of a feature. Therefore, a "nodeManagementSubscriptionRequestCall call" message must be used to create a full feature subscription. This means that if the Feature also contains data that is not relevant for the Use Cases supported by the subscriber, a subscriber might receive notifications with unsupported data. The subscriber shall ignore this data.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_CEM</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_HeatPump</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressDestination>
```

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```
      <msgCounter>3</msgCounter>
      <cmdClassifier>call</cmdClassifier>
      <ackRequest>true</ackRequest>
      <timestamp>2026-07-15T16:00:05.0Z</timestamp>
    </header>
    <payload>
      <cmd>
        <nodeManagementSubscriptionRequestCall>
          <subscriptionRequest>
            <clientAddress>
              <entity>1</entity>
              <feature>1</feature>
            </clientAddress>
            <serverAddress>
              <entity>1</entity>
              <feature>3</feature>
            </serverAddress>
          </subscriptionRequest>
        </nodeManagementSubscriptionRequestCall>
      </cmd>
    </payload>
  </datagram>
```

A.2.1.6 Example content of "resultData result [DeviceConfiguration subscription]"

DeviceConfiguration subscription result for Scenario 1.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressDestination>
    <msgCounter>3</msgCounter>
    <msgCounterReference>3</msgCounterReference>
    <cmdClassifier>result</cmdClassifier>
    <timestamp>2026-07-15T16:00:06.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <resultData>
        <errorNumber>0</errorNumber>
      </resultData>
    </cmd>
  </payload>
</datagram>
```

A.2.1.7 Example content of "nodeManagementSubscriptionRequestCall call [IncentiveTable]"

IncentiveTable subscription for Scenario 2, 3, 4.

Note: Currently, it is not possible to subscribe to parts of a feature. Therefore, a "nodeManagementSubscriptionRequestCall call" message must be used to create a full feature subscription. This means that if the Feature also contains data that is not relevant for the Use Cases

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supported by the subscriber, a subscriber might receive notifications with unsupported data. The subscriber shall ignore this data.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_CEM</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_HeatPump</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressDestination>
    <msgCounter>4</msgCounter>
    <cmdClassifier>call</cmdClassifier>
    <ackRequest>true</ackRequest>
    <timestamp>2026-07-15T16:00:07.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <nodeManagementSubscriptionRequestCall>
        <subscriptionRequest>
          <clientAddress>
            <entity>1</entity>
            <feature>1</feature>
          </clientAddress>
          <serverAddress>
            <entity>1</entity>
            <feature>2</feature>
          </serverAddress>
        </subscriptionRequest>
      </nodeManagementSubscriptionRequestCall>
    </cmd>
  </payload>
</datagram>
```

A.2.1.8 Example content of "resultData result [IncentiveTable subscription]"

IncentiveTable subscription result for Scenario 2, 3, 4.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressDestination>
    <msgCounter>4</msgCounter>
    <msgCounterReference>4</msgCounterReference>
    <cmdClassifier>result</cmdClassifier>
    <timestamp>2026-07-15T16:00:08.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <resultData>
        <errorNumber>0</errorNumber>
```

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```
      </resultData>
    </cmd>
  </payload>
</datagram>
```

A.2.1.9 Example content of "nodeManagementSubscriptionRequestCall call [TimeSeries]"

TimeSeries subscription for Scenario 2, 3, 5.

Note: Currently, it is not possible to subscribe to parts of a feature. Therefore, a "nodeManagementSubscriptionRequestCall call" message must be used to create a full feature subscription. This means that if the Feature also contains data that is not relevant for the Use Cases supported by the subscriber, a subscriber might receive notifications with unsupported data. The subscriber shall ignore this data.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_CEM</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_HeatPump</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressDestination>
    <msgCounter>5</msgCounter>
    <cmdClassifier>call</cmdClassifier>
    <ackRequest>true</ackRequest>
    <timestamp>2026-07-15T16:00:09.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <nodeManagementSubscriptionRequestCall>
        <subscriptionRequest>
          <clientAddress>
            <entity>1</entity>
            <feature>1</feature>
          </clientAddress>
          <serverAddress>
            <entity>1</entity>
            <feature>1</feature>
          </serverAddress>
        </subscriptionRequest>
      </nodeManagementSubscriptionRequestCall>
    </cmd>
  </payload>
</datagram>
```

A.2.1.10 Example content of "resultData result [TimeSeries subscription]"

TimeSeries subscription result for Scenario 2, 3, 5.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
```

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```
      <addressSource>
        <device>d:_i:46925_HeatPump</device>
        <entity>0</entity>
        <feature>0</feature>
      </addressSource>
      <addressDestination>
        <device>d:_i:46925_CEM</device>
        <entity>0</entity>
        <feature>0</feature>
      </addressDestination>
      <msgCounter>5</msgCounter>
      <msgCounterReference>5</msgCounterReference>
      <cmdClassifier>result</cmdClassifier>
      <timestamp>2026-07-15T16:00:10.0Z</timestamp>
    </header>
    <payload>
      <cmd>
        <resultData>
          <errorNumber>0</errorNumber>
        </resultData>
      </cmd>
    </payload>
  </datagram>
```

A.2.1.11 Example content of "nodeManagementBindingRequestCall call [IncentiveTable]"

IncentiveTable binding for Scenario 2, 3.

Note: Currently, it is not possible to bind to parts of a feature. Therefore, a "nodeManagementBindingRequestCall call" message must be used to create a full feature binding.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_CEM</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_HeatPump</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressDestination>
    <msgCounter>6</msgCounter>
    <cmdClassifier>call</cmdClassifier>
    <ackRequest>true</ackRequest>
    <timestamp>2026-07-15T16:00:11.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <nodeManagementBindingRequestCall>
        <bindingRequest>
          <clientAddress>
            <entity>1</entity>
            <feature>1</feature>
          </clientAddress>
          <serverAddress>
            <entity>1</entity>
            <feature>2</feature>
          </serverAddress>
        </bindingRequest>
      </nodeManagementBindingRequestCall>
    </cmd>
  </payload>
</datagram>
```

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A.2.1.12 Example content of "resultData result [IncentiveTable binding]"

IncentiveTable binding result for Scenario 2, 3.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>0</entity>
      <feature>0</feature>
    </addressDestination>
    <msgCounter>6</msgCounter>
    <msgCounterReference>6</msgCounterReference>
    <cmdClassifier>result</cmdClassifier>
    <timestamp>2026-07-15T16:00:12.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <resultData>
        <errorNumber>0</errorNumber>
      </resultData>
    </cmd>
  </payload>
</datagram>
```

A.2.2 Initial Scenario communication

A.2.2.1 Example content of "deviceConfigurationKeyValueDescriptionListData read"

deviceConfigurationKeyValueDescriptionListData read for Scenario 1.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>3</feature>
    </addressDestination>
    <msgCounter>100</msgCounter>
    <cmdClassifier>read</cmdClassifier>
    <timestamp>2026-07-15T16:20:04.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>deviceConfigurationKeyValueDescriptionListData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
        <deviceConfigurationKeyValueDescriptionListDataSelectors>
          <keyName>incentivesSimulationCyclesMax</keyName>
```

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```
        </deviceConfigurationKeyValueDescriptionListDataSelectors>
        <deviceConfigurationKeyValueDescriptionListDataSelectors>
          <keyName>incentivesSimulationConcurrent</keyName>
        </deviceConfigurationKeyValueDescriptionListDataSelectors>
        <deviceConfigurationKeyValueDescriptionListDataSelectors>
          <keyName>incentivesTimeoutIncentiveRequest</keyName>
        </deviceConfigurationKeyValueDescriptionListDataSelectors>
        <deviceConfigurationKeyValueDescriptionListDataSelectors>
          <keyName>incentivesWaitIncentiveWriteable</keyName>
        </deviceConfigurationKeyValueDescriptionListDataSelectors>
      </filter>
      <deviceConfigurationKeyValueDescriptionListData/>
    </cmd>
  </payload>
</datagram>
```

A.2.2.2 Example content of "deviceConfigurationKeyValueDescriptionListData reply"

deviceConfigurationKeyValueDescriptionListData reply for Scenario 1.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>3</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>60</msgCounter>
    <msgCounterReference>100</msgCounterReference>
    <cmdClassifier>reply</cmdClassifier>
    <timestamp>2026-07-15T16:20:05.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>deviceConfigurationKeyValueDescriptionListData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <deviceConfigurationKeyValueDescriptionListData>
        <deviceConfigurationKeyValueDescriptionData>
          <keyId>1</keyId>
          <keyName>incentivesSimulationCyclesMax</keyName>
          <valueType>integer</valueType>
        </deviceConfigurationKeyValueDescriptionData>
        <deviceConfigurationKeyValueDescriptionData>
          <keyId>2</keyId>
          <keyName>incentivesSimulationConcurrent</keyName>
          <valueType>boolean</valueType>
        </deviceConfigurationKeyValueDescriptionData>
        <deviceConfigurationKeyValueDescriptionData>
          <keyId>3</keyId>
          <keyName>incentivesTimeoutIncentiveRequest</keyName>
          <valueType>duration</valueType>
        </deviceConfigurationKeyValueDescriptionData>
        <deviceConfigurationKeyValueDescriptionData>
          <keyId>4</keyId>
          <keyName>incentivesWaitIncentiveWriteable</keyName>
          <valueType>duration</valueType>
        </deviceConfigurationKeyValueDescriptionData>
      </deviceConfigurationKeyValueDescriptionListData>
    </cmd>
  </payload>
</datagram>
```

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A.2.2.3 Example content of "deviceConfigurationKeyValueListData read"

deviceConfigurationKeyValueListData read for Scenario 1.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>3</feature>
    </addressDestination>
    <msgCounter>101</msgCounter>
    <cmdClassifier>read</cmdClassifier>
    <timestamp>2026-07-15T16:20:06.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>deviceConfigurationKeyValueListData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
        <deviceConfigurationKeyValueListDataSelectors>
          <keyId>1</keyId>
        </deviceConfigurationKeyValueListDataSelectors>
        <deviceConfigurationKeyValueListDataSelectors>
          <keyId>2</keyId>
        </deviceConfigurationKeyValueListDataSelectors>
        <deviceConfigurationKeyValueListDataSelectors>
          <keyId>3</keyId>
        </deviceConfigurationKeyValueListDataSelectors>
        <deviceConfigurationKeyValueListDataSelectors>
          <keyId>4</keyId>
        </deviceConfigurationKeyValueListDataSelectors>
      </filter>
      <deviceConfigurationKeyValueListData/>
    </cmd>
  </payload>
</datagram>
```

A.2.2.4 Example content of "deviceConfigurationKeyValueListData reply"

deviceConfigurationKeyValueListData reply for Scenario 1.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>3</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
```

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```
      <msgCounter>61</msgCounter>
      <msgCounterReference>101</msgCounterReference>
      <cmdClassifier>reply</cmdClassifier>
      <timestamp>2026-07-15T16:20:07.0Z</timestamp>
    </header>
    <payload>
      <cmd>
        <function>deviceConfigurationKeyValueListData</function>
        <filter>
          <cmdControl>
            <partial/>
          </cmdControl>
        </filter>
        <deviceConfigurationKeyValueListData>
          <deviceConfigurationKeyValueData>
            <keyId>1</keyId>
            <value>
              <integer>3</integer>
            </value>
          </deviceConfigurationKeyValueData>
          <deviceConfigurationKeyValueData>
            <keyId>2</keyId>
            <value>
              <boolean>true</boolean>
            </value>
          </deviceConfigurationKeyValueData>
          <deviceConfigurationKeyValueData>
            <keyId>3</keyId>
            <value>
              <duration>PT60S</duration>
            </value>
          </deviceConfigurationKeyValueData>
          <deviceConfigurationKeyValueData>
            <keyId>4</keyId>
            <value>
              <duration>PT180S</duration>
            </value>
          </deviceConfigurationKeyValueData>
        </deviceConfigurationKeyValueListData>
      </cmd>
    </payload>
  </datagram>
```

A.2.2.5 Example content of "incentiveTableDescriptionData read"

incentiveTableDescriptionData read for Scenario 2 (also basis for Scenario 4) and 3.

```
  <?xml version="1.0" encoding="UTF-8"?>
  <datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
    file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
    <header>
      <specificationVersion>1.3.0</specificationVersion>
      <addressSource>
        <device>d:_i:46925_CEM</device>
        <entity>1</entity>
        <feature>1</feature>
      </addressSource>
      <addressDestination>
        <device>d:_i:46925_HeatPump</device>
        <entity>1</entity>
        <feature>2</feature>
      </addressDestination>
      <msgCounter>110</msgCounter>
      <cmdClassifier>read</cmdClassifier>
      <timestamp>2026-07-15T16:20:30.0Z</timestamp>
    </header>
    <payload>
      <cmd>
        <function>incentiveTableDescriptionData</function>
        <filter>
          <cmdControl>
            <partial/>
          </cmdControl>
```

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```
        <incentiveTableDescriptionDataSelectors>
          <tariffDescription>
            <scopeType>simpleCommittedIncentiveTable</scopeType>
          </tariffDescription>
        </incentiveTableDescriptionDataSelectors>
        <incentiveTableDescriptionDataSelectors>
          <tariffDescription>
            <scopeType>simplePreliminaryIncentiveTable</scopeType>
          </tariffDescription>
        </incentiveTableDescriptionDataSelectors>
      </filter>
      <incentiveTableDescriptionData/>
    </cmd>
  </payload>
</datagram>
```

A.2.2.6 Example content of "incentiveTableDescriptionData reply"

incentiveTableDescriptionData reply for Scenario 2 (also basis for Scenario 4) and 3. This example assumes that neither a Committed Incentive Table nor a Preliminary Incentive Table has been written before by a CEM (i.e. the reply just contains non-writeable mandatory data) and that the Energy Consumer currently does NOT request towards Actor CEM to get a modified or new Committed Incentive Table ([ITPCM-007^{IT_NO_REQUEST}]) (see also section A.2.2.10).

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>2</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>70</msgCounter>
    <msgCounterReference>110</msgCounterReference>
    <cmdClassifier>reply</cmdClassifier>
    <timestamp>2026-07-15T16:20:31.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>incentiveTableDescriptionData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <incentiveTableDescriptionData>
        <incentiveTableDescription>
          <tariffDescription>
            <tariffId>1</tariffId>
            <tariffWriteable>true</tariffWriteable>
            <updateRequired>false</updateRequired>
            <scopeType>simpleCommittedIncentiveTable</scopeType>
            <slotIdSupport>true</slotIdSupport>
          </tariffDescription>
        </incentiveTableDescription>
        <incentiveTableDescription>
          <tariffDescription>
            <tariffId>2</tariffId>
            <tariffWriteable>true</tariffWriteable>
            <scopeType>simplePreliminaryIncentiveTable</scopeType>
            <slotIdSupport>true</slotIdSupport>
          </tariffDescription>
        </incentiveTableDescription>
```

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```
      </incentiveTableDescriptionData>
    </cmd>
  </payload>
</datagram>
```

A.2.2.7 Example content of "incentiveTableConstraintsData read"

incentiveTableConstraintsData read for Scenario 2 (also basis for Scenario 4) and 3.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>2</feature>
    </addressDestination>
    <msgCounter>111</msgCounter>
    <cmdClassifier>read</cmdClassifier>
    <timestamp>2026-07-15T16:20:32.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>incentiveTableConstraintsData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
        <incentiveTableConstraintsDataSelectors>
          <tariff>
            <tariffId>1</tariffId>
          </tariff>
        </incentiveTableConstraintsDataSelectors>
        <incentiveTableConstraintsDataSelectors>
          <tariff>
            <tariffId>2</tariffId>
          </tariff>
        </incentiveTableConstraintsDataSelectors>
      </filter>
      <incentiveTableConstraintsData/>
    </cmd>
  </payload>
</datagram>
```

A.2.2.8 Example content of "incentiveTableConstraintsData reply"

incentiveTableConstraintsData reply for Scenario 2 (also basis for Scenario 4) and 3.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>2</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
```

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```
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>71</msgCounter>
    <msgCounterReference>111</msgCounterReference>
    <cmdClassifier>reply</cmdClassifier>
    <timestamp>2026-07-15T16:20:33.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>incentiveTableConstraintsData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <incentiveTableConstraintsData>
        <incentiveTableConstraints>
          <tariff>
            <tariffId>1</tariffId>
          </tariff>
          <tariffConstraints>
            <maxTiersPerTariff>5</maxTiersPerTariff>
            <maxBoundariesPerTier>1</maxBoundariesPerTier>
            <maxIncentivesPerTier>3</maxIncentivesPerTier>
          </tariffConstraints>
          <incentiveSlotConstraints>
            <slotCountMax>1024</slotCountMax>
          </incentiveSlotConstraints>
        </incentiveTableConstraints>
        <incentiveTableConstraints>
          <tariff>
            <tariffId>2</tariffId>
          </tariff>
          <tariffConstraints>
            <maxTiersPerTariff>5</maxTiersPerTariff>
            <maxBoundariesPerTier>1</maxBoundariesPerTier>
            <maxIncentivesPerTier>3</maxIncentivesPerTier>
          </tariffConstraints>
          <incentiveSlotConstraints>
            <slotCountMax>1024</slotCountMax>
          </incentiveSlotConstraints>
        </incentiveTableConstraints>
      </incentiveTableConstraintsData>
    </cmd>
  </payload>
</datagram>
```

A.2.2.9 Example content of "incentiveTableData read"

incentiveTableData read for Scenario 2 (also basis for Scenario 4) and 3.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>2</feature>
    </addressDestination>
    <msgCounter>112</msgCounter>
    <cmdClassifier>read</cmdClassifier>
    <timestamp>2026-07-15T16:20:34.0Z</timestamp>
  </header>
  <payload>
    <cmd>
```

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```
      <function>incentiveTableData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
        <incentiveTableDataSelectors>
          <tariff>
            <tariffId>1</tariffId>
          </tariff>
        </incentiveTableDataSelectors>
        <incentiveTableDataSelectors>
          <tariff>
            <tariffId>2</tariffId>
          </tariff>
        </incentiveTableDataSelectors>
      </filter>
      <incentiveTableData/>
    </cmd>
  </payload>
</datagram>
```

A.2.2.10 Example content of "incentiveTableData reply"

incentiveTableData reply for Scenario 2 (also basis for Scenario 4) and 3. This example continues with the assumption of section A.2.2.6.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>2</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>72</msgCounter>
    <msgCounterReference>112</msgCounterReference>
    <cmdClassifier>reply</cmdClassifier>
    <timestamp>2026-07-15T16:20:35.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>incentiveTableData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <incentiveTableData/>
    </cmd>
  </payload>
</datagram>
```

A.2.2.11 Example content of "timeSeriesDescriptionListData read"

timeSeriesDescriptionListData read for Scenario 2, 3, 5.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
```

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```
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>120</msgCounter>
    <cmdClassifier>read</cmdClassifier>
    <timestamp>2026-07-15T16:20:40.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>timeSeriesDescriptionListData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
        <timeSeriesDescriptionListDataSelectors>
          <scopeType>committedPowerPlan</scopeType>
        </timeSeriesDescriptionListDataSelectors>
        <timeSeriesDescriptionListDataSelectors>
          <scopeType>preliminaryPowerPlan</scopeType>
        </timeSeriesDescriptionListDataSelectors>
      </filter>
      <timeSeriesDescriptionListData/>
    </cmd>
  </payload>
</datagram>
```

A.2.2.12 Example content of "timeSeriesDescriptionListData reply"

timeSeriesDescriptionListData reply for Scenario 2, 3, 5.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>80</msgCounter>
    <msgCounterReference>120</msgCounterReference>
    <cmdClassifier>reply</cmdClassifier>
    <timestamp>2026-07-15T16:20:41.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>timeSeriesDescriptionListData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <timeSeriesDescriptionListData>
        <timeSeriesDescriptionData>
          <timeSeriesId>1</timeSeriesId>
          <timeSeriesType>plan</timeSeriesType>
          <timeSeriesWriteable>false</timeSeriesWriteable>
          <measurementId>1</measurementId>
```

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```
          <unit>W</unit>
          <scopeType>committedPowerPlan</scopeType>
        </timeSeriesDescriptionData>
        <timeSeriesDescriptionData>
          <timeSeriesId>2</timeSeriesId>
          <timeSeriesType>plan</timeSeriesType>
          <timeSeriesWriteable>false</timeSeriesWriteable>
          <measurementId>1</measurementId>
          <unit>W</unit>
          <scopeType>preliminaryPowerPlan</scopeType>
        </timeSeriesDescriptionData>
      </timeSeriesDescriptionListData>
    </cmd>
  </payload>
</datagram>
```

A.2.2.13 Example content of "timeSeriesListData read"

timeSeriesListData read for Scenario 2, 3, 5.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>121</msgCounter>
    <cmdClassifier>read</cmdClassifier>
    <timestamp>2026-07-15T16:20:42.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>timeSeriesListData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
        <timeSeriesListDataSelectors>
          <timeSeriesId>1</timeSeriesId>
        </timeSeriesListDataSelectors>
        <timeSeriesListDataSelectors>
          <timeSeriesId>2</timeSeriesId>
        </timeSeriesListDataSelectors>
      </filter>
      <timeSeriesListData/>
    </cmd>
  </payload>
</datagram>
```

A.2.2.14 Example content of "timeSeriesListData reply"

timeSeriesListData reply for Scenario 2, 3, 5. This example continues with the assumption of section A.2.2.6: As no Incentive Table has been written before by a CEM, the Energy Consumer did not create proper Power Plans (though this is permissible for the Committed Power Plan). As a consequence, the content of Element "timeSeriesListData" below is empty.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
```

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```
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>81</msgCounter>
    <msgCounterReference>121</msgCounterReference>
    <cmdClassifier>reply</cmdClassifier>
    <timestamp>2026-07-15T16:20:43.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>timeSeriesListData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <timeSeriesListData>
      </timeSeriesListData>
    </cmd>
  </payload>
</datagram>
```

A.2.3 Runtime Scenario communication

A.2.3.1 Example content of "deviceConfigurationKeyValueDescriptionListData notify"

deviceConfigurationKeyValueDescriptionListData notify for Scenario 1.

This information usually does not change during runtime. The XML is listed here for the sake of completeness only.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>3</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>150</msgCounter>
    <cmdClassifier>notify</cmdClassifier>
    <timestamp>2026-07-15T16:20:50.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>deviceConfigurationKeyValueDescriptionListData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <deviceConfigurationKeyValueDescriptionListData>
```

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```
        <deviceConfigurationKeyValueDescriptionData>
          <keyId>1</keyId>
          <keyName>incentivesSimulationCyclesMax</keyName>
          <valueType>integer</valueType>
        </deviceConfigurationKeyValueDescriptionData>
        <deviceConfigurationKeyValueDescriptionData>
          <keyId>2</keyId>
          <keyName>incentivesSimulationConcurrent</keyName>
          <valueType>boolean</valueType>
        </deviceConfigurationKeyValueDescriptionData>
        <deviceConfigurationKeyValueDescriptionData>
          <keyId>3</keyId>
          <keyName>incentivesTimeoutIncentiveRequest</keyName>
          <valueType>duration</valueType>
        </deviceConfigurationKeyValueDescriptionData>
        <deviceConfigurationKeyValueDescriptionData>
          <keyId>4</keyId>
          <keyName>incentivesWaitIncentiveWriteable</keyName>
          <valueType>duration</valueType>
        </deviceConfigurationKeyValueDescriptionData>
      </deviceConfigurationKeyValueDescriptionListData>
    </cmd>
  </payload>
</datagram>
```

A.2.3.2 Example content of "deviceConfigurationKeyValueListData notify"

deviceConfigurationKeyValueListData notify for Scenario 1.

This information usually does not change during runtime. The XML is listed here for the sake of completeness only.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>3</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>151</msgCounter>
    <cmdClassifier>notify</cmdClassifier>
    <timestamp>2026-07-15T16:20:51.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>deviceConfigurationKeyValueListData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <deviceConfigurationKeyValueListData>
        <deviceConfigurationKeyValueData>
          <keyId>1</keyId>
          <value>
            <integer>3</integer>
          </value>
        </deviceConfigurationKeyValueData>
        <deviceConfigurationKeyValueData>
          <keyId>2</keyId>
          <value>
            <boolean>true</boolean>
          </value>
```

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```
        </deviceConfigurationKeyValueData>
        <deviceConfigurationKeyValueData>
          <keyId>3</keyId>
          <value>
            <duration>PT60S</duration>
          </value>
        </deviceConfigurationKeyValueData>
        <deviceConfigurationKeyValueData>
          <keyId>4</keyId>
          <value>
            <duration>PT180S</duration>
          </value>
        </deviceConfigurationKeyValueData>
      </deviceConfigurationKeyValueListData>
    </cmd>
  </payload>
</datagram>
```

A.2.3.3 Example content of "incentiveTableDescriptionData notify" for Scenario 2

incentiveTableDescriptionData notify for Scenario 2. This example corresponds to "Step 2" of section 2.4.2.1.2.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>2</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>160</msgCounter>
    <cmdClassifier>notify</cmdClassifier>
    <timestamp>2026-07-15T22:26:00.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>incentiveTableDescriptionData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <incentiveTableDescriptionData>
        <incentiveTableDescription>
          <tariffDescription>
            <tariffId>1</tariffId>
            <tariffWriteable>true</tariffWriteable>
            <updateRequired>true</updateRequired>
          </tariffDescription>
        </incentiveTableDescription>
      </incentiveTableDescriptionData>
    </cmd>
  </payload>
</datagram>
```

A.2.3.4 Example content of "incentiveTableDescriptionData write" for Scenario 2

incentiveTableDescriptionData write for Scenario 2. As the ackRequest is set, a result message with errorNumber "0" will be sent in response to this message if the write was successful.

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Note: Even though the reply example of section A.2.2.6 does not contain "tier" Elements, this example also demonstrates the use of the "delete-add" pattern (see section 3.4.2.4.1.4).

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>2</feature>
    </addressDestination>
    <msgCounter>200</msgCounter>
    <cmdClassifier>write</cmdClassifier>
    <ackRequest>true</ackRequest>
    <timestamp>2026-07-15T22:26:10.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>incentiveTableDescriptionData</function>
      <filter>
        <cmdControl><delete/></cmdControl>
        <incentiveTableDescriptionDataSelectors>
          <tariffDescription>
            <tariffId>1</tariffId>
          </tariffDescription>
        </incentiveTableDescriptionDataSelectors>
      </filter>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <incentiveTableDescriptionData>
        <incentiveTableDescription>
          <tariffDescription>
            <tariffId>1</tariffId>
          </tariffDescription>
            <tier>
              <tierDescription>
                <tierId>1</tierId>
                <tierType>dynamicCost</tierType>
              </tierDescription>
              <boundaryDescription>
                <boundaryId>1</boundaryId>
                <boundaryType>powerBoundary</boundaryType>
                <boundaryUnit>W</boundaryUnit>
              </boundaryDescription>
              <incentiveDescription>
                <incentiveId>1</incentiveId>
                <incentiveType>absoluteCost</incentiveType>
                <currency>EUR</currency>
                <unit>Wh</unit>
              </incentiveDescription>
              <incentiveDescription>
                <incentiveId>2</incentiveId>
                <incentiveType>renewableEnergyPercentage</incentiveType>
                <unit>pct</unit>
              </incentiveDescription>
            </tier>
            <tier>
              <tierDescription>
                <tierId>2</tierId>
                <tierType>dynamicCost</tierType>
              </tierDescription>
              <boundaryDescription>
                <boundaryId>1</boundaryId>
                <boundaryType>powerBoundary</boundaryType>
                <boundaryUnit>W</boundaryUnit>
```

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```
              </boundaryDescription>
              <incentiveDescription>
                <incentiveId>1</incentiveId>
                <incentiveType>absoluteCost</incentiveType>
                <currency>EUR</currency>
                <unit>Wh</unit>
              </incentiveDescription>
              <incentiveDescription>
                <incentiveId>2</incentiveId>
                <incentiveType>renewableEnergyPercentage</incentiveType>
                <unit>pct</unit>
              </incentiveDescription>
            </tier>
            <tier>
              <tierDescription>
                <tierId>3</tierId>
                <tierType>dynamicCost</tierType>
              </tierDescription>
              <boundaryDescription>
                <boundaryId>1</boundaryId>
                <boundaryType>powerBoundary</boundaryType>
                <boundaryUnit>W</boundaryUnit>
              </boundaryDescription>
              <incentiveDescription>
                <incentiveId>1</incentiveId>
                <incentiveType>absoluteCost</incentiveType>
                <currency>EUR</currency>
                <unit>Wh</unit>
              </incentiveDescription>
              <incentiveDescription>
                <incentiveId>2</incentiveId>
                <incentiveType>renewableEnergyPercentage</incentiveType>
                <unit>pct</unit>
              </incentiveDescription>
            </tier>
        </incentiveTableDescription>
      </incentiveTableDescriptionData>
    </cmd>
  </payload>
</datagram>
```

A.2.3.5 Example content of "incentiveTableData write" for Scenario 2

incentiveTableData write for Scenario 2. As the ackRequest is set, a result message with errorNumber "0" will be sent in response to this message if the write was successful.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>2</feature>
    </addressDestination>
    <msgCounter>201</msgCounter>
    <cmdClassifier>write</cmdClassifier>
    <ackRequest>true</ackRequest>
    <timestamp>2026-07-15T22:26:12.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>incentiveTableData</function>
      <filter>
        <cmdControl><delete/></cmdControl>
        <incentiveTableDataSelectors>
```

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```
            <tariff>
              <tariffId>1</tariffId>
            </tariff>
          </incentiveTableDataSelectors>
        </filter>
        <filter>
          <cmdControl>
            <partial/>
          </cmdControl>
        </filter>
        <incentiveTableData>
          <incentiveTable>
            <tariff>
              <tariffId>1</tariffId>
            </tariff>
            <incentiveSlot>
              <timeInterval>
                <timeSlotId>1</timeSlotId>
                <startTime>
                  <dateTime>2026-07-15T20:00:00.0Z</dateTime>
                </startTime>
              </timeInterval>
              <tier>
                <tier>
                  <tierId>3</tierId>
                </tier>
                <boundary>
                  <boundaryId>1</boundaryId>
                  <lowerBoundaryValue>
                    <number>0</number>
                    <scale>0</scale>
                  </lowerBoundaryValue>
                </boundary>
                <incentive>
                  <incentiveId>1</incentiveId>
                  <value>
                    <number>3</number>
                    <scale>-4</scale>
                  </value>
                </incentive>
                <incentive>
                  <incentiveId>2</incentiveId>
                  <value>
                    <number>4</number>
                    <scale>1</scale>
                  </value>
                </incentive>
              </tier>
            </incentiveSlot>
            <incentiveSlot>
              <timeInterval>
                <timeSlotId>2</timeSlotId>
                <startTime>
                  <dateTime>2026-07-16T10:00:00.0Z</dateTime>
                </startTime>
              </timeInterval>
              <tier>
                <tier>
                  <tierId>1</tierId>
                </tier>
                <boundary>
                  <boundaryId>1</boundaryId>
                  <lowerBoundaryValue>
                    <number>0</number>
                    <scale>0</scale>
                  </lowerBoundaryValue>
                </boundary>
                <incentive>
                  <incentiveId>1</incentiveId>
                  <value>
                    <number>0</number>
                    <scale>0</scale>
                  </value>
                </incentive>
                <incentive>
                  <incentiveId>2</incentiveId>
                  <value>
                    <number>1</number>
```

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```
                <scale>2</scale>
              </value>
            </incentive>
          </tier>
          <tier>
            <tier>
              <tierId>2</tierId>
            </tier>
            <boundary>
              <boundaryId>1</boundaryId>
              <lowerBoundaryValue>
                <number>5</number>
                <scale>2</scale>
              </lowerBoundaryValue>
            </boundary>
            <incentive>
              <incentiveId>1</incentiveId>
              <value>
                <number>25</number>
                <scale>-5</scale>
              </value>
            </incentive>
            <incentive>
              <incentiveId>2</incentiveId>
              <value>
                <number>1</number>
                <scale>2</scale>
              </value>
            </incentive>
          </tier>
          <tier>
            <tier>
              <tierId>3</tierId>
            </tier>
            <boundary>
              <boundaryId>1</boundaryId>
              <lowerBoundaryValue>
                <number>3</number>
                <scale>3</scale>
              </lowerBoundaryValue>
            </boundary>
            <incentive>
              <incentiveId>1</incentiveId>
              <value>
                <number>3</number>
                <scale>-4</scale>
              </value>
            </incentive>
            <incentive>
              <incentiveId>2</incentiveId>
              <value>
                <number>4</number>
                <scale>1</scale>
              </value>
            </incentive>
          </tier>
        </incentiveSlot>
        <incentiveSlot>
          <timeInterval>
            <timeSlotId>3</timeSlotId>
            <startTime>
              <dateTime>2026-07-16T16:30:00.0Z</dateTime>
            </startTime>
            <endTime>
              <dateTime>2026-07-17T10:00:00.0Z</dateTime>
            </endTime>
          </timeInterval>
          <tier>
            <tier>
              <tierId>2</tierId>
            </tier>
            <boundary>
              <boundaryId>1</boundaryId>
              <lowerBoundaryValue>
                <number>0</number>
                <scale>0</scale>
              </lowerBoundaryValue>
            </boundary>
```

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```
              <incentive>
                <incentiveId>1</incentiveId>
                <value>
                  <number>25</number>
                  <scale>-5</scale>
                </value>
              </incentive>
              <incentive>
                <incentiveId>2</incentiveId>
                <value>
                  <number>1</number>
                  <scale>2</scale>
                </value>
              </incentive>
            </tier>
            <tier>
              <tier>
                <tierId>3</tierId>
              </tier>
              <boundary>
                <boundaryId>1</boundaryId>
                <lowerBoundaryValue>
                  <number>35</number>
                  <scale>2</scale>
                </lowerBoundaryValue>
              </boundary>
              <incentive>
                <incentiveId>1</incentiveId>
                <value>
                  <number>3</number>
                  <scale>-4</scale>
                </value>
              </incentive>
              <incentive>
                <incentiveId>2</incentiveId>
                <value>
                  <number>4</number>
                  <scale>1</scale>
                </value>
              </incentive>
            </tier>
          </incentiveSlot>
        </incentiveTable>
      </incentiveTableData>
    </cmd>
  </payload>
</datagram>
```

A.2.3.6 Example content of "incentiveTableData notify" for Scenario 2

incentiveTableData notify for Scenario 2. This demonstrates the general notification to the subscribed CEM according to the write operation of section A.2.3.5.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>2</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>170</msgCounter>
    <cmdClassifier>notify</cmdClassifier>
    <timestamp>2026-07-15T22:26:14.0Z</timestamp>
  </header>
```

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```
<payload>
  <cmd>
    <function>incentiveTableData</function>
    <filter>
      <cmdControl><delete/></cmdControl>
      <incentiveTableDataSelectors>
        <tariff>
          <tariffId>1</tariffId>
        </tariff>
      </incentiveTableDataSelectors>
    </filter>
    <filter>
      <cmdControl>
        <partial/>
      </cmdControl>
    </filter>
    <incentiveTableData>
      <incentiveTable>
        <tariff>
          <tariffId>1</tariffId>
        </tariff>
        <incentiveSlot>
          <timeInterval>
            <timeSlotId>1</timeSlotId>
            <startTime>
              <dateTime>2026-07-15T20:00:00.0Z</dateTime>
            </startTime>
          </timeInterval>
          <tier>
            <tier>
              <tierId>3</tierId>
            </tier>
            <boundary>
              <boundaryId>1</boundaryId>
              <lowerBoundaryValue>
                <number>0</number>
                <scale>0</scale>
              </lowerBoundaryValue>
            </boundary>
            <incentive>
              <incentiveId>1</incentiveId>
              <value>
                <number>3</number>
                <scale>-4</scale>
              </value>
            </incentive>
            <incentive>
              <incentiveId>2</incentiveId>
              <value>
                <number>4</number>
                <scale>1</scale>
              </value>
            </incentive>
          </tier>
        </incentiveSlot>
        <incentiveSlot>
          <timeInterval>
            <timeSlotId>2</timeSlotId>
            <startTime>
              <dateTime>2026-07-16T10:00:00.0Z</dateTime>
            </startTime>
          </timeInterval>
          <tier>
            <tier>
              <tierId>1</tierId>
            </tier>
            <boundary>
              <boundaryId>1</boundaryId>
              <lowerBoundaryValue>
                <number>0</number>
                <scale>0</scale>
              </lowerBoundaryValue>
            </boundary>
            <incentive>
              <incentiveId>1</incentiveId>
              <value>
                <number>0</number>
                <scale>0</scale>
```

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```
              </value>
            </incentive>
            <incentive>
              <incentiveId>2</incentiveId>
              <value>
                <number>1</number>
                <scale>2</scale>
              </value>
            </incentive>
          </tier>
          <tier>
            <tier>
              <tierId>2</tierId>
            </tier>
            <boundary>
              <boundaryId>1</boundaryId>
              <lowerBoundaryValue>
                <number>5</number>
                <scale>2</scale>
              </lowerBoundaryValue>
            </boundary>
            <incentive>
              <incentiveId>1</incentiveId>
              <value>
                <number>25</number>
                <scale>-5</scale>
              </value>
            </incentive>
            <incentive>
              <incentiveId>2</incentiveId>
              <value>
                <number>1</number>
                <scale>2</scale>
              </value>
            </incentive>
          </tier>
          <tier>
            <tier>
              <tierId>3</tierId>
            </tier>
            <boundary>
              <boundaryId>1</boundaryId>
              <lowerBoundaryValue>
                <number>3</number>
                <scale>3</scale>
              </lowerBoundaryValue>
            </boundary>
            <incentive>
              <incentiveId>1</incentiveId>
              <value>
                <number>3</number>
                <scale>-4</scale>
              </value>
            </incentive>
            <incentive>
              <incentiveId>2</incentiveId>
              <value>
                <number>4</number>
                <scale>1</scale>
              </value>
            </incentive>
          </tier>
        </incentiveSlot>
        <incentiveSlot>
          <timeInterval>
            <timeSlotId>3</timeSlotId>
            <startTime>
              <dateTime>2026-07-16T16:30:00.0Z</dateTime>
            </startTime>
            <endTime>
              <dateTime>2026-07-17T10:00:00.0Z</dateTime>
            </endTime>
          </timeInterval>
          <tier>
            <tier>
              <tierId>2</tierId>
            </tier>
            <boundary>
```

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```
                <boundaryId>1</boundaryId>
                <lowerBoundaryValue>
                  <number>0</number>
                  <scale>0</scale>
                </lowerBoundaryValue>
              </boundary>
              <incentive>
                <incentiveId>1</incentiveId>
                <value>
                  <number>25</number>
                  <scale>-5</scale>
                </value>
              </incentive>
              <incentive>
                <incentiveId>2</incentiveId>
                <value>
                  <number>1</number>
                  <scale>2</scale>
                </value>
              </incentive>
            </tier>
            <tier>
              <tier>
                <tierId>3</tierId>
              </tier>
              <boundary>
                <boundaryId>1</boundaryId>
                <lowerBoundaryValue>
                  <number>35</number>
                  <scale>2</scale>
                </lowerBoundaryValue>
              </boundary>
              <incentive>
                <incentiveId>1</incentiveId>
                <value>
                  <number>3</number>
                  <scale>-4</scale>
                </value>
              </incentive>
              <incentive>
                <incentiveId>2</incentiveId>
                <value>
                  <number>4</number>
                  <scale>1</scale>
                </value>
              </incentive>
            </tier>
          </incentiveSlot>
        </incentiveTable>
      </incentiveTableData>
    </cmd>
  </payload>
</datagram>
```

A.2.3.7 Example content of "incentiveTableConstraintsData notify" for Scenario 2

incentiveTableConstraintsData notify for Scenario 2.

This information usually does not change during runtime. The XML is listed here for the sake of completeness only.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>2</feature>
    </addressSource>
```

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```
      <addressDestination>
        <device>d:_i:46925_CEM</device>
        <entity>1</entity>
        <feature>1</feature>
      </addressDestination>
      <msgCounter>171</msgCounter>
      <cmdClassifier>notify</cmdClassifier>
      <timestamp>2026-07-15T22:26:15.0Z</timestamp>
    </header>
    <payload>
      <cmd>
        <function>incentiveTableConstraintsData</function>
        <filter>
          <cmdControl>
            <partial/>
          </cmdControl>
        </filter>
        <incentiveTableConstraintsData>
          <incentiveTableConstraints>
            <tariff>
              <tariffId>1</tariffId>
            </tariff>
            <tariffConstraints>
              <maxTiersPerTariff>5</maxTiersPerTariff>
              <maxBoundariesPerTier>1</maxBoundariesPerTier>
              <maxIncentivesPerTier>3</maxIncentivesPerTier>
            </tariffConstraints>
            <incentiveSlotConstraints>
              <slotCountMax>1024</slotCountMax>
            </incentiveSlotConstraints>
          </incentiveTableConstraints>
        </incentiveTableConstraintsData>
      </cmd>
    </payload>
  </datagram>
```

A.2.3.8 Example content of "timeSeriesDescriptionListData notify" for Scenario 2

timeSeriesDescriptionListData notify for Scenario 2.

This information usually does not change during runtime. The XML is listed here for the sake of completeness only.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>180</msgCounter>
    <cmdClassifier>notify</cmdClassifier>
    <timestamp>2026-07-15T22:26:16.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>timeSeriesDescriptionListData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <timeSeriesDescriptionListData>
```

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```
        <timeSeriesDescriptionData>
          <timeSeriesId>1</timeSeriesId>
          <timeSeriesType>plan</timeSeriesType>
          <timeSeriesWriteable>false</timeSeriesWriteable>
          <measurementId>1</measurementId>
          <unit>W</unit>
          <scopeType>committedPowerPlan</scopeType>
        </timeSeriesDescriptionData>
      </timeSeriesDescriptionListData>
    </cmd>
  </payload>
</datagram>
```

A.2.3.9 Example content of "timeSeriesListData notify" for Scenario 2

timeSeriesListData notify for Scenario 2.

Note: Even though the reply example of section A.2.2.14 does not contain data for the requested Power Plan, this example also demonstrates the use of the "delete-add" pattern (see section 3.4.2.4.2.3), as this is also the usual approach for subsequent updates of this Power Plan.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>181</msgCounter>
    <cmdClassifier>notify</cmdClassifier>
    <timestamp>2026-07-15T22:26:17.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>timeSeriesListData</function>
      <filter>
        <cmdControl><delete/></cmdControl>
        <incentiveTableDataSelectors>
          <tariff>
            <tariffId>1</tariffId>
          </tariff>
        </incentiveTableDataSelectors>
      </filter>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <timeSeriesListData>
        <timeSeriesData>
          <timeSeriesId>1</timeSeriesId>
          <timeSeriesSlot>
            <timeSeriesSlotId>1</timeSeriesSlotId>
            <timePeriod>
              <startTime>2026-07-15T22:26:17.0Z</startTime>
            </timePeriod>
            <duration>PT1H17M3S</duration>
            <value>
              <number>0</number>
              <scale>0</scale>
            </value>
          </timeSeriesSlot>
```

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```
            <timeSeriesSlot>
              <timeSeriesSlotId>2</timeSeriesSlotId>
              <timePeriod>
                <startTime>2026-07-15T23:43:20.0Z</startTime>
              </timePeriod>
              <duration>PT27M</duration>
              <value>
                <number>17</number>
                <scale>2</scale>
              </value>
              <minValue>
                <number>0</number>
                <scale>0</scale>
              </minValue>
              <maxValue>
                <number>2</number>
                <scale>3</scale>
              </maxValue>
            </timeSeriesSlot>
            <timeSeriesSlot>
              <timeSeriesSlotId>3</timeSeriesSlotId>
              <timePeriod>
                <startTime>2026-07-16T00:10:00.0Z</startTime>
              </timePeriod>
              <duration>PT8M</duration>
              <value>
                <number>5</number>
                <scale>2</scale>
              </value>
              <minValue>
                <number>0</number>
                <scale>0</scale>
              </minValue>
              <maxValue>
                <number>1</number>
                <scale>3</scale>
              </maxValue>
            </timeSeriesSlot>
          </timeSeriesData>
        </timeSeriesListData>
      </cmd>
    </payload>
</datagram>
```

A.2.3.10 Example content of "incentiveTableDescriptionData notify" for Scenario 3

incentiveTableDescriptionData notify for Scenario 3. This example corresponds to "Step 2" of section 2.4.3.1.2.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>2</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>200</msgCounter>
    <cmdClassifier>notify</cmdClassifier>
    <timestamp>2026-07-15T22:50:00.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>incentiveTableDescriptionData</function>
      <filter>
```

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```
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <incentiveTableDescriptionData>
        <incentiveTableDescription>
          <tariffDescription>
            <tariffId>2</tariffId>
            <tariffWriteable>true</tariffWriteable>
          </tariffDescription>
        </incentiveTableDescription>
      </incentiveTableDescriptionData>
    </cmd>
  </payload>
</datagram>
```

A.2.3.11 Example content of "incentiveTableDescriptionData write" for Scenario 3

incentiveTableDescriptionData write for Scenario 3. As the ackRequest is set, a result message with errorNumber "0" will be sent in response to this message if the write was successful.

Note: Even though the reply example of section A.2.2.6 does not contain "tier" Elements, this example also demonstrates the use of the "delete-add" pattern (see section 3.4.3.4.1.3).

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>2</feature>
    </addressDestination>
    <msgCounter>300</msgCounter>
    <cmdClassifier>write</cmdClassifier>
    <ackRequest>true</ackRequest>
    <timestamp>2026-07-15T22:50:10.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>incentiveTableDescriptionData</function>
      <filter>
        <cmdControl><delete/></cmdControl>
        <incentiveTableDescriptionDataSelectors>
          <tariffDescription>
            <tariffId>2</tariffId>
          </tariffDescription>
        </incentiveTableDescriptionDataSelectors>
      </filter>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <incentiveTableDescriptionData>
        <incentiveTableDescription>
          <tariffDescription>
            <tariffId>2</tariffId>
          </tariffDescription>
            <tier>
              <tierDescription>
                <tierId>1</tierId>
                <tierType>dynamicCost</tierType>
              </tierDescription>
```

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```
              <boundaryDescription>
                <boundaryId>1</boundaryId>
                <boundaryType>powerBoundary</boundaryType>
                <boundaryUnit>W</boundaryUnit>
              </boundaryDescription>
              <incentiveDescription>
                <incentiveId>1</incentiveId>
                <incentiveType>absoluteCost</incentiveType>
                <currency>EUR</currency>
                <unit>Wh</unit>
              </incentiveDescription>
              <incentiveDescription>
                <incentiveId>2</incentiveId>
                <incentiveType>renewableEnergyPercentage</incentiveType>
                <unit>pct</unit>
              </incentiveDescription>
            </tier>
            <tier>
              <tierDescription>
                <tierId>2</tierId>
                <tierType>dynamicCost</tierType>
              </tierDescription>
              <boundaryDescription>
                <boundaryId>1</boundaryId>
                <boundaryType>powerBoundary</boundaryType>
                <boundaryUnit>W</boundaryUnit>
              </boundaryDescription>
              <incentiveDescription>
                <incentiveId>1</incentiveId>
                <incentiveType>absoluteCost</incentiveType>
                <currency>EUR</currency>
                <unit>Wh</unit>
              </incentiveDescription>
              <incentiveDescription>
                <incentiveId>2</incentiveId>
                <incentiveType>renewableEnergyPercentage</incentiveType>
                <unit>pct</unit>
              </incentiveDescription>
            </tier>
            <tier>
              <tierDescription>
                <tierId>3</tierId>
                <tierType>dynamicCost</tierType>
              </tierDescription>
              <boundaryDescription>
                <boundaryId>1</boundaryId>
                <boundaryType>powerBoundary</boundaryType>
                <boundaryUnit>W</boundaryUnit>
              </boundaryDescription>
              <incentiveDescription>
                <incentiveId>1</incentiveId>
                <incentiveType>absoluteCost</incentiveType>
                <currency>EUR</currency>
                <unit>Wh</unit>
              </incentiveDescription>
              <incentiveDescription>
                <incentiveId>2</incentiveId>
                <incentiveType>renewableEnergyPercentage</incentiveType>
                <unit>pct</unit>
              </incentiveDescription>
            </tier>
          </incentiveTableDescription>
        </incentiveTableDescriptionData>
      </cmd>
    </payload>
</datagram>
```

A.2.3.12 Example content of "incentiveTableData write" for Scenario 3

incentiveTableData write for Scenario 3. As the ackRequest is set, a result message with errorNumber "0" will be sent in response to this message if the write was successful.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
```

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```
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>2</feature>
    </addressDestination>
    <msgCounter>301</msgCounter>
    <cmdClassifier>write</cmdClassifier>
    <ackRequest>true</ackRequest>
    <timestamp>2026-07-15T22:50:11.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>incentiveTableData</function>
      <filter>
        <cmdControl><delete/></cmdControl>
        <incentiveTableDataSelectors>
          <tariff>
            <tariffId>2</tariffId>
          </tariff>
        </incentiveTableDataSelectors>
      </filter>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <incentiveTableData>
        <incentiveTable>
          <tariff>
            <tariffId>2</tariffId>
          </tariff>
          <incentiveSlot>
            <timeInterval>
              <timeSlotId>1</timeSlotId>
              <startTime>
                <dateTime>2026-07-15T20:00:00.0Z</dateTime>
              </startTime>
            </timeInterval>
            <tier>
              <tier>
                <tierId>3</tierId>
              </tier>
              <boundary>
                <boundaryId>1</boundaryId>
                <lowerBoundaryValue>
                  <number>0</number>
                  <scale>0</scale>
                </lowerBoundaryValue>
              </boundary>
              <incentive>
                <incentiveId>1</incentiveId>
                <value>
                  <number>3</number>
                  <scale>-4</scale>
                </value>
              </incentive>
              <incentive>
                <incentiveId>2</incentiveId>
                <value>
                  <number>4</number>
                  <scale>1</scale>
                </value>
              </incentive>
            </tier>
          </incentiveSlot>
          <incentiveSlot>
            <timeInterval>
              <timeSlotId>2</timeSlotId>
```

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```
            <startTime>
              <dateTime>2026-07-16T10:00:00.0Z</dateTime>
            </startTime>
          </timeInterval>
          <tier>
            <tier>
              <tierId>2</tierId>
            </tier>
            <boundary>
              <boundaryId>1</boundaryId>
              <lowerBoundaryValue>
                <number>0</number>
                <scale>0</scale>
              </lowerBoundaryValue>
            </boundary>
            <incentive>
              <incentiveId>1</incentiveId>
              <value>
                <number>25</number>
                <scale>-5</scale>
              </value>
            </incentive>
            <incentive>
              <incentiveId>2</incentiveId>
              <value>
                <number>1</number>
                <scale>2</scale>
              </value>
            </incentive>
          </tier>
          <tier>
            <tier>
              <tierId>3</tierId>
            </tier>
            <boundary>
              <boundaryId>1</boundaryId>
              <lowerBoundaryValue>
                <number>17</number>
                <scale>2</scale>
              </lowerBoundaryValue>
            </boundary>
            <incentive>
              <incentiveId>1</incentiveId>
              <value>
                <number>3</number>
                <scale>-4</scale>
              </value>
            </incentive>
            <incentive>
              <incentiveId>2</incentiveId>
              <value>
                <number>4</number>
                <scale>1</scale>
              </value>
            </incentive>
          </tier>
        </incentiveSlot>
        <incentiveSlot>
          <timeInterval>
            <timeSlotId>3</timeSlotId>
            <startTime>
              <dateTime>2026-07-16T16:30:00.0Z</dateTime>
            </startTime>
            <endTime>
              <dateTime>2026-07-17T10:00:00.0Z</dateTime>
            </endTime>
          </timeInterval>
          <tier>
            <tier>
              <tierId>1</tierId>
            </tier>
            <boundary>
              <boundaryId>1</boundaryId>
              <lowerBoundaryValue>
                <number>0</number>
                <scale>0</scale>
              </lowerBoundaryValue>
            </boundary>
```

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```
              <incentive>
                <incentiveId>1</incentiveId>
                <value>
                  <number>0</number>
                  <scale>0</scale>
                </value>
              </incentive>
              <incentive>
                <incentiveId>2</incentiveId>
                <value>
                  <number>1</number>
                  <scale>2</scale>
                </value>
              </incentive>
            </tier>
            <tier>
              <tier>
                <tierId>2</tierId>
              </tier>
              <boundary>
                <boundaryId>1</boundaryId>
                <lowerBoundaryValue>
                  <number>11</number>
                  <scale>2</scale>
                </lowerBoundaryValue>
              </boundary>
              <incentive>
                <incentiveId>1</incentiveId>
                <value>
                  <number>25</number>
                  <scale>-5</scale>
                </value>
              </incentive>
              <incentive>
                <incentiveId>2</incentiveId>
                <value>
                  <number>1</number>
                  <scale>2</scale>
                </value>
              </incentive>
            </tier>
            <tier>
              <tier>
                <tierId>3</tierId>
              </tier>
              <boundary>
                <boundaryId>1</boundaryId>
                <lowerBoundaryValue>
                  <number>35</number>
                  <scale>2</scale>
                </lowerBoundaryValue>
              </boundary>
              <incentive>
                <incentiveId>1</incentiveId>
                <value>
                  <number>3</number>
                  <scale>-4</scale>
                </value>
              </incentive>
              <incentive>
                <incentiveId>2</incentiveId>
                <value>
                  <number>4</number>
                  <scale>1</scale>
                </value>
              </incentive>
            </tier>
          </incentiveSlot>
        </incentiveTable>
      </incentiveTableData>
    </cmd>
  </payload>
</datagram>
```

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A.2.3.13 Example content of "incentiveTableData notify" for Scenario 3

incentiveTableData notify for Scenario 3. This demonstrates the general notification to the subscribed CEM according to the write operation of section A.2.3.12.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>2</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>210</msgCounter>
    <cmdClassifier>notify</cmdClassifier>
    <timestamp>2026-07-15T22:50:12.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>incentiveTableData</function>
      <filter>
        <cmdControl><delete/></cmdControl>
        <incentiveTableDataSelectors>
          <tariff>
            <tariffId>2</tariffId>
          </tariff>
        </incentiveTableDataSelectors>
      </filter>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <incentiveTableData>
        <incentiveTable>
          <tariff>
            <tariffId>2</tariffId>
          </tariff>
          <incentiveSlot>
            <timeInterval>
              <timeSlotId>1</timeSlotId>
              <startTime>
                <dateTime>2026-07-15T20:00:00.0Z</dateTime>
              </startTime>
            </timeInterval>
            <tier>
              <tier>
                <tierId>3</tierId>
              </tier>
              <boundary>
                <boundaryId>1</boundaryId>
                <lowerBoundaryValue>
                  <number>0</number>
                  <scale>0</scale>
                </lowerBoundaryValue>
              </boundary>
              <incentive>
                <incentiveId>1</incentiveId>
                <value>
                  <number>3</number>
                  <scale>-4</scale>
                </value>
              </incentive>
              <incentive>
                <incentiveId>2</incentiveId>
                <value>
                  <number>4</number>
                  <scale>1</scale>
```

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```
              </value>
            </incentive>
          </tier>
        </incentiveSlot>
        <incentiveSlot>
          <timeInterval>
            <timeSlotId>2</timeSlotId>
            <startTime>
              <dateTime>2026-07-16T10:00:00.0Z</dateTime>
            </startTime>
          </timeInterval>
          <tier>
            <tier>
              <tierId>2</tierId>
            </tier>
            <boundary>
              <boundaryId>1</boundaryId>
              <lowerBoundaryValue>
                <number>0</number>
                <scale>0</scale>
              </lowerBoundaryValue>
            </boundary>
            <incentive>
              <incentiveId>1</incentiveId>
              <value>
                <number>25</number>
                <scale>-5</scale>
              </value>
            </incentive>
            <incentive>
              <incentiveId>2</incentiveId>
              <value>
                <number>1</number>
                <scale>2</scale>
              </value>
            </incentive>
          </tier>
          <tier>
            <tier>
              <tierId>3</tierId>
            </tier>
            <boundary>
              <boundaryId>1</boundaryId>
              <lowerBoundaryValue>
                <number>17</number>
                <scale>2</scale>
              </lowerBoundaryValue>
            </boundary>
            <incentive>
              <incentiveId>1</incentiveId>
              <value>
                <number>3</number>
                <scale>-4</scale>
              </value>
            </incentive>
            <incentive>
              <incentiveId>2</incentiveId>
              <value>
                <number>4</number>
                <scale>1</scale>
              </value>
            </incentive>
          </tier>
        </incentiveSlot>
        <incentiveSlot>
          <timeInterval>
            <timeSlotId>3</timeSlotId>
            <startTime>
              <dateTime>2026-07-16T16:30:00.0Z</dateTime>
            </startTime>
            <endTime>
              <dateTime>2026-07-17T10:00:00.0Z</dateTime>
            </endTime>
          </timeInterval>
          <tier>
            <tier>
              <tierId>1</tierId>
            </tier>
```

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```
            <boundary>
              <boundaryId>1</boundaryId>
              <lowerBoundaryValue>
                <number>0</number>
                <scale>0</scale>
              </lowerBoundaryValue>
            </boundary>
            <incentive>
              <incentiveId>1</incentiveId>
              <value>
                <number>0</number>
                <scale>0</scale>
              </value>
            </incentive>
            <incentive>
              <incentiveId>2</incentiveId>
              <value>
                <number>1</number>
                <scale>2</scale>
              </value>
            </incentive>
          </tier>
          <tier>
            <tier>
              <tierId>2</tierId>
            </tier>
            <boundary>
              <boundaryId>1</boundaryId>
              <lowerBoundaryValue>
                <number>11</number>
                <scale>2</scale>
              </lowerBoundaryValue>
            </boundary>
            <incentive>
              <incentiveId>1</incentiveId>
              <value>
                <number>25</number>
                <scale>-5</scale>
              </value>
            </incentive>
            <incentive>
              <incentiveId>2</incentiveId>
              <value>
                <number>1</number>
                <scale>2</scale>
              </value>
            </incentive>
          </tier>
          <tier>
            <tier>
              <tierId>3</tierId>
            </tier>
            <boundary>
              <boundaryId>1</boundaryId>
              <lowerBoundaryValue>
                <number>35</number>
                <scale>2</scale>
              </lowerBoundaryValue>
            </boundary>
            <incentive>
              <incentiveId>1</incentiveId>
              <value>
                <number>3</number>
                <scale>-4</scale>
              </value>
            </incentive>
            <incentive>
              <incentiveId>2</incentiveId>
              <value>
                <number>4</number>
                <scale>1</scale>
              </value>
            </incentive>
          </tier>
        </incentiveSlot>
      </incentiveTable>
    </incentiveTableData>
  </cmd>
```

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```
  </payload>
</datagram>
```

A.2.3.14 Example content of "incentiveTableConstraintsData notify" for Scenario 3

incentiveTableConstraintsData notify for Scenario 3.

This information usually does not change during runtime. The XML is listed here for the sake of completeness only.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>2</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>211</msgCounter>
    <cmdClassifier>notify</cmdClassifier>
    <timestamp>2026-07-15T22:50:13.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>incentiveTableConstraintsData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <incentiveTableConstraintsData>
        <incentiveTableConstraints>
          <tariff>
            <tariffId>2</tariffId>
          </tariff>
          <tariffConstraints>
            <maxTiersPerTariff>5</maxTiersPerTariff>
            <maxBoundariesPerTier>1</maxBoundariesPerTier>
            <maxIncentivesPerTier>3</maxIncentivesPerTier>
          </tariffConstraints>
          <incentiveSlotConstraints>
            <slotCountMax>1024</slotCountMax>
          </incentiveSlotConstraints>
        </incentiveTableConstraints>
      </incentiveTableConstraintsData>
    </cmd>
  </payload>
</datagram>
```

A.2.3.15 Example content of "timeSeriesDescriptionListData notify" for Scenario 3

timeSeriesDescriptionListData notify for Scenario 3.

This information usually does not change during runtime. The XML is listed here for the sake of completeness only.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

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```
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>220</msgCounter>
    <cmdClassifier>notify</cmdClassifier>
    <timestamp>2026-07-15T22:50:20.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>timeSeriesDescriptionListData</function>
      <filter>
        <cmdControl>
          <partial/>
        </cmdControl>
      </filter>
      <timeSeriesDescriptionListData>
        <timeSeriesDescriptionData>
          <timeSeriesId>2</timeSeriesId>
          <timeSeriesType>plan</timeSeriesType>
          <timeSeriesWriteable>false</timeSeriesWriteable>
          <measurementId>1</measurementId>
          <unit>W</unit>
          <scopeType>preliminaryPowerPlan</scopeType>
        </timeSeriesDescriptionData>
      </timeSeriesDescriptionListData>
    </cmd>
  </payload>
</datagram>
```

A.2.3.16 Example content of "timeSeriesListData notify" for Scenario 3

timeSeriesListData notify for Scenario 3.

Note: Even though the reply example of section A.2.2.14 does not contain data for the requested Power Plan, this example also demonstrates the use of the "delete-add" pattern (see section 3.4.3.4.2.2), as this is also the usual approach for subsequent updates of this Power Plan.

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
  file:../../XSDs/EEBus_SPINE_TS_Datagram.xsd">
  <header>
    <specificationVersion>1.3.0</specificationVersion>
    <addressSource>
      <device>d:_i:46925_HeatPump</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressSource>
    <addressDestination>
      <device>d:_i:46925_CEM</device>
      <entity>1</entity>
      <feature>1</feature>
    </addressDestination>
    <msgCounter>221</msgCounter>
    <cmdClassifier>notify</cmdClassifier>
    <timestamp>2026-07-15T22:50:21.0Z</timestamp>
  </header>
  <payload>
    <cmd>
      <function>timeSeriesListData</function>
```

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```
        <filter>
          <cmdControl><delete/></cmdControl>
          <timeSeriesListDataSelectors>
            <timeSeriesId>2</timeSeriesId>
          </timeSeriesListDataSelectors>
        </filter>
        <filter>
          <cmdControl>
            <partial/>
          </cmdControl>
        </filter>
        <timeSeriesListData>
          <timeSeriesData>
            <timeSeriesId>2</timeSeriesId>
            <timeSeriesSlot>
              <timeSeriesSlotId>1</timeSeriesSlotId>
              <timePeriod>
                <startTime>2026-07-15T22:26:00.0Z</startTime>
              </timePeriod>
              <duration>PT8H2M</duration>
              <value>
                <number>0</number>
                <scale>0</scale>
              </value>
            </timeSeriesSlot>
            <timeSeriesSlot>
              <timeSeriesSlotId>2</timeSeriesSlotId>
              <timePeriod>
                <startTime>2026-07-16T06:28:00.0Z</startTime>
              </timePeriod>
              <duration>PT27M</duration>
              <value>
                <number>19</number>
                <scale>2</scale>
              </value>
              <minValue>
                <number>0</number>
                <scale>0</scale>
              </minValue>
              <maxValue>
                <number>2</number>
                <scale>3</scale>
              </maxValue>
            </timeSeriesSlot>
            <timeSeriesSlot>
              <timeSeriesSlotId>3</timeSeriesSlotId>
              <timePeriod>
                <startTime>2026-07-16T06:55:00.0Z</startTime>
              </timePeriod>
              <duration>PT8M</duration>
              <value>
                <number>45</number>
                <scale>1</scale>
              </value>
              <minValue>
                <number>0</number>
                <scale>0</scale>
              </minValue>
              <maxValue>
                <number>1</number>
                <scale>3</scale>
              </maxValue>
            </timeSeriesSlot>
          </timeSeriesData>
        </timeSeriesListData>
      </cmd>
    </payload>
  </datagram>
```

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